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Motivations & Context

Objectifs du travail

Organisation of the thesis

Chapter 1

Metamaterials using lossy media for acoustic absorption

Contents

The present chapter is intended as an introduction to the fundamental concepts deemed necessary for the good understanding of the manuscript. Some theoretical aspects will be presented as well as some bibliography elements, which are useful for the context the present research takes place in.

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historique ? DBM -> JCA -> Biot dissipation mechanisms in porous layers

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thèses: natacha aberkane-gauthier, théo cavalieri, macaglione ?

Chapter

2 Template d'article en tant que chapitre

Contents

This chapter is based on the material published in REF. Minor modifications to the text and figures have been made to fit the format and notation of this thesis.

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The objective of this work is to model the acoustic response of multilayered structures composed of both homogeneous layers and periodic cells to obtain their scattering properties (i.e. reflection and transmission coefficients). Physical constituents are elastic, fluid, and poroelastic materials. A method for predicting the transfer matrix of a periodic cell is first proposed. This method was implemented to be compatible with higher-order finite-elements. Subsequently, two modelling methods from the literature, the Global Method and the Transfer Matrix Method, initially dedicated to stacks of homogeneous layers, are extended to enable the modelling of acoustic grating stacks.

The methods are then tested, and results confirm that they are both suited to higher-order finite elements. The results from these are also tested on examples from the literature or adaptations thereof, in order to have a complete panel of test configurations. Stability issues are then studied and were shown to arise for the Transfer Matrix Method in some cases. The Global Method remains stable for the studied configurations. Finally, a method for the determination of the series truncation of the Bloch modes is proposed.

2.5 Introduction

Acoustic sound packages, used for both absorption and transmission problems, are generally stacks made up of several layers of acoustic structures, some of them having dissipative properties. Originally, dissipation was provided by porous materials that transform mechanical energy into heat through viscous, thermal, and structural damping [1–3]. However, the performance of these materials is relatively limited at low frequencies, which is especially detrimental because this frequency range is the one that needs to be controlled in practical applications. In the last 15 years metaporous materials have been developed to overcome this limitation [4–9]. These materials are based on the combination of a porous core with periodic inclusions, some of which are possibly resonant structures [10–14]. Other configurations have been studied, among them the periodic partitioning of the metaporous layer [9] or an embedded labyrinthine channel [15].

The efficient design of these structures is of utmost importance in noise control engineering in both the transport and the building industry. Then it is necessary to develop accurate models to predict their behaviour to optimise their performance [16–19]. For multilayered structures composed only of homogeneous layers, plane wave methods are the most common [1]. Among them, we can highlight two of them. The first is the Transfer Matrix Method (TMM) [20–22], which involves constructing a matrix (called the Transfer Matrix) linking the state vectors at the interfaces of each layer, and then the state vectors at the interfaces of the stack are linked by interface relations. A second method is the Global Method (GM) [23], in which the state vectors at the interfaces are written in terms of the amplitudes of the waves propagating inside the layers. The interface relations are then written, and a global linear system is obtained that links all layer amplitudes. This second method has been shown to be robust, especially when the spatial origin of the waves is properly chosen. However, the TMM is inherently unstable, especially for layers of great thickness or with high attenuation rates. So, although theoretically exact, the TMM leads to instabilities when implemented in finite-precision arithmetic. Methods have been proposed to overcome this instability, in both acoustics and optics [24–26]. Among them, Dazel *et al.* [27] proposed a Recursive Method (RM) to fix the stability issues of the TMM. This method is based on reformulation of the state vector as a product of a translation matrix by an information vector and an adequate factorisation of the growing exponential terms of the wave numbers in the propagation direction. These vectors and matrices are then updated recursively starting from the top interface and going down to the bottom one. The final linear system to solve is then relative to the incident interface and is then quite small compared to the one of the GM. This method then mixes the advantages of the TMM and the GM without their drawbacks.

For periodic cells, it is not possible to use these methods directly because the concept of waves is not applicable. In addition, contrary to the case of stacks of homogeneous structures, periodicity induces the presence of not only a single mode but also an infinity (called Bloch modes). Then a common solution can be to predict the response using the Finite Element Method (FEM) [28–30]. The key advantage of FEM is its ability to handle complex geometries and material properties. Additionally, FEM is particularly well-suited to Floquet-Bloch relations. In recent years, higher-order FEM has been studied [31] and provides interesting capabilities in acoustics [32, 33] and especially for poroelastic materials [34]. However, one major drawback of this method is its computational cost. This is particularly true for the problems we are interested in, since a full FEM model of a sound package generally also requires discretization of the homogeneous layers and the incident and transmitting media (which in most cases must themselves be coupled to PMLs) [7, 17, 19, 35]. This not only generates a significant numerical overhead but also does not concentrate the degrees of freedom (and then the computational resources) on the periodic cells.

For periodic structures, several works have focused on obtaining dispersion curves, whether using SAFE or WFEM. These works have focused on mechanical structures [36–39], even if some contributions relate to porous materials with [40, 41] or without inclusions [42], or also periodic cells of alternating materials [43]. A phononic crystal made up of a poroelastic host was investigated with cylindrical poroelastic inclusions [44] or with other inclusion material [45]. However, these methods do not allow us to obtain the response of the structure directly because of the underlying eigenvalue problem formulation, which is often used for wave dispersion solving, and none of these works has focused on stacks. Recently, pioneering contributions have proposed methods to determine the transfer matrix of periodic cells on the basis of a finite element model[46, 47]. These contributions have paved the way for the modelling of stacks with periodic cells and form the starting point for this work. They focus on the periodic cell, and the discretisation is done only by a quadratic element. Hence, strategies to couple them to homogeneous layers are not studied as well as stability issues.

The aim of this paper is therefore to extend this work to the case of stacks. More specifically, we will first propose an approach slightly different from the one presented in [46, 47]. The proposed approach is compatible with high-order finite elements. Secondly, we propose extensions for the GM and TMM methods for the case of grating stacks. The modularity of the method allows for the modelling of a wide range of configurations (arbitrary stack of layers with scatterers of arbitrary geometry). In Section 2, the problem will be presented along with the different quantities of interest. Section 3 focusses on the modelling of homogeneous and periodic cells and the coupling between them. The key to this coupling is to adapt the FE discretization of the problem so that the appropriate state vector can be expressed by relating the acoustic properties from one end of the layer to the other. The extensions for the two methods resulting are further presented. The validation results are then presented in Section 4 putting in perspective the GM and TMM performance with examples covering the range of configurations described in the literature. Section 5 is a conclusion.

2.6 Position of the problem

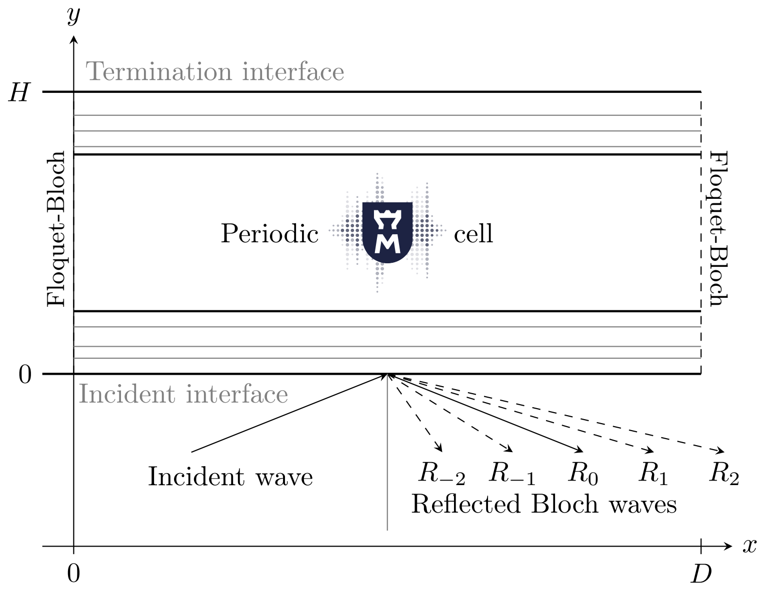


Figure 2.1: Problem of interest

We are interested in the problem presented in Fig. 2.1 corresponding to the acoustic response of a flat multilayered structure composed of homogeneous layers and periodic cells. Let D be the period of the cell. The structure (of total height H) is assumed to be infinite in the perpendicular direction and subjected to an incident plane wave at pulsation ω with an angle θ to the normal of the panel. On top of the structure, a termination condition is imposed. It can be a rigid boundary or radiation in an infinite air medium. In all this paper, a positive Fourier transform convention is used and the common $e^{j\omega t}$ (with $j^2 = -1$) term is omitted in all expressions. The incident pressure is expressed as:

$$p_i(x, y) = e^{-j(k_x x + k_y y)} \tag{2.1}$$

The amplitude of the incident field is unity, as the physical system is linear. The wave numbers in the x and y directions are defined by

$$k_x = \frac{\omega}{c_0}\sin(\theta)\,,\ k_y = \frac{\omega}{c_0}\cos(\theta), \tag{2.2}$$

with c_0 the celerity of acoustic waves in the air. The reflected field p_r is, due to periodicity of the structure, composed of a countable superposition of Bloch modes:

$$p_r(x, y) = \sum_{w\in\mathbb{Z}} R_w e^{-j[k_x^{(w)}x - k_y^{(w)}y]} \tag{2.3}$$

with

$$k_x^{(w)} = k_x + \frac{w\pi}{D}\,,\ k_y^{(w)} = \sqrt{k_0^2 - k_x^{(w)2}}. \tag{2.4}$$

R_w corresponds to the reflection coefficient of wave w. Note that the origin of the reflected waves is chosen at the incident interface. The separation of the x and y dependences always plays a role in these approaches. Let e_w be the function of x that represents the dependence on x. One has the following:

$$e_w(x) = e^{-jk_x^{(w)}x}, \; p_r(x, y) = \sum_{w \in \mathbb{Z}} R_w \, e^{jk_y^{(w)}y} e_w(x) \tag{2.5}$$

One key property of these functions is their orthogonality:

$$\forall\, w, w' \in \mathbb{Z}, \; < e_w, e_{w'} >= \int_0^D \overline{e_w(x)} e_{w'}(x) dx = D\delta_{ww'}. \tag{2.6}$$

The overline bar represents the complex conjugation.

On the top interface, a termination condition is imposed. It can be a rigid boundary or a radiation condition in an infinite medium. The rigid termination condition corresponds to zero displacement. For a transmission condition, the transmitted pressure is defined as

$$p_t(x, y) = \sum_{w \in \mathbb{Z}} T_w \, e^{-jk_y^{(w)}(y-H)} \, e_w(x), \tag{2.7}$$

where T_w are the transmission coefficients. Their origin is chosen at the termination interface.

The number of Bloch modes is infinite. In practice, a truncation is always performed. An integer N will be defined and all sums will be truncated to $-N \leqslant w \leqslant N$. The total number of Bloch modes is then

$$n_B = 2N + 1. \tag{2.8}$$

A discussion on the choice of N is presented in section 2.8.4.

2.7 Models for the propagation in multilayers

In this section, we present the different methods. They are the GM whose unknowns are the amplitudes of the waves propagating in the different layers and the TMM. These methods were initially developed for multilayers that comprise only homogeneous layers. We will therefore modify them slightly to take into account the presence of Bloch modes in homogeneous layers. section 2.7.1 presents the modifications in the case of homogeneous layers. For periodic layers, in section 2.7.2 we propose an original method to relate physical quantities at the interfaces. The interface relations are presented in section 2.7.3 and the methods for obtaining the multilayer response are then presented in section 2.7.4 .

One of the main characters of this work is the state vector $\boldsymbol{S}$, which gathers a list of relevant physical fields that represent the dynamics of the layer. In the context of a homogeneous layer and in the absence of Bloch modes, the number of components of this vector is $2n_w$ where n_w is the number of waves in the layer. The second and third columns of Table 2.1 show the value of n_w and a selection of fields in $\boldsymbol{S}$ for elastic, fluid, and poroelastic materials. Chapter 11 of [1] contains more details on state vectors.

The interface relations will relate the state vectors at the interfaces between two layers. In order to avoid confusion with $\boldsymbol{S}$, which is a function of space, a specific notation is considered for the values of state variables at the interface: $\boldsymbol{\Sigma}$. Throughout this section, we will use the superscripts t and b to

Medium	n_w	State variables $\boldsymbol{S}$	Primal variable $\boldsymbol{\pi}$	Dual variable $\boldsymbol{b}$
Fluid	1	$\{u_y, p\}$	p	$\dfrac{1}{\rho\omega^2}\dfrac{\partial p}{\partial n}$
Elastic	2	$\{\sigma_{xy}, u_y, \sigma_{yy}, u_x\}$	$\boldsymbol{u}^e$	$\boldsymbol{\sigma}^e.\boldsymbol{n}$
Poroelastic	3	$\left\{\hat{\sigma}_{xy}, u^s_y, u^t_y, \hat{\sigma}_{yy}, p, u^s_x\right\}$	$\{\mathbf{u}^s, p\}$	$\left\{\hat{\boldsymbol{\sigma}}^s.\boldsymbol{n}, \dfrac{1}{\rho\omega^2}\dfrac{\partial p}{\partial n}\right\}$

Table 2.1: Variables for plane waves and finite-element methods

designate the quantities at the top and bottom interfaces of each layer (e.g. y^t and y^b will designate the coordinates of the upper and lower interfaces). We will also define a number of quantities whose expressions on the top and bottom interfaces only differ by superscript. In order not to overload the reader, we have decided to present only the expressions associated with the top interface.

2.7.1. Homogeneous layers

The response of homogeneous layers in multilayer structures comprising periodic cells differs relatively little from the case of homogeneous multilayers. The only difference is that we have to take the Bloch modes into account. So for a given angle of incidence, we don't have just one state vector but an infinite number: one per Bloch mode. However, because of their decoupling, the relationships between the state vectors can be written separately for each wave. So, in addition to extending the method to the problem under consideration, the aim of this section is mainly to define the quantities and definitions that will be used throughout the work.

Inside the layer, the state vector $\boldsymbol{S}$ is defined as a function of the x and y coordinates and is the superposition of the contributions of the different Bloch modes:

$$\boldsymbol{S}(x,y) = \sum_{w=-N}^{N} e_w(x)\boldsymbol{S}_w(y) \tag{2.9}$$

$\boldsymbol{S}_w$ is associated to a Bloch mode w and can be written as a superposition of forward and backward waves travelling in the y direction:

$$\boldsymbol{S}_w(y) = \sum_{\tau\in\mathcal{T}} \boldsymbol{S}^+_{w,\tau} e^{-jk_\tau^{(w)}(y-y_b)}\, q^+_{w,\tau} + \boldsymbol{S}^-_{w,\tau} e^{jk_\tau^{(w)}(y-y_t)}\, q^-_{w,\tau}. \tag{2.10}$$

$\mathcal{T}$ is the set of plane waves (compressional, shear ...) travelling in the considered layer. $\boldsymbol{S}^\pm_{w,\tau}$ is the vector of polarisations associated with the forward wave of type τ and the Bloch mode w. $k_\tau^{(w)}$ is its associated projection y of the wave vector and $q^\pm_{w,\tau}$ corresponds to the amplitudes. Similar definitions are introduced for backward waves but with the exponent + replaced by −. Note that the origin of the forward waves (resp. backward) is chosen at $y = y_b$ (resp. $y = y_t$) so that this expression only involves decaying exponential terms in the y direction.

It is more convenient to write eq. (2.10) in matrix form as:

$$\boldsymbol{S}_w(y) = [S_w]\left[E_w(y)\right]\boldsymbol{q}_w. \tag{2.11}$$

$[S_w]$ is a matrix that gathers the polarisation vectors, $\left[E_w(y)\right]$ is a diagonal matrix associated with the exponential terms, and $\boldsymbol{q}_w$ is the vector of the amplitudes of waves associated with Bloch mode w. At the top interface, one has:

$$\Sigma^t_w = \boldsymbol{S}_w(y^t) = [S_w]\left[E_w(y^t)\right]\boldsymbol{q}_w \tag{2.12}$$

The transfer matrix $[T_w]$ associated with the Bloch mode w relates the state vector at the top interface to the state vector at the bottom interface:

$$\Sigma_w^b = [T_w]\Sigma_w^t \tag{2.13}$$

Its expression is as follows.

$$[T_w] = [S_w]\left[E_w(y_b)\right]\left[E_w(y_t)\right]^{-1}[S_w]^{-1} \tag{2.14}$$

Global state vectors can be defined gathering the state vectors associated to selected Bloch modes:

$$\Sigma^t = \left\{\Sigma_{-N}^b, \dots, \Sigma_0^b, \dots, \Sigma_N^b\right\} \in \mathbb{C}^{n_\Sigma}, \; n_\Sigma = 2n_w n_B. \tag{2.15}$$

They can be also expressed as a function of the amplitudes of the global waves amplitudes $\boldsymbol{q}$ in a form similar to eq. (2.12):

$$\Sigma^t = \left[\Xi^t\right]\boldsymbol{q}\,,\; \boldsymbol{q} = \left\{q_{-N}, \dots, q_0, \dots, q_N\right\} \tag{2.16}$$

Note that $\left[\Xi^t\right]$ is diagonal by block, each block being $[S_w]\left[E_w(y^t)\right]$ so that this matrix gathers both polarisations and exponential terms. Σ^b and Σ^t are related by a global transfer matrix $[T]$ that is diagonally orientated by blocks, each block $[T_w]$ (for $w = -N, \dots, N$) being defined as in eq. (2.13):

$$\Sigma_w^b = [T_w]\Sigma_w^t \quad \Longrightarrow \quad \Sigma^b = [T]\Sigma^t. \tag{2.17}$$

Note that formally, one has:

$$[T] = \left[\Xi^b\right]\left[\Xi^t\right]^{-1}. \tag{2.18}$$

2.7.2. Case of periodic cells

The case of periodic cells is now presented. We will couple FEM to state vectors Σ^t and Σ^b on the top and bottom interfaces. The cell is associated with a volume Ω bounded by a rectangle of the boundary Γ. Ω can be filled by an aggregate of different media but we will consider that each cell is composed of a core material which covers the upper and lower boundaries. This weak form involves two terms: a volume term associated with the primal variable $\boldsymbol{\pi}$ and a boundary term associated with the dual variable $\boldsymbol{b}$.

The key idea of the proposed method is that state variables allows to corresponds to the aggregate of primal and dual variables, as illustrated in Table 2.1. First, the state vector $\boldsymbol{S}^t$ is a function of x whose relation with Σ^t is:

$$\boldsymbol{S}^t(x) = \sum_{w=-N}^{N} e_w(x)\Sigma_w^t \quad \Longrightarrow \quad \boldsymbol{S}^t(x) = [\Lambda(x)]\Sigma^t \tag{2.19}$$

$[\Lambda(x)]$ is a matrix $2n_w \times 2n_w n_B$ that gathers the exponential terms $e_w(x)$. It reads:

$$[\Lambda(x)] = \left[e_{-N}(x)[\boldsymbol{I}] \;\; \dots \;\; e_N(x)[\boldsymbol{I}]\right] = \boldsymbol{e}(x) \otimes [\boldsymbol{I}] \text{ with } \boldsymbol{e}(x) = \left[e_{-N}(x), \dots, e_N(x)\right] \tag{2.20}$$

where $[\boldsymbol{I}]$ is the identity matrix of dimension $2n_w$.

Second, two boolean matrices $[\boldsymbol{\pi}]$ and $[\boldsymbol{b}]$ of dimension $n_w \times 2n_w$ can be defined to relate the FEM variables to the state variables.

$$\boldsymbol{\pi}(x, y = y_t) = [\boldsymbol{\pi}][\Lambda(x)]\Sigma^t\,, \tag{2.21a}$$

$$\boldsymbol{b}(x, y = y_t) = [\boldsymbol{b}][\boldsymbol{\Lambda}(x)]\boldsymbol{\Sigma}^t \tag{2.21b}$$

Note that $\boldsymbol{\pi}$ and $\boldsymbol{b}$ are considered vectors insofar as the primal variables can be associated with several physical fields.

The weak form associated with Ω can be written in the form:

$$\forall \delta\boldsymbol{\pi} \in V\,,\; a(\boldsymbol{\pi}, \delta\boldsymbol{\pi}) = \int_\Gamma \delta\boldsymbol{\pi}(M)^T \boldsymbol{b}(M) dS \tag{2.22}$$

a is a bilinear form that is defined in the space V of admissible functions. $\delta\boldsymbol{\pi} \in V$ is a test function, and ^{T} is the real transposition. Γ can be decomposed into four edges Γ_t, Γ_b, Γ_l and Γ_r, respectively, associated with the top, bottom, left and right edges of the cell so that eq. (2.22) can be written as:

$$a(\boldsymbol{\pi}, \delta\boldsymbol{\pi}) - \int_{\Gamma_l \cup \Gamma_r} \delta\boldsymbol{\pi}(y)^T \boldsymbol{b}(y) dy = \int_{\Gamma_t} \delta\boldsymbol{\pi}(x)^T [\boldsymbol{b}][\boldsymbol{\Lambda}(x)]\boldsymbol{\Sigma}^t dx + \int_{\Gamma_b} \delta\boldsymbol{\pi}(x)^T [\boldsymbol{b}][\boldsymbol{\Lambda}(x)]\boldsymbol{\Sigma}^b dx \tag{2.23}$$

Imposition of eq. (2.21a) is done in a weak sense by projection on the e_w functions:

$$\int_{\Gamma_t} \overline{e_w(x)}\boldsymbol{\pi}(x) dx = \int_{\Gamma_t} \overline{e_w(x)}[\boldsymbol{\pi}][\boldsymbol{\Lambda}(x)]\boldsymbol{\Sigma}^t dx \tag{2.24}$$

Due to the orthogonality of the functions e_w, one as:

$$\int_{\Gamma_t} \overline{e_w(x)}[\boldsymbol{\pi}][\boldsymbol{\Lambda}(x)]\boldsymbol{\Sigma}^t dx = D[\boldsymbol{\pi}]\boldsymbol{\Sigma}^t_w. \tag{2.25}$$

This relationship therefore corresponds to the calculation of the contributions of the Bloch modes using the primal variables of the finite element model. Hence, for each primal variable $\pi_k \in \boldsymbol{\pi}$ and each Bloch mode w:

$$\int_{\Gamma_t} \overline{e_w(x)}\pi_k(x) dx = D\Sigma^t_{k,w}, \tag{2.26}$$

where $\Sigma^t_{k,w}$ is the component of the state vector associated with the Bloch mode w and field π_k.

Let us come to the discretization of the system made up of equations (2.23) and (2.26). The first one is discretised with higher order finite-elements. The primal variable is approximated as:

$$\boldsymbol{\pi}(M) = [\boldsymbol{N}(M)]\boldsymbol{X}\,,\; \delta\boldsymbol{\pi}(M) = [\boldsymbol{N}(M)]\delta\boldsymbol{X} \tag{2.27}$$

where $[\boldsymbol{N}(M)] \in \mathbb{R}^{n_w \times n}$ is the matrix of shape function and $\boldsymbol{X} \in \mathbb{R}^n$ is the vector of the finite-element unknowns. The restriction of the variable π_k on the upper interface leads to:

$$\pi_{k|y=y_t}(M) = \left[\boldsymbol{N}_t^k(M)\right]\boldsymbol{X}\,,\; \left[\boldsymbol{N}_t^k(M)\right] \in \mathbb{R}^{1\times n}. \tag{2.28}$$

The full vector of the primal variables can then be discretised as:

$$\boldsymbol{\pi}_{|y=y_t}(M) = [\boldsymbol{N}_t(M)]\boldsymbol{X}\,,\; \delta\boldsymbol{\pi}_{|y=y_t}(M) = [\boldsymbol{N}_t(M)]\delta\boldsymbol{X} \tag{2.29}$$

This method has now been widely used in the literature; see, for instance, [28, 30, 31], and is then not detailed in the present work. The term a associated with the volumic term leads to a matrix $[\boldsymbol{D}_{\pi\pi}] \in \mathbb{C}^{n\times n}$. The term of (2.23) associated to Γ_t is discretised as:

$$\int_{\Gamma_t} \delta\boldsymbol{\pi}(x)^T [\boldsymbol{b}][\boldsymbol{\Lambda}(x)]\boldsymbol{\Sigma}^t dx \approx \delta\boldsymbol{X} \underbrace{\int_{\Gamma_t} [\boldsymbol{N}_t(x)]^T [\boldsymbol{b}][\boldsymbol{\Lambda}(x)] dx}_{[\boldsymbol{D}_{\pi t}]} \boldsymbol{\Sigma}^t \tag{2.30}$$

$[\boldsymbol{D}_{\pi t}]$ is a matrix of dimension $n \times n_\Sigma$ that is not classical in the sense of the FEM, as the integrand is associated to the product of polynomial by exponential functions. Its terms are based on those of $[\phi] \in \mathbb{C}^{n \times 2N+1}$ with

$$[\phi] = \int_{\Gamma_t} [\boldsymbol{N}_t(x)]^T [\boldsymbol{e}(x)] dx \tag{2.31}$$

which contains all the integration of the products of the traces of finite element shape functions by exponential Bloch modes functions. Note that the terms of the latter matrix can be computed analytically. The discretization of (2.26) leads to $n_w n_B$ equations linking $\boldsymbol{X}$ and $\boldsymbol{\Sigma}^t$. The right-hand side is discretised as:

$$\int_{\Gamma_t} \overline{e_w(x)} \pi_k(x) dx \approx \underbrace{\int_{\Gamma_t} \overline{e_w(x)} \left[\boldsymbol{N}_t^k(x)\right] dx}_{\left[\boldsymbol{D}_{t\pi}^{k,w}\right]} \boldsymbol{X} \tag{2.32}$$

$\left[\boldsymbol{D}_{t\pi}^{k,w}\right] \in \mathbb{C}^{1 \times n}$ is a row matrix whose terms are conjugates of those associated with mode w in $[\phi]$. By gathering all the orthogonality relations, it is possible to write the relation between $\boldsymbol{\Sigma}^t$ and $\boldsymbol{X}$ as:

$$[\boldsymbol{D}_{tt}]\boldsymbol{\Sigma}^t + [\boldsymbol{D}_{t\pi}]\boldsymbol{X} = \boldsymbol{0}. \tag{2.33}$$

$[\boldsymbol{D}_{t\pi}]$ is built after the aggregation of the row matrices $\left[\boldsymbol{D}_{t\pi}^{k,w}\right]$ and $[\boldsymbol{D}_{tt}] \in \mathbb{R}^{n_w n_B \times n_w n_B}$ is the scalar matrix whose entries on the diagonal are all equal to D.

The last terms to be considered are terms associated with Γ_l and Γ_r in eq. (2.23). Due to the periodicity condition, they can be cancelled by a master-slave approach that consists of imposing that both degrees of freedom and boundary terms on the right edge are equal to those on the left edge multiplied by a phase factor $e^{jk_x D}$. This is not detailed here for the sake of concision. The final linear system associated with the periodic cell is then written as:

$$\begin{bmatrix} [\boldsymbol{D}_{tt}] & [\boldsymbol{D}_{t\pi}] & [0] \\ [\boldsymbol{D}_{\pi t}] & [\boldsymbol{D}_{\pi\pi}] & [\boldsymbol{D}_{\pi b}] \\ [0] & [\boldsymbol{D}_{b\pi}] & [\boldsymbol{D}_{bb}] \end{bmatrix} \begin{Bmatrix} \boldsymbol{\Sigma}^t \\ \boldsymbol{X} \\ \boldsymbol{\Sigma}^b \end{Bmatrix} = \boldsymbol{0}. \tag{2.34}$$

Let us now deduce relations between the state vectors at the top and bottom interfaces. The finite-element degrees of freedom can first be condensed as:

$$\boldsymbol{X} = [\boldsymbol{R}_t]\boldsymbol{\Sigma}^t + [\boldsymbol{R}_b]\boldsymbol{\Sigma}^b \tag{2.35}$$

with

$$[\boldsymbol{R}_t] = -[\boldsymbol{D}_{\pi\pi}]^{-1}[\boldsymbol{D}_{\pi t}],\ [\boldsymbol{R}_b] = -[\boldsymbol{D}_{\pi\pi}]^{-1}[\boldsymbol{D}_{\pi b}] \tag{2.36}$$

We then have:

$$\underbrace{\begin{bmatrix} [\boldsymbol{D}_{t\pi}][\boldsymbol{R}_b] \\ [\boldsymbol{D}_{bb}] + [\boldsymbol{D}_{b\pi}][\boldsymbol{R}_b] \end{bmatrix}}_{[\boldsymbol{M}_b]} \boldsymbol{\Sigma}^b + \underbrace{\begin{bmatrix} [\boldsymbol{D}_{tt}] + [\boldsymbol{D}_{t\pi}][\boldsymbol{R}_t] \\ [\boldsymbol{D}_{b\pi}][\boldsymbol{R}_t] \end{bmatrix}}_{[\boldsymbol{M}_t]} \boldsymbol{\Sigma}^t = \boldsymbol{0} \tag{2.37}$$

The transfer matrix of the periodic layer is then:

$$\boldsymbol{\Sigma}^b = [\boldsymbol{T}]\boldsymbol{\Sigma}^t \text{ with } [\boldsymbol{T}] = -[\boldsymbol{M}_b]^{-1}[\boldsymbol{M}_t]. \tag{2.38}$$

2.7.3. Interfaces relations

Let us now look at the relationship between the fields at the interfaces between layers. First, these are independent of whether the layers are homogeneous or periodic. To avoid confusion with what we have presented earlier, we will consider that the layer above will be marked with the subscript a and the layer below with the subscript b. We therefore have two state vectors $\mathbf{\Sigma}_+$ and $\mathbf{\Sigma}_-$, which are composed, for each layer, of the aggregation of the contributions of each Bloch mode to the state vector. As these are orthogonal, the interface relationships are decoupled from each other.

The state vectors are, respectively, of length $2n_a n_B$ and $2n_b n_B$ where n_a and n_b are the number of waves n_w shown in Table 2.1. The number of interface relations is equal to $n_a + n_b$ and can be written in the form:

$$[\boldsymbol{C_a}]\mathbf{\Sigma}_a + [\boldsymbol{C_b}]\mathbf{\Sigma}_b = \mathbf{0}, \tag{2.39}$$

where $[\boldsymbol{C_a}] \in \mathbb{R}^{n_a+n_b\times 2n_a}$ and $[\boldsymbol{C_b}] \in \mathbb{C}^{n_a+n_b\times 2n_b}$ are interface matrices. These matrices express the continuities between the stresses and displacements of the media in the two layers at the interface or the possible nullities of some fields. The expressions of these matrices are not given, but the reader can in particular consult Table IV of [27].

2.7.4. Methods to model the acoustics response of the multilayer structure

The two methods we are going to compare are now be presented. The proposed approach is a general one, but in the context of this section we will limit ourselves to present it in the case of a bilayer structure composed of an homogeneous layer (labeled 1) on top of a periodic cell (labeled 2) as presented in Fig. 2.2.

A state vector $\mathbf{\Sigma}_- \in \mathbb{C}^{2n_B}$ is defined on the lower side of the incident interface. It can be written as a sum of a vector $\boldsymbol{E}$ associated to the incident field and a contribution of the reflected waves:

$$\mathbf{\Sigma}_- = \left[\boldsymbol{\rho}\right]\mathcal{R} + \boldsymbol{E}\,,\; \mathcal{R} = \{R_{-N}, \dots, R_N\} \tag{2.40}$$

$\left[\boldsymbol{\rho}\right] \in \mathbb{C}^{2\times n_B}$. It is diagonal by block, each column is associated to a Bloch mode and provides the displacement along y and the pressure of the considered Bloch mode as a function of the reflexion coefficient R_w of wave w. $\mathcal{R} \in \mathbb{C}^{n_B}$ is the vector of reflexion coefficients.

The termination condition can be either transmission in a semi-infinite media or a rigid wall. The first case is similar to the one of the incident interface. The state vector $\mathbf{\Sigma}_+$ is defined from the transmission coefficient vector $\mathcal{T}$ as follows:

$$\mathbf{\Sigma}_+ = [\boldsymbol{\tau}]\mathcal{T}\,,\; \mathcal{T} = \{T_{-N}, \dots, T_N\} \tag{2.41}$$

$[\boldsymbol{\tau}] \in \mathbb{C}^{2\times n_B}$ is a diagonal by block matrix that relates the displacement along y and the pressure of the considered Bloch mode as a function of the transmission coefficient T_w of wave w. For that termination condition, the total number of relations at interfaces is:

$$n_t = n_B(1+n_1) + n_B(n_1+n_2) + n_B(n_2+1) = (2+2n_1+2n_2)n_B. \tag{2.42}$$

In the case of a rigid wall, the nullity of the displacement in $\mathbf{\Sigma}_1^t$ is imposed as

$$[\boldsymbol{D}]\mathbf{\Sigma}_1^t = [\mathbf{0}]\,,\; [\boldsymbol{D}] = \left[\hat{\boldsymbol{D}}\right] \otimes \left[\boldsymbol{I}_{n_B}\right] \tag{2.43}$$

$\left[\hat{\boldsymbol{D}}\right] \in \mathbb{R}^{n_1\times 2n_1}$ is a boolean matrix whose non-zero entries are associated to the displacements in the state vector. The total number of boundary and interface relations is then:

$$n_t = n_B n_1 + n_B(n_1+n_2) + n_B(n_2+1) = (1+2n_1+2n_2)n_B \tag{2.44}$$

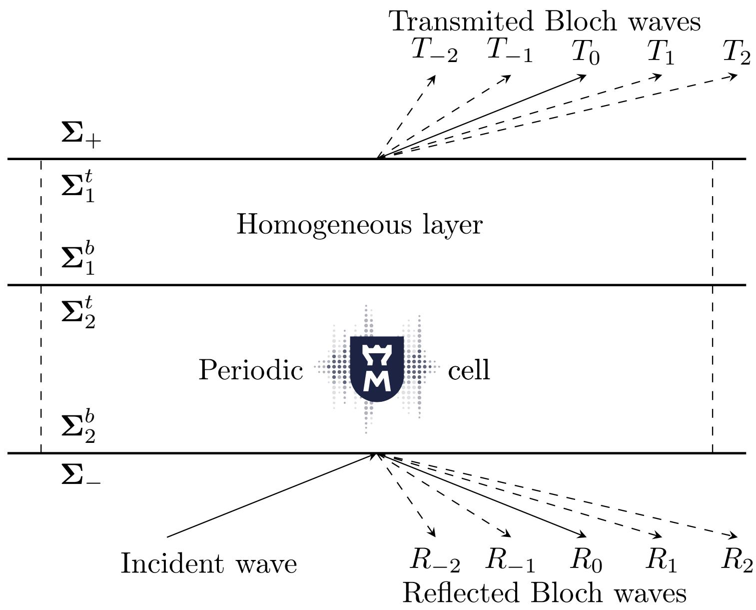


Figure 2.2: Two layers problem

2.7.5. Global method

The unknonws of the global method are the amplitudes of the waves for homogeneous layers and the top and bottom state vectors for periodic layers. In the case of a transmission condition, the system of equations is written as follows:

$$\begin{bmatrix} [C_+][\tau] & [C_1^t][\Xi^t] & [0] & [0] & [0] \\ [0] & [C_1^b][\Xi^b] & [C_2^t] & [0] & [0] \\ [0] & [0] & [M_t] & [M_b] & [0] \\ [0] & [0] & [0] & [C_2^b] & [C_-][\rho] \end{bmatrix} \begin{Bmatrix} \mathcal{T} \\ \boldsymbol{q}_1 \\ \boldsymbol{\Sigma}_2^t \\ \boldsymbol{\Sigma}_2^b \\ \mathcal{R} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ -\left[C_b^1\right] \boldsymbol{Inc} \end{Bmatrix} \tag{2.45}$$

The number of unknowns is n_B for both $\mathcal{T}$ and $\mathcal{R}$, $2n_1n_B$ for $\boldsymbol{q}_1$ and $2n_2n_B$ for $\boldsymbol{\Sigma}_2^t$ and $\boldsymbol{\Sigma}_2^b$. The first and second block rows correspond to $n_B(1+n_1)$ (resp. $n_B[n_1+n_2]$) continuity relations at the transmition (resp. second interface). The third block row corresponds to the $2n_Bn_2$ relations of eq. (2.37) and the last block row to the $n_B(n_2+1)$ relation at the incident interface. In the case of a rigid termination, $\boldsymbol{T}$ is no longer in the unknowns, and the first block column disapears. In addition, $\left[C_b^1\right][\Xi^t]$ is replaced by $[D][\Xi^t]$ corresponding to n_1n_B relations. Hence, in both cases, the relations are independent, and their number is equal to the number of unknowns. As the origin of waves is at the adequate interface, this system only involves decaying exponentials.

2.7.6. Transfer Matrix Method

The unkowns of the TMM are the state vectors on the top interfaces of each layer, $\mathcal{R}$ (and $\mathcal{T}$ in the case of a transmission medium). The expression of state vectors on the bottom interfaces are obtained by a multiplcation by the transfer matrices of each layer. The system of equations is then written as follows:

$$\begin{bmatrix} [C_+][\tau] & [C_1^t] & [0] & [0] \\ [0] & [C_1^b][T_1] & [C_2^t] & [0] \\ [0] & [0] & [C_2^b][T_2] & [C_-][\rho] \end{bmatrix} \begin{Bmatrix} \mathcal{T} \\ \Sigma_1^t \\ \Sigma_2^t \\ \mathcal{R} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ -\left[C_b^1\right] Inc \end{Bmatrix} \tag{2.46}$$

When the multilayer comprises a succession of several materials of the same type, the lower interface corresponds to that of the last layer and the transfer matrix corresponds to the product of the matrices of the successive layers. The succession is thus treated as a single layer.

2.8 Results

2.8.1. Validation on homogeneous multilayer structures

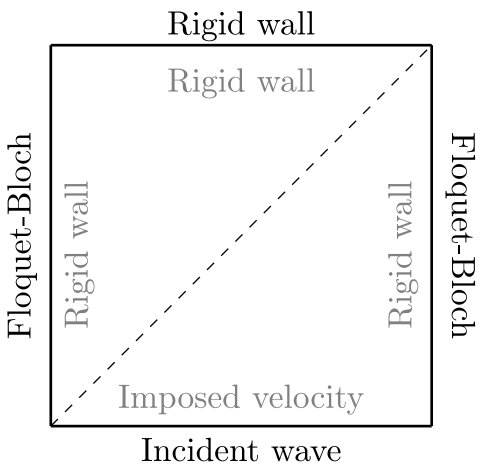
The main goal of these first two examples is to compare the different methods applied to a homogeneous multilayered structures case and to show that the errors for each method are identical. The first problem is that of a 1 meter square air cavity, excited by a plane wave at normal incidence ($k_x = 0$). The cavity, described as a periodic cell with periodicity $D = 1$ m and thickness $H = 1$ m, is modeled using FEM and discretized by 2 triangular elements. A rigid boundary condition is applied on top. For this simple case, it is easy to obtain the reflection coefficient for this problem, which serves as the reference solution:

$$R = \frac{Z_s - 1}{Z_s + 1}, \; Z_s = -j \cot(kH), \; k = \frac{\omega}{c_0}. \tag{2.47}$$

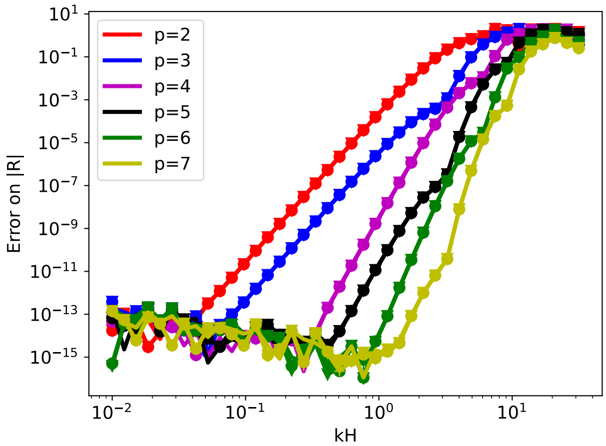
Three methods are compared in this example and are all associated to the same mesh: the GM and TMM methods presented in section 2.7.4 as well as a FEM resolution in which a unit velocity is imposed at the bottom boundary, the three other ones are associated to rigid walls. The surface impedance Z_s can then be deduced from the pressure on this boundary and finally the reflection coefficient. This method serves as an indicator on the convergence of the FEM. The diagram of the configuration is depicted in Fig. 2.3a and the error for each method is given in Fig. 2.3b. There, the convergence for each method is shown depending on the adimensional parameter kH and for different interpolation orders $p = 2$ to 7. At each order, the errors of each of the four methods are perfectly superimposed (except when the error reaches the machine precision). All methods lead to the same error as the direct FEM resolution, thus indicating that there is no other additional source of error than the FEM approximation.

Parameter	PEM 1	PEM 2	Rubber	Steel
Porosity ϕ [$\emptyset$]	0.989	0.95	/	/
Resistivity σ [$N.s.m^{-4}$]	8060	42 000	/	/
Tortuosity α_∞ [$\emptyset$]	1.0	1.1	/	/
Viscous characteristic length Λ [μm]	214	15	/	/
Thermal characteristic length Λ' [μm]	214	45	/	/
Density ρ [$kg.m^{-3}$]	6.1	126	1800	7800
Young's modulus E [kPa]	56.5	694.4	1900	49.7×10^6
Poisson's ratio ν [$\emptyset$]	0.24	0.24	0.48	0.45
Frame loss model	structural	structural	rubber	none
Structural loss factor η [$\emptyset$]	0.02	0.05	796	0

Table 2.2: Physical parameters of the different media



(a) Geometry of the problem. Boundary conditions in black (resp.gray) are for the proposed methods (resp. the FEM reference).



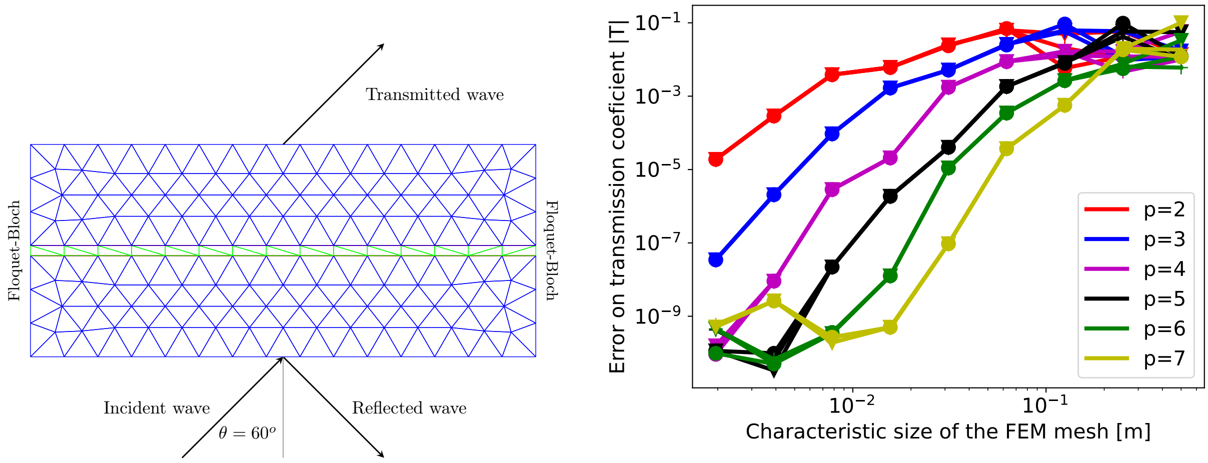
(b) solid line: FEM reference: +: Global Method, ▿: TMM

Figure 2.3: Convergence of the reflection coefficient for an air cavity excited by a plane wave under normal incidence for several interpolation orders

A second validation case is presented in Fig. 2.4a with the transmission problem of sandwich structure of two rubber coatings ($h = 0.2$ mm) surrounding a poroelastic layer of PEM 1 ($h = 2$ cm) immerged in two semi-infinite air media. Note that in Fig. 2.4a, the rubber coatings are hardly visible due to their very thin thickness. The excitation angle is 60^o. In this example, we have chosen to model the air layers (with a thickness of 20 cm) for several reasons: to ensure that the method is properly implemented (especially the elastic-fluid coupling), to check that the phase shift induced by the propagation in the two air layers are properly captured, and because in several choices of FEM implementation, they need to be modelled (and are generally coupled to PML). We point out that the objective of this subsection is to test the method rather than to obtain the most efficient model. The reference reference solution is computed through the RM.

This structure can then be considered as a five-layer stack, and three different approaches are considered to test the implementation. The first one (called entire structure) is to consider that this is a stack of five periodic cells. The second approach (called sandwich structure) accounts for the modelling of the two air layers as homogeneous materials and the 3 layer stack as a periodic cell. Note that in that case the FEM model of the periodic cell is composed of two different materials (PEM 1 and rubber). In the last case (called the poroelastic core), only the porous material is modelled as a periodic cell, and the rubber and air layers are considered as homogeneous materials. The period D is 1 meter and the periodic cell meshes are created independently of each other, with a characteristic size h. Note that the different meshes may not be compatible at the interfaces, which is not a problem for the proposed methods.

The error on the modulus of the transmission coefficient is presented in Fig. 2.4b. In order not to overload the figure, only the results with the GM are presented but the TMM matches at machine precision. For each order the errors of the three models are perfectly superposed (except for coarse meshes where the FE models are not converges and for fine meshes and order higher than 6 where the systems at ill-conditioned). This is validate the approach but also indicate that, in that problem, the error is governed by the one of the poroelastic core. We can also draw the conclusion that it is obviously superfluous to discretize homogeneous layers by finite elements. In the following cases, only periodic layers will be modelled using finite elements and the homogeneous layers will be modelled using plane waves.



(a) Coupled vibroacoustic sandwich problem

(b) (+) poroelastic core, (▽) sandwich structure, (o) entire structure

Figure 2.4: Convergence curve for the global method for the problem of a sandwich structure

2.8.2. Validation on metaporous structures

In this subsection, we will compare the two methods with examples inspired by the literature on metaporous structures. The first is a poroelastic core ($D = H = 2$ cm) with an embedded steel inclusions (of radius 8 mm) with a rigid backing studied in [48] and depicted in Fig. 2.5a. The second is a two side rubber-coated poroelastic slab with a resonant air-filled rubber shell inclusion (case b presented in figure 2 of [49]) (Fig. 2.5c).

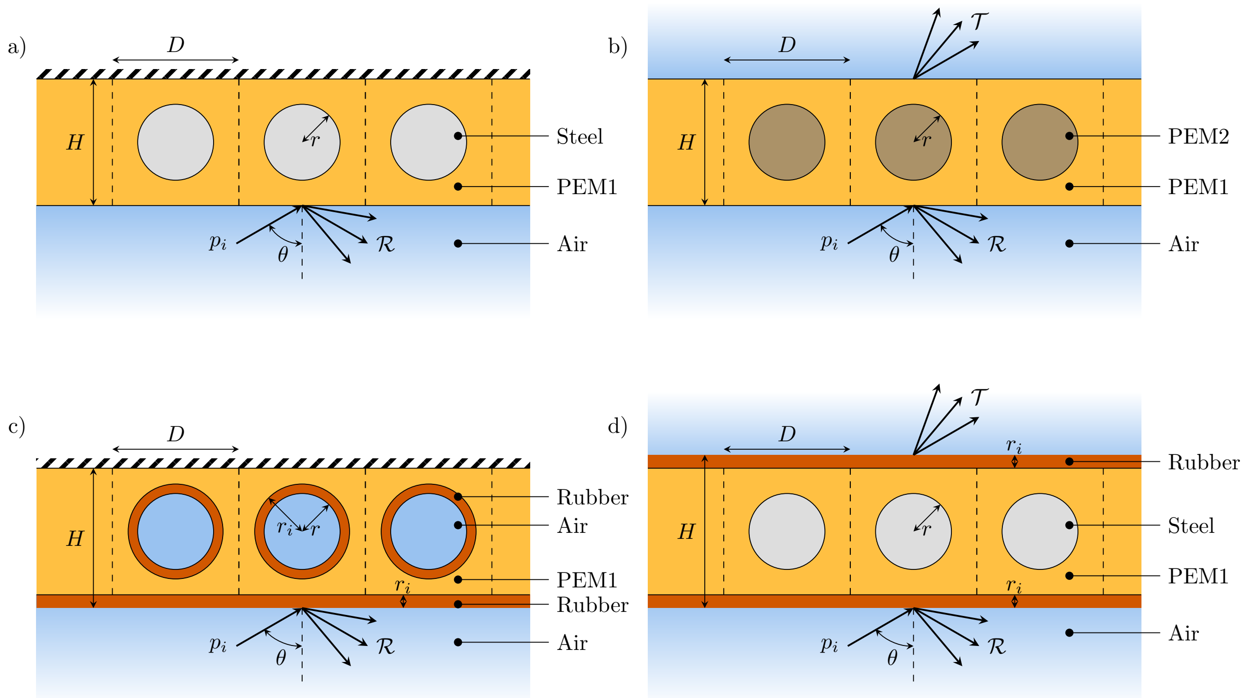
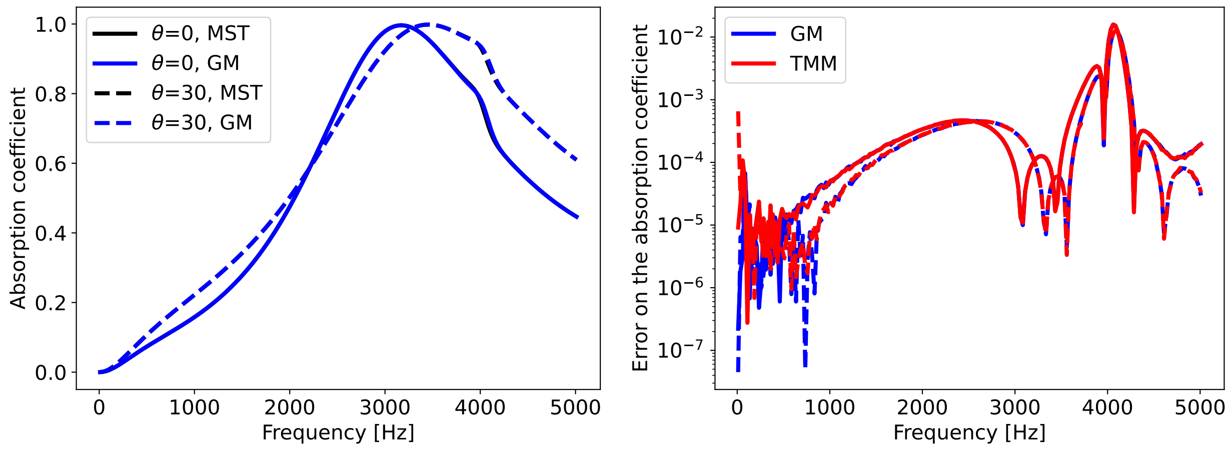


Figure 2.5: Diagram for the presented cases, each case is a poroelastic host layer, a) rigidly-backed and steel inclusions, b) transmission problem with embedded poroelastic inclusions, c) rigidly-backed with a rubber-coating and resonant air-filled rubber shell, d) transmission problem with a rubber-coating on each side and steel inclusion

Fig. 2.6 presents the results for the first case. The number of Bloch waves is chosen as $N = 2$ and the size of the mesh is $h = D/5 = 4$ mm. The results are presented for two incident angles ($\theta = 0^o$ and $\theta = 30^o$). The absorption curves presented in Fig. 2.6a are superimposed to the multiple scattering solutions which is considered as the reference. In this figure, only the results obtained by the Global Method are presented but the TMM results are very close. More precisely, their error to the reference solution are presented in Fig. 2.6b. They coincide for both angles, except at low frequencies where errors are around 10^{-5}. It is worth noticing that the maximum error is around 4000 Hz (around 10^{-3} and is due to Biot effects. It can be reduced by refining the mesh.

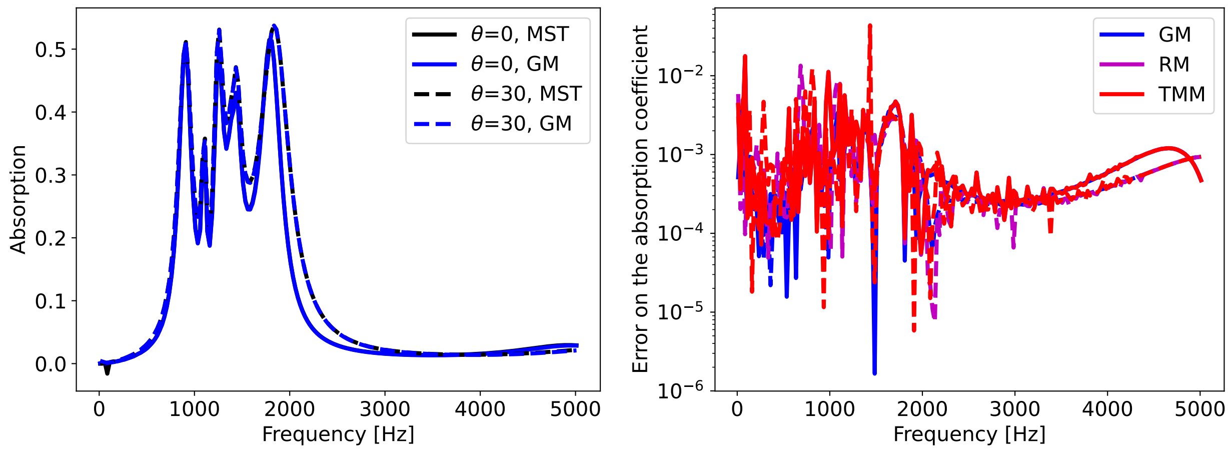


(a) Absorption coefficient at two angles

(b) Errors of the methods.

Figure 2.6: Case of a elastic inclusion in a porous material [48]. Solid (resp. dashed) lines correspond to $\theta = 0^o$ (resp. $\theta = 30^o$)

Similar conclusions can be observed in the second case presented in Fig. 2.7. The number of Bloch waves is $N = 3$ and the characteristic size of the mesh is $h = D/40 = 0.5$ mm so as to capture the resonance of the inclusion. The level of error presented in Fig. 2.7b is very low indicating the good prediction obtained by the three methods. Nevertheless, a closer look at the errors above 3000 Hz shows small oscillations in the TMM, suggesting stability issues that we will investigate now more in detail.



(a) Absorption coefficient at two angles

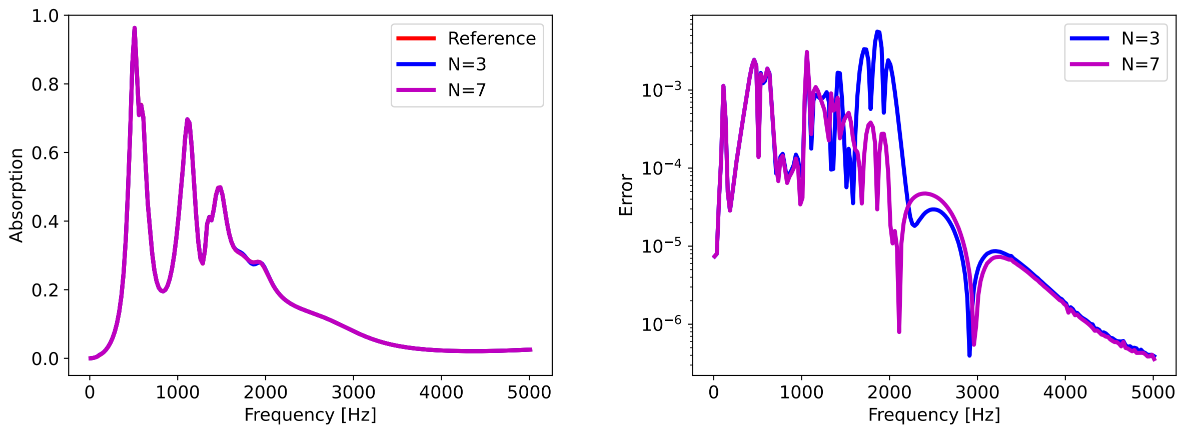
(b) Errors of the three methods

Figure 2.7: Case of a two-sided coated of a resonant inclusion in two side coated material. Solid (resp. dashed) lines correspond to $\theta = 0^o$ (resp. $\theta = 30^o$)

2.8.3. Stability issues

This part is dedicated to highlighting the stability issues that arise with the TMM. The configuration presented in Fig. 2.5d is used. Fig. 2.7a is showing matching results to the reference for the absorption curves computed using GM and two Bloch truncation values $N = 3$ and $N = 7$. The error profile also shows that the error is the same for both truncations at low frequencies. However, in the mid-frequency range, greater precision is achieved by accounting more Bloch modes since higher frequencies generally require the progressive addition of Bloch modes to the series. This behaviour shows a satisfying stability for the GM, seemingly able to account for a large amount of Bloch waves without loss of precision.

However, the TMM results compared to the same reference is showing large discrepancies on its absorption profile at lower frequencies in the case $N = 3$. These results are made worse with a growing amount of Bloch waves taken into account $N = 7$, where the absorption response does not match the reference over the whole frequency range, as shown also with the condition number gaining around 10 orders of magnitude for the TMM when going from 3 to 7 Bloch waves in the truncation. This also shows that the conditioning of the GM global matrices remains tame with an increasing amount of Bloch waves, as observed in the previous paragraph. The inherent instability of the TMM shows some limitations in this particular configuration as highlighted by this result. This has also been observed for a wide range of numerical experiments that we have performed and not shown in the present work. Moreover, unstability of the TMM is a common feature [23]. We point out that in our implementation of the GM for periodic cell, the stablility is preserved as both $\mathbf{\Sigma}^t$ and $\mathbf{\Sigma}^b$ are unknowns of eq. (2.45) and not just one of them.



(a) Absorption coefficient

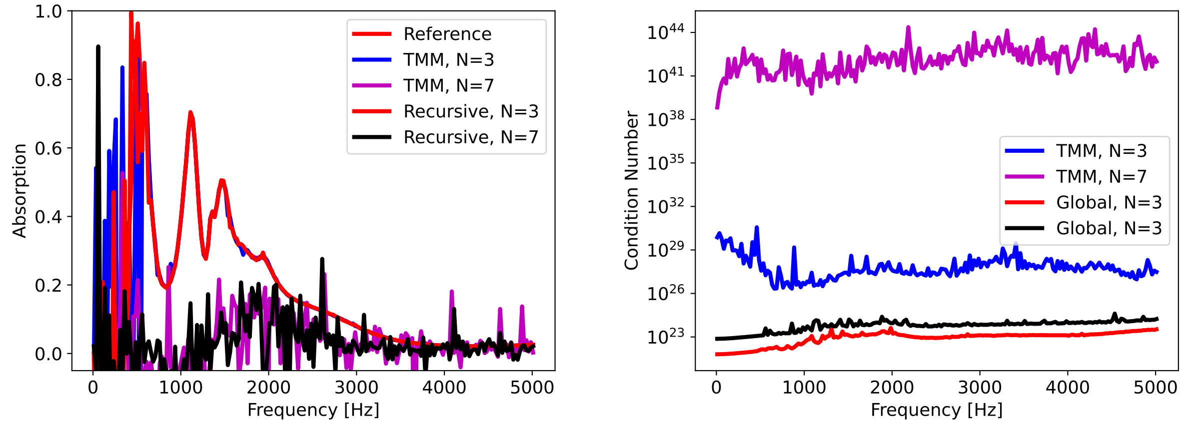
(b) Error on the absorption

Figure 2.8: Prediction of the GM for two truncation of the Bloch modes

2.8.4. Automatic selection of Bloch wave numbers

In this last part, the concern of the number of Bloch modes required for an accurate solution evaluation will be dealt with. Only results from the GM are presented since it was identified as the most stable method among the different numerical strategies discussed in previous results. The multiple scattering solution will be used as reference, its Bloch modes truncation not being at stake in this discussion. In the literature, the truncating of the series uses an empirical formula that is detailed e.g. in Appendix D from Ref. [50]. The influence on the error of the GM by varying the amount of Bloch modes accounted is shown in Fig. 2.10a. The error is also presented for different orders of interpolation ranging from p = 2 to 5 and a varying mesh characteristic size. Refining the mesh quickly leads to a plateau, showing that the FEM has converged with regards to the interpolation order and mesh size. By increasing the Bloch wave truncation, the plateau decreases significantly going from N=0 to 3. The main influence of the error is thus not due to the FEM discretization but rather is mainly due to the Bloch wave truncation.

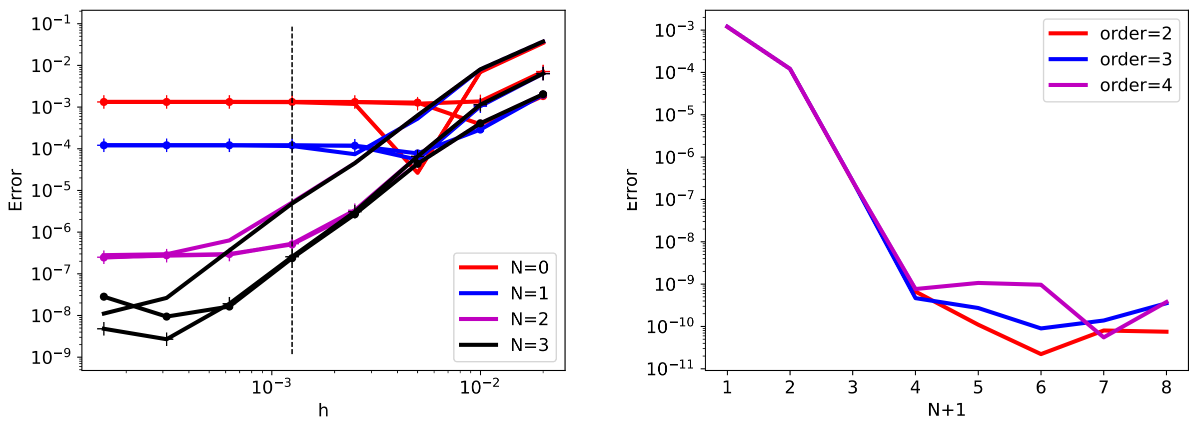
Finally, the influence of the number of accounted Bloch modes is investigated in the present



(a) Absorption coefficient **(b)** Condition number

Figure 2.9: Prediction of the RM and TMM for two truncation of the Bloch modes

section. By selecting the solution along the dashed line in Fig. 2.10a, the variation of the relative error comparing a computation that accounts N Bloch waves to a computation using an additional mode at $N+1$ Bloch modes is presented in Fig. 2.10b. A plateau is reached around $N = 4$, signifying that the convergence of the Bloch wave series for all shown orders of interpolation has been reached. This could be implemented as an automatic truncation criterion, as the evaluation of the full system accounting for additional Bloch waves is much less computationnaly intensive than evaluating the full system: the main FEM matrix does not have to be re-evaluated in this case and the LU factors of the main FEM matrix $[\boldsymbol{D}]_{\pi\pi}$ could be stored to evaluate the $[\boldsymbol{R}_b]$ and $[\boldsymbol{R}_t]$ from eq. (2.36) with at a low computational cost, ensuring that the convergence is reached for the Bloch wave truncation.



(a) Influence of the number of Bloch modes **(b)** Differences on the errors

Figure 2.10: Automatic selection of the number of Bloch modes

2.9 Conclusion

A stable and robust approach for the modeling of the acoustic response of grating stacks with homogeneous layers and periodic cells has been proposed in this paper. Each layer in the stack is coupled to its neighbours using the appropriate state vectors, thus allowing for the coupling of different materials as well as non-coinciding finite elements meshes. This method has also been benchmarked against another more classical numerical approach, that is the TMM. While the TMM has been successfully coupled to high-order finite elements when modelling some of the tackled configurations spanning previous results from the literature, they exhibited limitations due to their instability for a few of the selected cases. On the other end, the GM produced satisfactory results for all presented cases, demonstrating its robustness. A criterion for the automatic evaluation of the number of required Bloch modes has been proposed as well.

This work could be extended by proposing a full 3D extension for the problem formulation. A similar method for the stabilization of the recursive method could also be interesting, but more work is needed. Even though for this work the same extension as TMM was operated on the RM, the observed results remained unsatisfactory compared to the GM. Other prospects for this work include the declination of the methods to other physics: electromagnetics to compute the scattering properties of different types of grating stacks.

Chapter 3

A general spectral collocation method for computing the dispersion relations of guided acoustic waves in multilayer dissipative structures

Contents

Authored by: *Mathieu Maréchal, Alan Geslain, Jean-Philippe Groby, Vicent Romero-García and Olivier Dazel*.

This chapter is based on the material published in [0]. Minor modifications to the text and figures have been made to fit the format and notation of this thesis.

3.1 Abstract

A spectral collocation method is proposed to compute the complex wavenumber-real frequency dispersion relations of guided acoustic waves in multilayer structures involving dissipative materials. The nature of these dissipative materials is initially considered to be arbitrary, i.e., poroelastic, viscoelastic, or viscoacoustic. For a given frequency, the complex wavenumbers as well as the physical fields, which are further used to evaluate the Poynting vectors and analyze the energy flux, are obtained by solving a generalized eigenvalue problem. The latter arises from a set of discretized equations of motion and appropriate boundary (coupling) conditions. These equations of motion and boundary (coupling) conditions are imposed by the nature of the material composing each layer of the structure. A focus is made on poroelastic layers. The dispersion relation of a two-layer elastic-poroelastic structure is analyzed, as well as the energy flows in the structure. The results as calculated with the present spectral collocation method are validated against those obtained with a classical complex root-finding (Müller) method and experiments.

3.2 Introduction

Dealing with guided acoustic waves in multilayer stuctures of dissipative (viscoelastic, viscothermal, or poroelastic) material requires the evaluation of complex wavenumber-real frequency dispersion relations. Solving for these dispersion relations consists in finding complex roots of complex characteristic equations, which often arise from complex matrix determinants and can rarely be solved analytically. Various methods [51, 52], among which the Müller algorithm [53], are commonly used to achieve this challenging task. These methods usually rely on preliminary semi-analytical derivations, require multiple initial guesses, and are iterative, so they can get stuck in local minima. Computing the complex wavenumber-real frequency dispersion relation in a reliable and robust way is thus of primary importance. Commercial software such as Disperse [54] or open-source software such as Dispersion Calculator [55, 56] are available for calculating dispersion curves. They use mode tracing or root-finding methods and focus mainly on multilayer elastic structures. Alternatively, several numerical methods can be used, such as the Finite Element Method [57], the Semi-Analytical Finite Element method (SAFE) [58–60], or spectral methods [61]. Spectral methods are especially relevant in this framework. They are numerical approximation techniques that seek the solution of a differential equation using a finite series of infinitely differentiable basis functions with *spectral accuracy* [62].

The propagation and dispersion of waves in multilayer systems have long been investigated [63, 64]. Spectral methods have been developed to cope with wave radiation problems, but have also emerged in the last decades to solve dispersion relations [65]. Effectively, the numerical scheme boils down to discretizing the geometry along the layer thickness and solving an eigenvalue problem. Here, a collocation method is used to discretize the geometry. A finite-dimension polynomial basis is chosen and the collocation points are distributed according to the roots of this polynomial basis. Leaky modes appear in practice when multilayer structures are surrounded by one or two half-spaces. Accounting for this phenomenon brings us closer to practical considerations regarding these systems, as they would radiate part of the acoustic energy back to the surrounding medium, although it adds hardships to the modeling of the system; the waves modeled in these half-spaces should satisfy some radiation condition. These can be imposed by implementing perfectly matched layers [66], by using a bounded mapping, and applying a Spectral Collocation Method (SCM) to all layers of the extended system [67], or by using a suitable change of variable and solving a non-linear eigenvalue problem [68]. This last method inspires the current work. We propose a SCM suitable for solving complex wavenumber-real frequency dispersion relations in multilayer systems involving dissipative media. As a way of example, we will focus on multilayer poroelastic systems. Mechanical wave propagation in bulk poroelastic materials are well understood and modeled thanks to the seminal contributions of M.A. Biot [69–72] and subsequent contributions, notably by D.L. Johnson *et al.* [73] and J.-F. Allard *et al.* [74, 75] concerning the modeling of viscothermal losses. Although poroelastic materials are bounded in practice and are usually encapsulated in more complex multilayer structures combining layers of materials of different nature in geophysics and acoustics, there are only a few studies on the dispersion relation of guided acoustic waves in poroelastic [76–78], porous [79, 80] or anisotropic poroelastic layers [81]. The SCM has already been used to solve the dispersion relation of guided acoustic waves in elastic layered rings [82, 83] or anisotropic elastic multilayer systems [84], but has never been used to cope with multilayer systems involving dissipative materials of arbitrary nature.

This work aims at filling this gap and is divided into two sections. First, the SCM is derived for generic multilayer configurations possibly coupled to two surrounding identical fluid half-spaces. A two-layer elastic-poroelastic structure is then considered by way of example. The dispersion relation computed with the SCM method is compared to that obtained with a classical root-finding (Müller) method and that measured. The energy flow is further analyzed for this two-layer configuration, with very little additional computational cost. In the last section, we experimentally analyze the two-layer structure by retrieving its complex dispersion relation from displacement measurements using the SLaTCoW (Spatial LAplace Transform for COmplex Wavenumber recovery). The previously obtained numerical results can thus be correlated with the data of the dispersion relation.

3.3 Solving the dispersion relations for multilayer structures

3.3.1. Spectral Collocation Method

Figure 3.1a depicts the general geometry of the multilayer structure considered in this article. It is composed of N homogeneous layers of arbitrary nature. The n-th layer has a thickness of $h^{(n)}$ and occupies the domain $\Omega^{(n)}$. The lower and upper interfaces are denoted $\Gamma^{(n-1)}$ and $\Gamma^{(n)}$. The

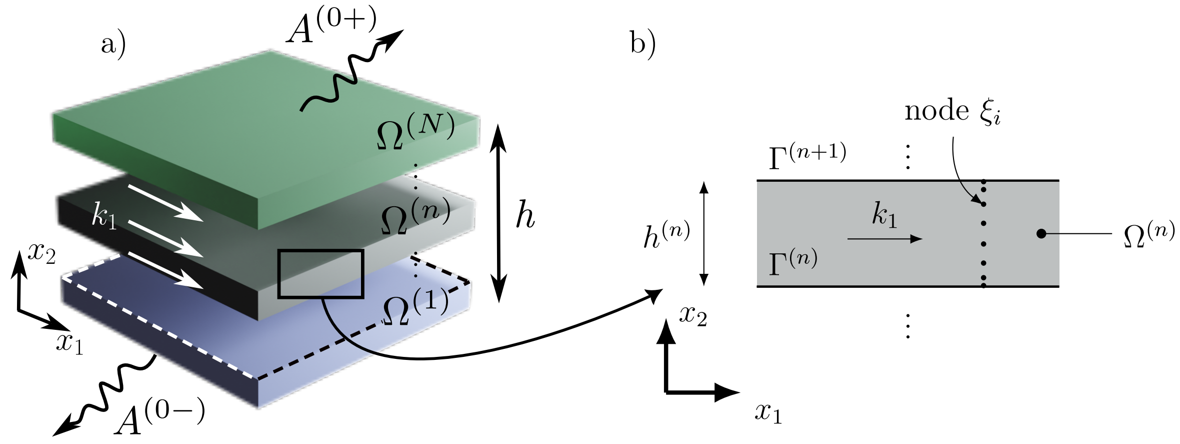


Figure 3.1: a) Sketch of the geometry of the multilayer system and b) zoom on the n-th layer.

multilayer structure is assumed to be two-dimensional, of infinite extent along x_1, and bounded along the x_2-axis between 0 and the total structure thickness $h = \sum_{n=1}^{N} h^{(n)}$. Outgoing waves radiate above and below the structure in two fluid half-spaces $\Omega^{(0^\pm)}$, occupied by the same fluid.

We assume an implicit time dependence $e^{-\mathrm{i}\omega t}$. The physical fields associated to the guided waves in the n-th layer propagating along the positive x_1-axis take the form

$$\boldsymbol{\Theta}^{(n)}(\boldsymbol{x}) = \boldsymbol{s}^{(n)}(k_1, x_2)e^{\mathrm{i}k_1x_1}, \tag{3.1}$$

where $\boldsymbol{s}^{(n)}(k_1, x_2)$ is the spatial Laplace transform of $\boldsymbol{\Theta}^{(n)}(\boldsymbol{x})$ and k_1 is the complex valued x_1-component of the wavenumber. Note that the spatial Laplace transform is used instead of the spatial Fourier transform because k_1 is complex-valued.because k_1, the x_1 component of the wavenumber is complex-valued. The physical fields that compose $\boldsymbol{\Theta}^{(n)}(\boldsymbol{x})$ depend on the nature of the material occupying the n-th layer. Introducing $\boldsymbol{\Theta}^{(n)}(\boldsymbol{x})$ in the corresponding wave equations (of second order in space in this work) reduces to a second-order differential operator $\mathcal{L}^{(n)}$ in x_2, that depends on the material properties and can further be developed as a second-order polynomial in k_1

$$\mathcal{L}^{(n)}(x_2)\boldsymbol{s}^{(n)}(k_1, x_2) = \left(k_1^2\mathcal{L}_2^{(n)}(x_2) + k_1\mathcal{L}_1^{(n)}(x_2) + \mathcal{L}_0^{(n)}(x_2)\right)\boldsymbol{s}^{(n)}(k_1, x_2) = \boldsymbol{0}, \tag{3.2}$$

where $\mathcal{L}_j^{(n)}$ are the coefficients of the polynomial expansion $j = 0, 1, 2$. Please note that this operator is nothing but the spatial Laplace transform of the equations of motion.

The interface (boundary) conditions at $\Gamma^{(n)}$ between the $n+1$-th and the n-th layers again depend on the nature of these two layers, but can be formally written in the form

$$C^{(n)-}\ \boldsymbol{s}^{(n)}(k_1)\Big|_{\Gamma^{(n)}} - C^{(n)+}\ \boldsymbol{s}^{(n+1)}(k_1)\Big|_{\Gamma^{(n)}} = \boldsymbol{0}, \tag{3.3}$$

where $C^{(n)\pm}$ are interfaces operators, which involve differential operators of maximum second-order in x_2. Following this procedure and introducing $\boldsymbol{S} = \left(\boldsymbol{s}^{(0^-)}\dots\boldsymbol{s}^{(n)}\dots\boldsymbol{s}^{(0^+)}\right)^\mathsf{T}$, with $^\mathsf{T}$ denoting transposition, the problem can be cast in matrix form, whose coefficients are differential operators in

x_2,

$$\boldsymbol{K}\boldsymbol{S}(k_1,x_2) = \begin{pmatrix} C^{(0^-)} & -C^{(0^+)} & 0 & & & & \\ 0 & \mathcal{L}^{(1)}(x_2) & 0 & & & & \\ 0 & C^{(1^-)} & -C^{(1^+)} & 0 & & & \\ & 0 & \mathcal{L}^{(2)}(x_2) & 0 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 0 & \mathcal{L}^{(N)}(x_2) & 0 \\ & & & 0 & C^{(N^-)} & -C^{(N^+)} \end{pmatrix} \boldsymbol{S}(k_1,x_2) = \boldsymbol{0}. \tag{3.4}$$

Note that the equations of motion in both upper and lower half-spaces are not required in Eq.(3.4). This system only depends on the values of the two spatial Laplace transforms $s^{(0^\pm)}$ at the upper and lower interfaces. Instead, we specify the nature of the two half-spaces, i.e. a fluid medium, and make use of the Sommerfeld condition, to explicitly give the form of the spatial Laplace transforms $p^{(0^\pm)}(k_1,x_2)$ of the pressure fields $p^{(0^\pm)}(\boldsymbol{x})$

$$\begin{aligned} p^{(0^-)}(k_1,x_2) &= A^{(0-)}e^{-ik_2^{(0)}x_2}, \in \Omega^{(0^-)}, \\ p^{(0^+)}(k_1,x_2) &= A^{(0+)}e^{ik_2^{(0)}(x_2-h)}, \in \Omega^{(0^+)}, \end{aligned} \tag{3.5}$$

with $k_2^{(0)} = \sqrt{\left(k^{(0)}\right)^2 - k_1^2}$, such that $\mathrm{Re}\left(k_2^{(0)}\right) \geq 0$, where $k^{(0)} = \omega/c_f$ is the wavenumber in the fluid and c_f the wave velocity. Thus, $\boldsymbol{s}^{(0^\pm)}$ appearing in Eq.(3.4) reduces to $s^{(0^\pm)} = A^{(0\pm)}$. $C^{(0^-)}$ and $C^{(N^+)}$ are therefore no longer operators and reduce to $C^{(0^-)} = \boldsymbol{C}^{(0^-)}$ and $C^{(N^+)} = \boldsymbol{C}^{(N^+)}$. In our fluid case, the interface conditions at $\Gamma^{(0)}$ and $\Gamma^{(N)}$ involve the spatial Laplace transform of the pressure and its first spatial derivative with respect to x_2 whatever the nature of the 1-st and N-th layers. Both matrices $\boldsymbol{C}^{(j)}$, $j = (0^-), (N^+)$, can thus be formally written as

$$\boldsymbol{C}^{(j)} = k_1\boldsymbol{C}_1^{(j)} + \boldsymbol{C}_0^{(j)} + k_2^{(0)}\boldsymbol{C}_{1'}^{(j)}, \tag{3.6}$$

with the subscript denoting the submatrices rearranged according to the power order $k1$. Index $1'$ corresponds to the term in $k_2^{(0)}$. Introducing Eqs.(3.2) and (3.6) in Eq. (3.4) leads to

$$\left(k_1^2\boldsymbol{K}_2 + k_1\boldsymbol{K}_1 + \boldsymbol{K}_0 + ik_2^{(0)}\boldsymbol{K}_{1'}\right)\boldsymbol{S} = \boldsymbol{0}. \tag{3.7}$$

This eigenvalue problem is non-linear because of the presence of the term $k_2^{(0)}$. Making use of the following changes of variable $k_1 = k^{(0)}\left(\gamma + \gamma^{-1}\right)/2$ and $k_2^{(0)} = k^{(0)}\left(\gamma - \gamma^{-1}\right)/2i$ [85, 86] and companion linearization [87], a generalized eigenvalue problem is formed and solved for γ, from which the dispersion relation can be calculated. More details on this procedure can be found in [68]. Note that the first or the last column of Eq.(3.4) vanishes when radiation is absent on either side of the multilayer system. The situation is different when both fluid half-spaces disappear. The eigenvalue problem Eq.(3.7) is no longer non-linear and the problem can directly be solved after companion linearization. Note also that fluid half-spaces are only considered for the sake of simplicity and that this procedure can be adapted to any kind of material occupying the half-spaces, as described recently [88]. Note finally that this procedure avoids discretization of these two half-spaces.

To numerically solve the problem, the spatial Laplace transform of the physical fields in each n-th layer are expanded as

$$\boldsymbol{s}^{(n)}(x_2) \approx \underline{s}^{(n)}(x_2) = \sum_{m=0}^{M} \alpha_m^{(n)} \underline{\psi}_m(\xi^{(n)}), \tag{3.8}$$

where α_m are coefficients of the polynomial expansion, $\underline{\psi}_m$ are the Chebyshev polynomial of order m, and $\xi^{(n)} = 2(x_2 - \sum_{j=1}^{n-1} h^{(j)})/h^{(n)} - 1, \ \in [-1,1]$. Underlined variables represent discrete vectors and double-underlined variables discrete matrices. The goal of this approximation is to discretize the differential operators in x_2 appearing in the problem. To do so, the SCM is employed. $\xi^{(n)}$ is discretized on the roots of the Chebyshev polynomials, $\underline{\xi}_j^{(n)} = \cos\left(\frac{j\pi}{M^{(n)}}\right)$ with $j = 0 \ldots M^{(n)}$. Please note that $\underline{\xi}_j^{(n)}$ discretely runs from 1 to -1 with increasing j, while $\xi^{(n)}$ continuously runs over the same interval but in the opposite direction, i.e. from -1 to 1. These collocation points/nodes form a non-uniform grid along the thickness of each layer. This discretization is usually preferred because the collocation points are clustered at the interfaces [62]. The locations of these nodes are presented in Fig. 3.1b for the n-th layer with an arbitrary value $M^{(n)} = 8$. The differential operators in the n-th layer are thus represented by differentiation matrices (DMs), such that

$$
\begin{aligned}
s^{(n)}(x_2) \rightarrow \underline{\underline{I}}\mathbf{:}\underline{s}^{(n)}(\underline{\xi}_j), \qquad \partial_2 s^{(n)}(x_2) \rightarrow \left(\frac{2}{h^{(n)}}\right)\underline{\underline{D_2}}\mathbf{:}\underline{s}^{(n)}(\underline{\xi}_j) = \underline{\underline{D_2}}^{(n)}\mathbf{:}\underline{s}^{(n)}(\underline{\xi}_j), \\
\partial_{22} s^{(n)}(x_2) \rightarrow \left(\frac{2}{h^{(n)}}\right)^2 \underline{\underline{D}}_{22}\mathbf{:}\underline{s}^{(n)}(\underline{\xi}_j) = \underline{\underline{D}}_{22}^{(n)}\mathbf{:}\underline{s}^{(n)}(\underline{\xi}_j).
\end{aligned}
\tag{3.9}
$$

where $\underline{\underline{I}}$ is the identity matrix and $\underline{\underline{D}}_2$ and $\underline{\underline{D}}_{22}$ are respectively the first-order and second-order normalized DMs. Contrary to an interpolation with Chebyshev polynomials, DMs interpolate the functions and their spatial derivatives directly and thus do not involve the polynomial expansion coefficients per se. [89]. Note that the normalized DMs are denoted as $\underline{\underline{D}}_2^{(n)}$ and $\underline{\underline{D}}_{22}^{(n)}$ in the following . These matrices are square and have a size of $(M \times M)$. This size also directly depends on the order of the considered Chebyshev polynomials. The formulation prevents the occurrence of roundoff errors, which is of particular importance when SCM is applied to solve eigenvalue problems [90].

The linear operator $\mathcal{L}^{(n)}$ is discretized as a matrix $\underline{\underline{L}}^{(n)}$ with size $(P^{(n)}(M^{(n)}-1), P^{(n)}(M^{(n)}+1))$, where $P^{(n)}$ is the number of physical fields required to describe the wave propagation in the layer and again depends on the nature of the material this layer is composed of. The first and the last rows of the DMs are removed because they are used to apply the interface conditions at $\Gamma^{(n+1)}$ and $\Gamma^{(n)}$ respectively. The interface operators $C^{(n)\pm}$ are effectively expressed in terms of this first row of the DMs of the n-th layer and this last row of the DMs of the $(n+1)$-th layer. These rows respectively correspond to the first, i.e. $\underline{\xi}_{M^{(n)}} = -1$ and $\xi^{(n)} = 1$, and last collocation nodes of the n-th and $(n+1)$-th layers, i.e. $\underline{\xi}_0 = 1$ and $\xi^{(n+1)} = -1$. These vectors are subsequently denoted $\underline{I}^{(n+1)-}$, $\underline{I}^{(n)+}$, $\underline{D}_2^{(n+1)-}$, $\underline{D}_2^{(n)+}$, $\underline{D}_{22}^{(n+1)-}$, and $\underline{D}_{22}^{(n)+}$. The interface matrix $\underline{\underline{C}}^{(n)^-}$ is of size $(P^{(n)} + P^{(n+1)}, P^{(n)}(M^{(n)} + 1))$ and the interface matrix $\underline{\underline{C}}^{(n)^+}$ is of size $(P^{(n)} + P^{(n+1)}, P^{(n+1)}(M^{(n+1)} + 1))$.

Finally, the full $(\sum_{n=0}^{N} M^{(n)}P^{(n)}+2, \sum_{n=0}^{N} M^{(n)}P^{(n)}+2)$-matrix system for the general multilayer system is

$$
\underline{\underline{K}}\mathbf{:}\underline{S} = \begin{pmatrix}
\underline{\underline{C}}^{(0^-)} & -\underline{\underline{C}}^{(0^+)} & 0 & & & & \\
0 & \underline{\underline{L}}^{(1)} & 0 & & & & \\
0 & \underline{\underline{C}}^{(1^-)} & -\underline{\underline{C}}^{(1^+)} & 0 & & & \\
& 0 & \underline{\underline{L}}^{(2)} & 0 & & & \\
& & \ddots & \ddots & \ddots & & \\
& & & 0 & \underline{\underline{L}}^{(N)} & 0 & \\
& & & 0 & \underline{\underline{C}}^{(N^-)} & -\underline{\underline{C}}^{(N^+)} &
\end{pmatrix}
\begin{pmatrix}
\underline{A}^{(0-)} \\ \underline{s}^{(1)} \\ \underline{s}^{(2)} \\ \vdots \\ \underline{s}^{(N)} \\ \underline{A}^{(0+)}
\end{pmatrix} = \underline{0}.
\tag{3.10}
$$

This system is cast in the form of a generalized eigenvalue problem that is solved by traditional eigenvalue solvers at each frequency. The dispersion relation for the acoustic guided wave in the dissipative multilayer system is calculated, repeating the procedure for each frequency ω.

3.3.2. Solving the dispersion relation with Müller algorithm

The dispersion relation of multi-layer dissipative structure are more commonly calculated with secant-based algorithm as the Müller algorithm. The latter relies on a semi-analytic description of the fields. These fields are best written in each layer, independently of the nature of the material it is composed of, using the potentials, i.e. decomposing the field $\theta = \nabla\varphi + \nabla\times\psi$ into an irrotational component φ and a transverse component $\psi = \psi e_3$. The spatial Laplace transform of the potentials in the n-th layer are entirely described by

$$\theta_n(k_1, x_2) = A_X^{(n)+} e^{\mathrm{i}k_{2,X}^{(n)}(x_2-h^{(n)})} + A_X^{(n)-} e^{-\mathrm{i}k_{2,X}^{(n)}(x_2-h^{(n)})}, \tag{3.11}$$

with $A_X^{(n)\pm}$ the up and down-going wave amplitudes, $k_{2,X}^{(n)} = \sqrt{\left(k_X^{(n)}\right)^2 - k_1^2}$, such that $\mathrm{Re}\left(k_{2,X}^{(n)}\right) \geq 0$, and X refers to the type of wave, i.e. shear, compressional or acoustic. If necessary, these Laplace transform representations are complemented by those of the pressure field in the fluid half-spaces provided in Eq.(3.5).

The boundary conditions are applied at each interface and the set of equations is cast into the following matrix form $\boldsymbol{M}(k_1)\boldsymbol{q} = 0$, with $\boldsymbol{q}$ a vector containing the amplitudes of each field/potential. The modes of the system correspond to the wavenumbers k_1 for which the matrix $\boldsymbol{M}(k_1)$ is singular, i.e.

$$\det\left(\boldsymbol{M}(k_1)\right) = 0. \tag{3.12}$$

These wavenumbers are the complex roots of an implicit complex-valued equation. They are evaluated by running our own implementation of the Müller algorithm [53] from 3 initial guesses. This equation is solved iteratively for each frequency ω, giving the dispersion relation.

In addition, $\boldsymbol{q}$ can be obtained by computing the nullspace of $\boldsymbol{M}$ [91] (This null space is computed using singular value decomposition as embedded in the `linalg.null_space` function from the Python package SciPy). The amplitudes of the potentials in the layers are thus evaluated, enabling all the physical fields that depend on these potentials to be calculated. However, the $\boldsymbol{M}$ matrix becomes increasingly tedious to write as the number of layers increases, and even more so when it comes to calculating the roots of its determinant and nullspaces. Nevertheless, this approach remains a fairly effective way of providing a reference solution for validating numerical results when studying a two-layer system.

3.4 Application to a two-layer poroelastic-elastic structure

In this section, the calculation of the dispersion relation for guided waves in a two-layer structure, i.e. $N = 2$, is used as an example of the application of the present SCM. This two-layer structure consists of a $h^{(1)}$-thick poroelastic layer, coated on one side by a $h^{(2)}$-thick aluminum plate, that is radiating in a fluid half-space, and rigidly backed on the other side as depicted in Fig. 3.2a. The SCM results are analyzed and compared to those obtained with the Müller root-finding method.

3.4.1. Numerical scheme

The pressure field $p^{(1)}$ and the two components of the frame displacement, $u_{1,s}^{(1)}$ and $u_{2,s}^{(1)}$, i.e. $P^{(1)} = 3$, as well as the two-components of the elastic displacement, $u_1^{(2)}$ and $u_2^{(2)}$, i.e. $P^{(2)} = 2$, are used to model the propagation in the poroelastic and in the elastic layers respectively. The $\{\mathbf{u}_s, p\}$ formulation is used [92] for the poroelastic medium. The more usual displacement formulations ($\{\mathbf{u}_s, \mathbf{u}_f\}$ [69] or

$\{\mathbf{u}_s, \mathbf{w}\}$ [71]) cannot be used in the present case, because the number of physical fields they rely on, i.e. 4, is different from that of the potentials in the layer, i.e. 3.

In total, 5 physical fields inside the two-layer structure and an additional amplitude for the acoustic wave radiated towards $x_2 \to \infty$ are needed to solve for the dispersion relation. Each domain is discretized on $M^{(j)}$, $j = 1, 2$, collocation points.

In more details, the $\{\boldsymbol{u}_s, p\}$ formulation of the Biot theory [92] is employed to model the propagation in the homogeneous isotropic poroelastic layer. The coupled equations of motion read as

$$\begin{cases} \nabla \cdot \hat{\boldsymbol{\sigma}}_s^{(1)} + \tilde{\rho}\omega^2 \boldsymbol{u}_s^{(1)} + \tilde{\beta}\boldsymbol{\nabla} p^{(1)} = 0, \\ \dfrac{\nabla^2 p^{(1)}}{\tilde{\rho}_{22}\omega^2} - \dfrac{\tilde{\beta}}{\phi^2}\nabla \cdot \boldsymbol{u}_s^{(1)} + \dfrac{1}{\tilde{R}} p^{(1)} = 0, \end{cases} \tag{3.13}$$

where $\hat{\boldsymbol{\sigma}}_s^{(1)} = \hat{A}\nabla \cdot \boldsymbol{u}_s^{(1)} \boldsymbol{I} + 2N_s \boldsymbol{\epsilon}_s$ is the *in vacuo* solid stress tensor, with $\boldsymbol{\epsilon}_s$, the solid-phase strain tensor. The parameters are generally expressed in terms of the elastic coefficients P, Q and R and the effective densities ρ_{11}, ρ_{12} and ρ_{22} initially introduced by Biot [69, 93] as

$$\hat{A} = P - 2N_s - \frac{Q^2}{R}, \quad \tilde{\rho} = \rho_{11} - \frac{\rho_{12}^2}{\rho_{22}}, \quad \tilde{\beta} = \phi\left(\frac{\rho_{12}}{\rho_{22}} - \frac{Q}{R}\right). \tag{3.14}$$

These parameters depend on the frame properties and on the effective density $\rho_{eq}(\omega)$ and bulk modulus $K_{eq}(\omega)$ of the fluid phase as reminded in 3.6.A. In short, a poroelastic material is entirely described by the porosity ϕ, the shear modulus of the frame N_s, the Poisson ratio ν, the density of the frame ρ_1, the viscous and thermal characteristic lengths Λ and Λ', tortuosity α_∞, and the flow resistivity R_f when saturated by a light fluid [69, 73, 74]. All these parameters are real valued except for the shear modulus, which incorporates a viscoelastic damping factor.

Introducing $\boldsymbol{s}^{(1)} = (\tilde{u}_1^s, \tilde{u}_2^s, \tilde{p})^{\mathsf{T}}$, the spatial Laplace transform of Eq. (3.13) is expanded in the form of Eq. (3.2) with

$$\begin{aligned} \mathcal{L}_0^{(1)} &= \begin{pmatrix} N_s\partial_{22} + \tilde{\rho}\omega^2 & 0 & 0 \\ 0 & \hat{P}\partial_{22} + \tilde{\rho}\omega^2 & \tilde{\beta}\partial_2 \\ 0 & -\tilde{\beta}/\phi^2\partial_2 & 1/R + \partial_{22}/(\rho_{22}\omega^2) \end{pmatrix}, \\ \mathcal{L}_1^{(1)} &= \begin{pmatrix} 0 & \mathrm{i}\hat{P}\partial_2 & \mathrm{i}\tilde{\beta} \\ \mathrm{i}(\hat{A} + N_s)\partial_2 & 0 & 0 \\ -\mathrm{i}\tilde{\beta}/\phi^2 & 0 & 0 \end{pmatrix}, \text{ and } \mathcal{L}_2^{(1)} = \begin{pmatrix} \hat{P} & 0 & 0 \\ 0 & N_s & 0 \\ 0 & 0 & 1/(\rho_{22}\omega^2) \end{pmatrix}. \end{aligned} \tag{3.15}$$

Finally, the discretized equations of motion read as

$$\underline{\underline{L}}^{(1)}\underline{s}^{(1)} = \begin{pmatrix} (\omega^2\tilde{\rho} - k_1^2\hat{P})\underline{\underline{I}} + N_s\underline{\underline{D}}_{22} & \mathrm{i}k_1(\hat{P} - N_s)\underline{\underline{D}}_{22} & -\mathrm{i}k_1\tilde{\beta}\underline{\underline{I}} \\ \mathrm{i}k_1(\hat{P} - N_s)\underline{\underline{D}}_2 & \hat{P}\underline{\underline{D}}_{22} + (\omega^2\tilde{\rho} - k_1^2 N_s)\underline{\underline{I}} & \tilde{\beta}\underline{\underline{D}}_2 \\ \dfrac{-\mathrm{i}\tilde{\beta}\omega^2}{\phi^2}\underline{\underline{I}} & \dfrac{-\tilde{\beta}\omega^2}{\phi^2}\underline{\underline{D}}_2 & \dfrac{k_1^2}{\rho_{22}}\underline{\underline{I}} + \dfrac{\omega^2}{R}\underline{\underline{I}} \end{pmatrix}\underline{s}^{(1)} = 0. \tag{3.16}$$

The equation of motion in the elastic medium is derived from the fundamental elasticity equations [94] and reads as,

$$\rho\omega^2\boldsymbol{u}^{(2)} + (\lambda + \mu)\boldsymbol{\nabla}(\nabla \cdot \boldsymbol{u}^{(2)}) + \mu\nabla^2\boldsymbol{u}^{(2)} = 0, \tag{3.17}$$

where ρ is the density and λ and μ are the Lamé coefficients. The corresponding discretized equation of motion is

$$\underline{\underline{L}}^{(2)}\underline{s}^{(2)} = \left(\frac{\rho^{(2)}}{\mu}\omega^2\underline{\underline{I}} + \begin{pmatrix} -k_1^2(\lambda/\mu + 2)\underline{\underline{I}} + \underline{\underline{D}}_{22} & \mathrm{i}k_1(\lambda/\mu + 1)\underline{\underline{D}}_2 \\ \mathrm{i}k_1(\lambda/\mu + 1)\underline{\underline{D}}_2 & (\lambda/\mu + 2)\underline{\underline{D}}_{22} - k_1^2\underline{\underline{I}} \end{pmatrix}\right)\underline{s}^{(2)} = 0. \tag{3.18}$$

In addition, the same procedure can be followed for the equation of motion of a wave propagating in a layer of fluid material. We give its expression in order to provide an exhaustive list of the types of layers that can be modeled using the present method. The Helmholtz equation, $(\nabla^2 + k_0^2)p = 0$, with k_0 the bulk wavenumber and p the pressure field in the fluid material, is discretized into the form,

$$\left(\underline{\underline{D}}_{22} - k_1^2 + k_0^2\right)\underline{p} = 0. \tag{3.19}$$

The interface and boundary conditions depend on the vectors $\underline{I}^{\pm}$, $\underline{D}_2^{\pm}$ and $\underline{D}_{22}^{\pm}$ which corresponds to the last (subscript +) and first row (subscript −) of the DMs. From the bottom to the top of the two-layer system, they are

- Rigid boundary condition at $\Gamma^{(0)}$ which leads to

$$\boldsymbol{u}^{(1)}(x_2 = 0) = 0; \quad \boldsymbol{u}_f^{(1)}(x_2 = 0)\cdot\boldsymbol{n} - \boldsymbol{u}_s^{(1)}(x_2 = 0)\cdot\boldsymbol{n} = 0, \tag{3.20}$$

  the discretized form of which is,

$$\underline{\underline{C}}^{(0)}\underline{s}^{(1)} = \begin{pmatrix} \underline{0} & \underline{I}^+ & \underline{0} \\ \underline{0} & \underline{0} & \underline{I}^+ \\ \dfrac{\phi}{\rho_{22}}\underline{D}_2^+ & \underline{0} & -\left(1+\dfrac{\rho_{12}}{\rho_{22}}\right)\omega^2\underline{I}^+ \end{pmatrix}\begin{pmatrix} \underline{p}^{(1)} \\ \underline{u}_1^{(1)} \\ \underline{u}_2^{(1)} \end{pmatrix} = \underline{0}. \tag{3.21}$$

- Continuity between the poroelastic and the elastic media [95] at $\Gamma^{(1)}$ which leads to

$$\begin{gathered}\boldsymbol{\sigma}^{t(1)}(x_2 = h^{(1)}) - \boldsymbol{\sigma}^{(2)}(x_2 = h^{(1)}) = 0; \quad \boldsymbol{u}^{(2)}(x_2 = h^{(1)}) = \boldsymbol{u}_s^{(1)}(x_2 = h^{(1)}); \\ \boldsymbol{u}_f^{(1)}(x_2 = h^{(1)})\cdot\boldsymbol{n} - \boldsymbol{u}_s^{(1)}(x_2 = h^{(1)})\cdot\boldsymbol{n} = 0,\end{gathered} \tag{3.22}$$

  where $\boldsymbol{\sigma}^{t(1)} = \hat{\boldsymbol{\sigma}}_s - \phi\left(1 + Q/R\right)$ is the total stress tensor in the poroelastic layer. The corresponding discretized interface matrices, arising from

$$\underline{\underline{C}}^{(1)+}\underline{s}^{(2)} - \underline{\underline{C}}^{(1)-}\underline{s}^{(1)} = \underline{\underline{C}}^{(1)}\begin{pmatrix}\underline{p}^{(1)} & \underline{u}_1^{(1)} & \underline{u}_2^{(1)} & \underline{u}_1^{(2)} & \underline{u}_2^{(2)}\end{pmatrix}^{\mathsf{T}} = 0, \tag{3.23}$$

  are cast as

$$\underline{\underline{C}}^{(1)} = \left(\begin{array}{ccc|cc} -N_s\underline{D}_2^+ & -\mathrm{i}k_1 N_s\underline{I} & \underline{0} & \mu\underline{D}_2^- & \mathrm{i}k_1\mu\underline{I}^- \\ -\mathrm{i}k_1\hat{A}\underline{I}^+ & -\hat{P}\underline{D}_2^+ & \phi\left(1+\dfrac{Q}{R}\right)\underline{I}^+ & \mathrm{i}k_1\lambda\underline{I}^- & (\lambda+2\mu)\underline{D}_2^- \\ \underline{0} & -\underline{I}^+ & \underline{0} & \underline{I}^- & \underline{0} \\ \underline{0} & \underline{0} & -\underline{I}^+ & \underline{0} & \underline{I}^- \\ -\dfrac{\phi}{\rho_{22}}:\underline{D}_2^+ & \left(1+\dfrac{\rho_{12}}{\rho_{22}}\right)\omega^2\underline{I}^+ & \underline{0} & \underline{0} & \underline{0} \end{array}\right), \tag{3.24}$$

  with the left-hand side corresponding to $\underline{\underline{C}}^{(1)-}$ and the right-hand side to $\underline{\underline{C}}^{(1)+}$.

- Continuity between the elastic and the fluid media at $\Gamma^{(2)}$ which leads to

$$\begin{gathered}\sigma_{12}^{(2)}(x_2 = h) = 0; \quad \sigma_{22}^{(2)}(x_2 = h) = -p^{(0+)}; \\ u_2^{(2)}(x_2 = h) = u_2^{(0+)}(x_2 = h).\end{gathered} \tag{3.25}$$

  The corresponding discretized interface matrix $\underline{\underline{C}}^{(2)}\begin{pmatrix}\underline{u}_1^{(2)} & \underline{u}_2^{(2)} & A^{(0+)}\end{pmatrix}^{\mathsf{T}} = 0$ is

$$\underline{\underline{C}}^{(2)} = \left(\begin{array}{cc|c} \underline{D}_2^+ & \mathrm{i}k_1\underline{I}^+ & 0 \\ \mathrm{i}k_1\lambda\underline{I}^+ & (\lambda+2\mu)\underline{D}_2^+ & 1 \\ \underline{0} & \omega^2\underline{I}^+ & \dfrac{\mathrm{i}k_2^{(0)}}{\rho_f} \end{array}\right). \tag{3.26}$$

Combining the discretized interface conditions, Eqs.(3.21), (3.24) and, (3.26), with the discretized equations of motion, Eqs.(3.16) and (3.18), leads to the following system

$$\begin{pmatrix} \underline{\underline{C}}^{(0)} & 0 & 0 \\ \underline{\underline{L}}^{(1)} & 0 & 0 \\ \underline{\underline{C}}^{(1)-} & -\underline{\underline{C}}^{(1)+} & 0 \\ 0 & \underline{\underline{L}}^{(2)} & 0 \\ 0 & \underline{\underline{C}}^{(2)-} & -\underline{\underline{C}}^{(2)+} \end{pmatrix} \begin{pmatrix} \underline{s}^{(1)} \\ \underline{s}^{(2)} \\ \underline{A}^{(0+)} \end{pmatrix} = \underline{0}. \tag{3.27}$$

This system is further expanded in the form of Eq.(3.7) and solved for a set of eigenvalues γ and associated eigenvectors $\boldsymbol{S}$ at each frequency ω. Wavenumbers k_1 are then obtained from γ. The dispersion curve is thus calculated over a given frequency range. However, the condition $\mathrm{Re}\left(k_2^0\right) \geq 0$ should be imposed to sort out the solution that do not meet it.

Although usual and simple, this two-layer structure is challenging from a numerical point of view because it involves materials with very large impedance contrasts and layers whose thicknesses differ by several orders of magnitude. These problems are partially solved by normalizing each line and rearranging the terms of the matrix Eq. (3.7), leading to a better numerical conditioning of the system.

3.4.2. Dispersion relation

The material properties of the $h^{(1)} = 52$ mm-thick poroelastic (melanime) layer are listed in Table 3.1, while the properties of the $h^{(2)} = 1$ mm-thick aluminium plate are the density $\rho = 2700$ kg.m^{-3} and the Lamé coefficients $\lambda = 60.75$ GPa and $\mu = 26.03$ GPa. The saturating fluid and that occupying the half-space is air with density $\rho_f = 1.213$ kg.m^{-3}, the heat capacity ratio $\kappa = 1.4$, kinematic compressibility $\mu_f = 1.839 \times 10^{-5}$ Pa.s, the Prandtl number Pr $= 0.71$, and adiabatic bulk modulus κP_0, where the atmospheric pressure is $P_0 = 1.013 \times 10^5$ Pa. The total two-layer thickness is thus $h = h^{(1)} + h^{(2)} = 53$ mm. $M^{(1)} = 5$ and $M^{(2)} = 11$ collocation points are employed respectively in the poroelastic and elastic layers.

The 3D view of the complex dispersion diagram is depicted Fig. 3.2b. To ease readability, real and imaginary parts are depicted separately in Fig. 3.2c. Please note that this last representation can be misleading because only the modes that lie in the complex wavenumber range depicted in Fig. 3.2b are represented in Fig. 3.2c. Cut-on frequencies are thus fictitious. The results calculated using the SCM method, plotted in blue markers, are compared with those obtained using the Müller method, plotted with black open markers. Both methods provide identical results, up to around four digits, therefore validating the present approach. Note that the SCM solutions are used as initial guesses of the Müller algorithm to shorten computational time. In this way, fewer iterations are needed for the results of the root-finding method to converge. The main advantages of SCM are that no initial guesses are needed, i.e. as long as the discretization is sufficient to model the full wave behavior in the structure, the dispersion relation is guaranteed to be calculated. In addition, the matrices used for the numerical model are very small compared to those required by other methods such as FEM, and are therefore much less demanding in terms of calculation.

Wavenumbers are perfectly symmetric with both $\mathrm{Re}\left(k_1 h/2\pi\right) = 0$, and $\mathrm{Im}\left(k_1 h/2\pi\right) = 0$ axis, which is a feature of reciprocal systems. Wavenumbers having a very large slope and a low imaginary part correspond to modes mostly propagating in the aluminium plate. They are very close to the bulk longitudinal k_l and tranverse k_t wavenumbers in the aluminum. This is due to the very large contrast between the elastic properties of the aluminum and those of the poroelastic frame. The other branches asymptotically tend toward the bulk acoustic, k_a, compression, k_p, or shear, k_s,

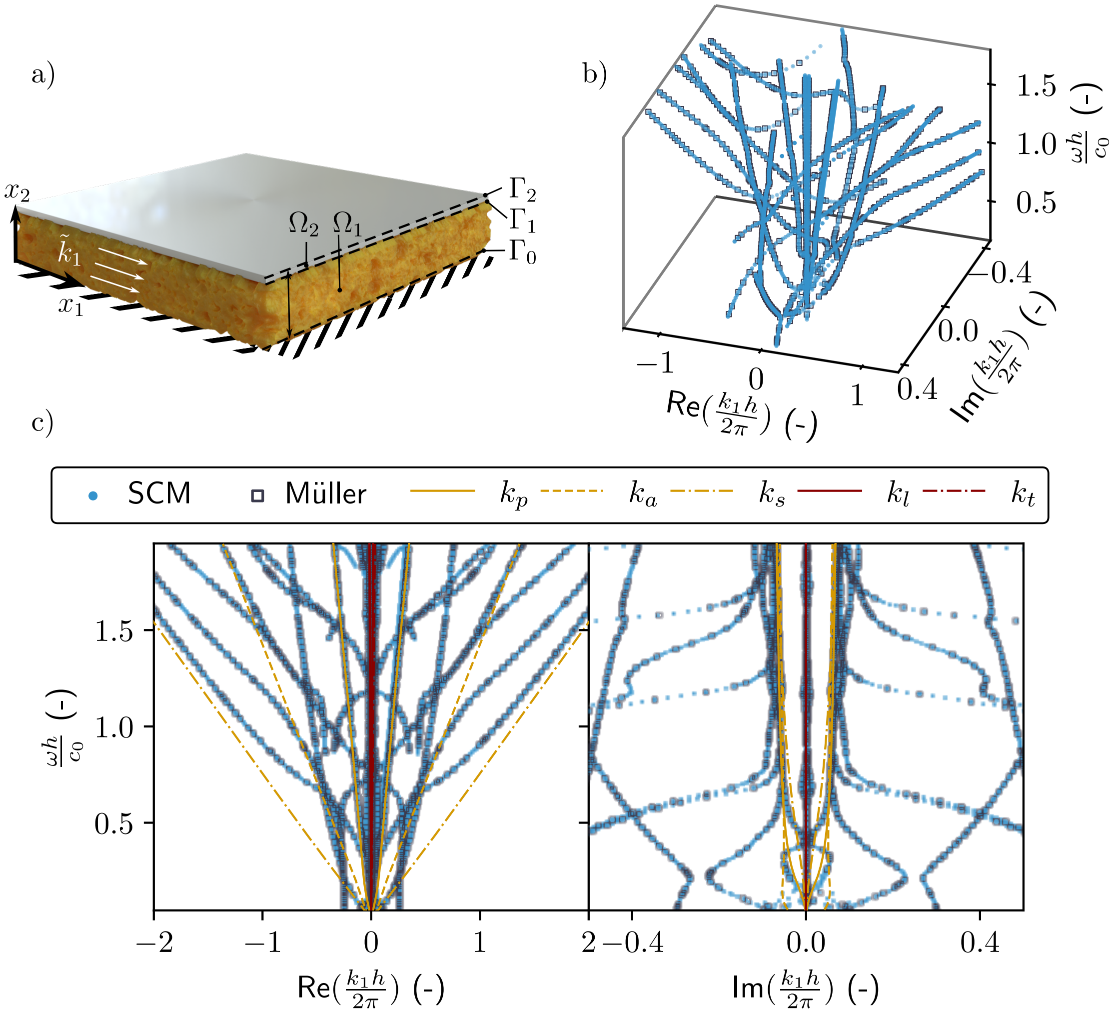


Figure 3.2: a) Sketch of the two-layer structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. Blue dots represents the results as calculated with the SCM and black open markers represents those as calculated with the Müller method. Bulk wavenumbers in the poroelatic and elastic materials are highlighted in yellow and dark red respectively.

Table 3.1: Parameters of the poroelastic material (melamine foam)

Parameter	ϕ	ρ_1	R_f	Λ	Λ'	α_∞	ν	N_s
Unit	-	kg.m^{-3}	kPa.s.m^{-2}	μm	μm	-	-	kPa
Melamine	0.98	6.5	5.6	214	214	1	0.24	11.96 (1 + i0.07)

wavenumbers of the poroelastic medium at high frequency. At low frequencies, these modes are highly dispersive.

If we compare these results with the dispersion relation depicted in 3.6.Bc) for the guided waves in a single poroelastic plate twice as thick and in the absence of the elastic plate, we find some essential differences . The branches associated with the elastic plate and the branch clearly associated with the coupling between the elastic and poroelastic layers (referenced as (1) in Fig.3.6.Bc)) do not exist. The other branches are similar to those of this single poroelastic layer, although slightly modified by the presence of the elastic plate. This proves a weak coupling between the elastic and poroelastic layers, particularly at high frequencies. At lower frequencies, the coupling between the two layers is stronger. Effectively, the wavelength is much larger than the thickness of at least one of the two layers in these frequency ranges, which favors coupling. The A_0 branch observed in the case of a single poroelastic layer disappear in the two-layer system rigidly backed, because this rigid-backing does not allow the existence of such a mode.

3.4.3. Mode shapes and energy fluxes

Beyond the SCM numerical efficiency, a significant advantage of the method is that the eigenvectors associated to each eigenvalue directly provides the associated mode shape. Since these mode shapes, i.e. eigenvectors, are calculated from an eigenvalue problem, their amplitude are defined up to a constant. They are normalized to the value of the pressure field at the bottom of the poroelastic layer, i.e. at the location of the rigid backing, because this condition imposes $\boldsymbol{u}^{(1)}(0) = 0$ (and in particular $u_2^{(1)}(0) = 0$), which is equivalent to a maximum pressure field. From these normalized eigenvectors, any physical fields can be calculated as detailed in 3.6.C.

Let us consider two solutions $k_1^{\pm} = \pm|\mathrm{Re}\,(k_1)\,|$ symmetric in the dispersion diagram with respect to the axis $\mathrm{Re}\,(k_1) = 0$ as shown in Figs. 3.3a-b. The displacement fields and total energy fluxes in each direction are depicted in Fig. 3.3c-f. The results calculated using the SCM method, plotted in continuous lines, match those obtained using the Müller method, plotted with marker lines. These two modes are identical but propagate in opposite directions, i.e., the mode k_1^+ propagates towards positive x_1 while the mode k_1^- propagates towards negative x_1. Displacement fields and energy fluxes are identical along the x_2 axis but are of equal modulus but opposite sign along the x_1 axis (see Fig. 3.3c-d and e-f). Both displacement field and energy flux amplitudes are two to three orders of magnitude smaller in the elastic plate (for $h_1 < x_2 < h$) than in the poroelastic plate. The aluminum coating is almost acting as a rigid boundary condition, as also testified by the fact that the x_1 components of the displacement field and energy fluxes are almost symmetric with respect to the axis $x_2 = h_1/2$, while their x_2 components are almost antisymmetric along the x_2 axis. Note that the fluid phase Poynting vectors depicted in Fig. 3.3f are of equal modulus but opposite sign, symmetric with respect to the axis $x_2 = h_1/2$, and more than a hundred times smaller than total energy flux in amplitude. The energy of these modes are thus mainly localized in the solid phase of the poroelastic plate.

To go a step further, we evaluated the energy transport velocity $\overline{V_e} = \overline{P}_t/\overline{U}$ as the ratio between the average energy flux in each layer $\overline{P}_t$ over the mean total energy $\overline{U}$. The energy transport velocity of the mode highlighted in orange in Fig. 3.3a-b is compared to the group velocity $v_g = \partial\omega/\partial\mathrm{Re}\,(k_1)$ [96] in Fig. 3.3g. It is numerically computed by doing finite differences on a branch of the dispersion relation. Both velocities match for frequency ranges where the imaginary part of k_1 is low, i.e., for weak attenuation modes [97–100], but are different for strong attenuation. In particular, the group velocity diverges around the cut-on frequency. This is the reason why the energy transport velocity is usually preferred to study absorbing structures. The energy transport velocity $\overline{V_e}$ of the mode highlighted in green Fig. 3.3a-b is depicted in Fig. 3.3h. This energy transport velocity is always positive, while k_1 crosses the $\mathrm{Re}\,(k_1) = 0$ axis. When $\mathrm{Re}\,(k_1) < 0$ the phase velocity is negative, while the $\overline{V_e} > 0$. This branch (and the symmetric one) has also a negative group velocity as also testified

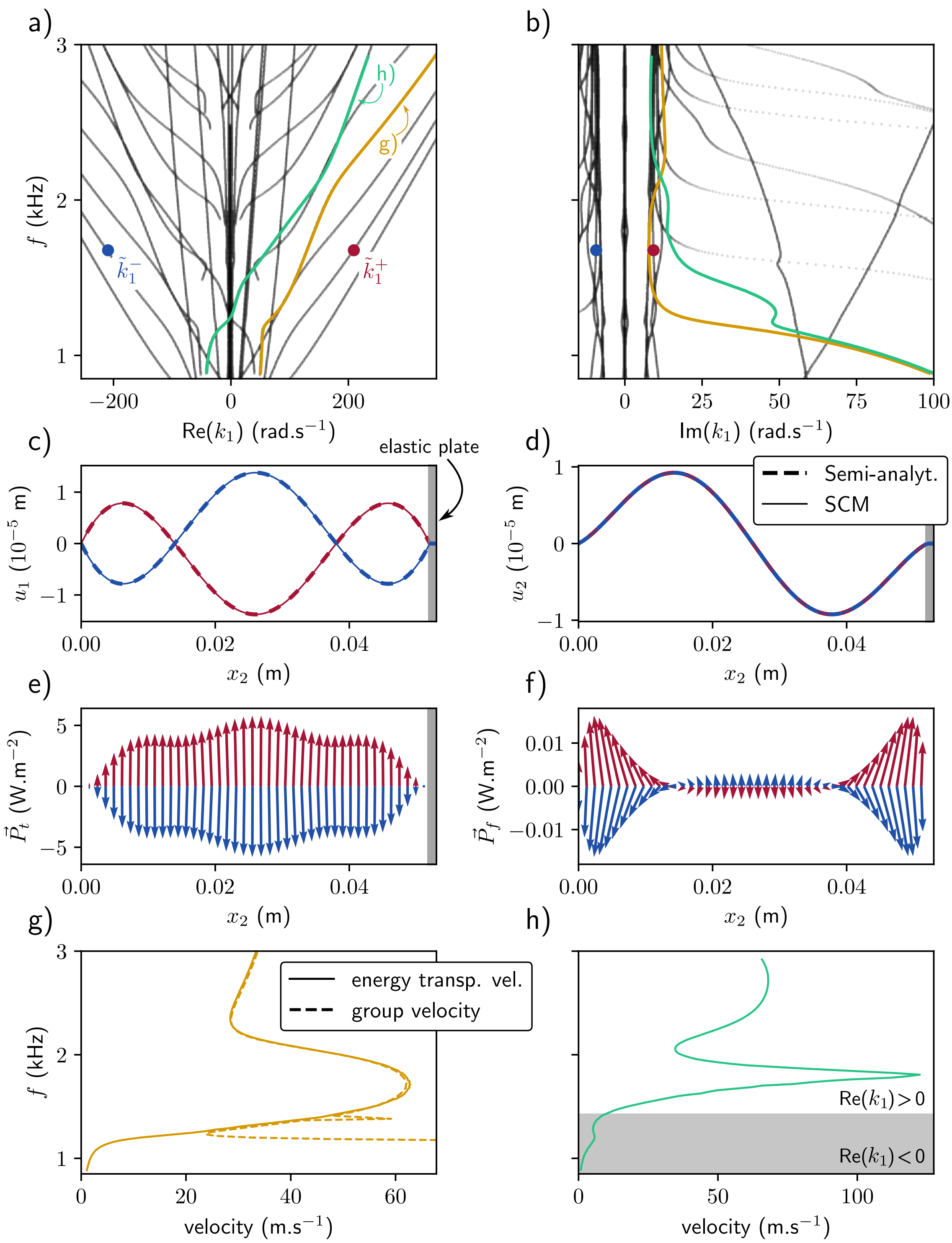


Figure 3.3: a) and b) represent the real and imaginary parts of the dispersion relation of the two-layer structure. Two branches are highlighted in yellow and green, as well as two specific solutions, k_1^- (blue) and k_1^+ (red), symmetric with respect to the Re $(k_1) = 0$ axis. Displacement fields in the x_1 c) and x_2 direction d) associated with the k_1^+ and k_1^- solutions displayed. Total Poynting e) vector and fluid Poynting vector f) with the length of the arrow indicating the amplitude and angle indicating the direction of the vector field. g) Group and energy transport velocities for the yellow branch. h) Energy transport velocity for the green branch. The gray area represents the frequency range over which this velocity is negative.

by the fact Re $(k_1) < 0$ while Im $(k_1) > 0$ in the shaded area in Fig. 3.3h. This branch has also an exponentially growing amplitude.

3.4.4. Experimental validation

The experimental set-up is depicted in Fig. 3.4a. The sample is the two-layer structure considered in the previous subsection. The aluminum plate is glued on top of the poroelastic layer. The latter layer is glued on a 3 cm-thick aluminium plate, which mimic the rigid backing. The three elements are 45 cm wide and 85 cm long. The surrounding medium and the saturating fluid is air, at regular pressure and temperature conditions. The sample is positioned perpendicularly to the ground on its longest side as depicted in Fig. 3.4a to ease the mode excitation in the poroelastic layer. The excitation is provided by a shaker (Brüel and Kjaer type 4810), which is rigidly attached to the sample with a threaded steel rod fixed to the shaker on one side and glued on a 1mm thick aluminum plate of width 10 cm and height 1.5 cm. This plate is cut at the edge opposite to the threaded steel rod and glued to the porous sample, creating a line source at the edge of the sample. The resonance of this part was measured to be 4500 Hz, which is the upper limit of the measured frequency range. The excitation signal is a swept-sine function with 400 points ranging from 1 kHz to 3 kHz. The general layout of the experiments is similar to that used originally in [101]. The key difference in our case is that the excitation is operated directly to the poroelastic layer and not to the top of the aluminum plate, because the modes that corresponds to propagating solutions in the poroelastic layer are encapsulated in between two almost rigid plates (see Section 3.4.3). The normal displacement is measured with a laser Doppler vibrometer (Polytec VibroFlex Neo) on a line of length $L = 40$ cm, along the side of the thickness of the porous layer. These measurements are averaged 100 times. The excitation signal and the measured field are interfaced through a computer via a Zürich Instruments acquisition card.

The SLaTCoW method is used to analyze the space-frequency displacement measurements and recover the complex wavenumber-real frequency dispersion relation. This method relies on the Laplace spatial transform of these measurements, which gives access to the real and imaginary components of the complex wavenumber. The difference between this Laplace transform and a correctly chosen ansatz function is finally minimized for each frequency, to recover the dispersion relation. More details on this method are given in Appendix 3.6.E as well as in the original article [101]. The complex wavenumber-real frequency dispersion relation in Fig. 3.4b-c depicts the wavenumbers recovered experimentally k_s and simulated k_n. Four modes were sought during the minimization with the SLaTCow method: three of them correspond to the three bulk waves due to excitation and the fourth is depicted. The experimentally recovered mode jumps from one branch to the other because of the large modal density. The cut-on frequencies of the modes are clearly visible, especially on the imaginary part of the wavenumber. The lower real wavenumber parts of the numerical dispersion relation are not retrieved with the experimental results. This is because they correspond to a high attenuation of the mode, a part that is inherently difficult to measure in this experimental conditions. Generally, the recovered branches are those around the bulk solid compression wavenumber, corresponding to the wave that we most probably excite in the structure. The difficulties in efficiently exciting all the guided waves supported by this configuration largely explain the missing curves in the dispersion relation. The location and orientation of the excitation line limit possible symmetries and the generation of potential shear waves. This choice was made to avoid the main excitation of guided waves in the elastic plate when exitation and displacement measurement are performed only on this element. In addition, the modes mainly supported by the fluid phase should also be better excited by a loudspeaker and therefore better measured by a microphone. However, we have chosen to focus on the modes mainly supported by the solid phase. Nevertheless, the agreement between experimental results and simulation is quite good, thus validating the method.

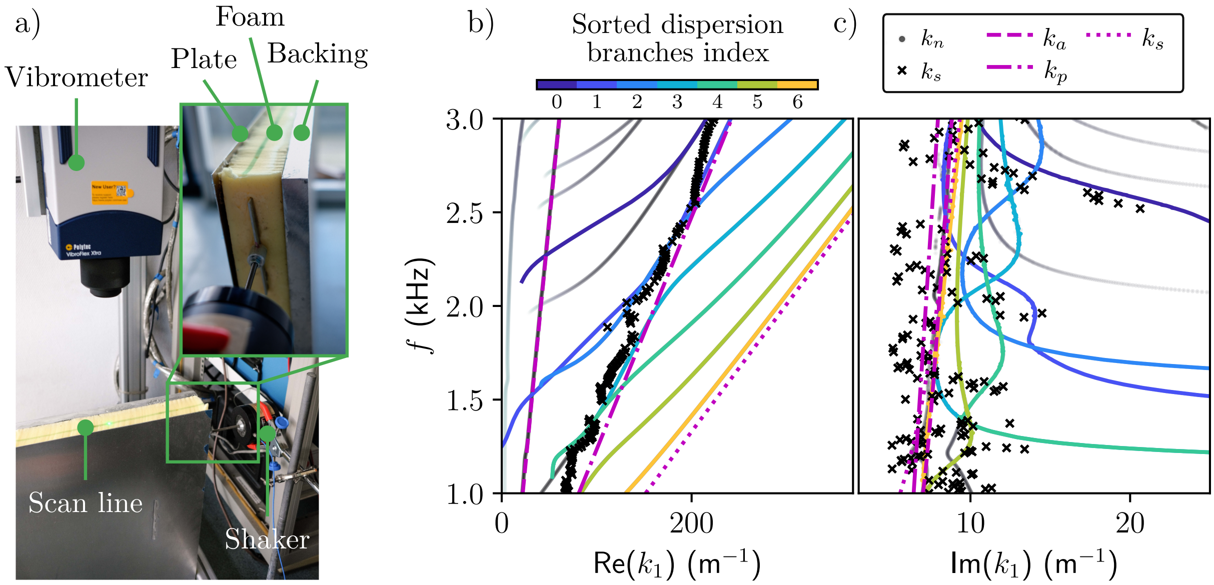


Figure 3.4: a) Experimental set-up used for the measurement of the normal displacement field of the two-layer structure, with a close-up of the sample excitation. b) Real and c) imaginary parts of the dispersion relations recovered experimentally from the SLaTCoW method k_s (black crosses) and calculated from the SCM k_s (colored lines and black-to-grey dots) and the bulk wavenumbers in the poroelastic material (magenta lines).

3.5 Conclusion

A spectral collocation method (SCM) is proposed to calculate dispersion relations for guided acoustic wave propagation in multilayer structures. The formalism is general enough to be able to consider arbitrary arrangements of (visco-) elastic, poroelastic, and (viscothermal-) fluid layers. Outgoing fluid radiation, leading to Leaky modes, can be accounted for. The fields in each layer is approximated by Chebyshev polynomials and discretized on the corresponding nodes. Interface conditions are implemented to couple layers together. The solution provides the full complex wavenumber-real frequency dispersion relation of the considered structure. In addition, the physical fields in each layer are directly evaluated since they are the eigenvectors associated with the eigenvalue problem giving the dispersion relation. The dispersion diagram calculated with the present SCM is validated against experimental results and that calculated with a root-finding method (Müller method) in a simple two-layer structure consisting of a rigidly backed poroelastic layer covered by a thin aluminum plate radiating in a fluid half-space. The semi-analytical approach underlying the root-finding method is taken a step further to solve for the wave amplitudes as well, providing a reference for the mode shapes calculated with the SCM. The dispersion properties in such structures, as well as the direction of the energy density in each phase of the poroelastic layers using a decoupled expression for the Poynting vectors is analyzed. Results on energy transport velocity are also reported.

This method paves the way of complex wavenumber-real frequency dispersion relation calculation and analysis of more complex structures, like layers with embedded periodic inclusions (elastic or resonant), i.e. metaporolastic surfaces.

3.6 Appendices

3.6.A. Expression of the Biot elastic coefficients and effective parameters in poroelastic materials

The complex and frequency dependent density and bulk modulus of the fluid phase, which accounts respectively for the viscous and thermal losses are [73, 74] as,

$$
\begin{aligned}
\rho_{\mathrm{eq}}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi}\left(1+\frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty\omega}\sqrt{1-\mathrm{i}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda}\right)^2}\right),\\
K_{\mathrm{eq}} &= \frac{\kappa P_0}{\phi}\left(\kappa-(\kappa-1)\Big/\left(1+\frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty\mathrm{Pr}\omega}\sqrt{1-\mathrm{iPr}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda'}\right)^2}\right)\right)^{-1},
\end{aligned}
\tag{3.28}
$$

where ϕ is the porosity, α_∞ is the tortuosity, Λ and Λ' are respectively the viscous and thermal characteristic lengths, R_f is the flow resistivity, μ_f is the dynamic viscosity, κ is the heat capacity ratio, ρ_f the density of the saturating fluid, and P_0 is the ambient pressure.

The Biot elastic coefficients, P, Q, and R, are more commonly expressed in terms of the frame, K_b, effective fluid, K_{eq}, and shear, N_s, moduli when saturated by a light fluid as

$$
P = K_b + \frac{4}{3}N_s + (1-\phi)^2\frac{K_{\mathrm{eq}}}{\phi}; \quad Q = K_f(1-\phi), \quad R = \phi K_{\mathrm{eq}}, \tag{3.29}
$$

while the apparent densities are

$$
\rho_{22} = \phi^2\rho_{eq}, \quad \rho_{12} = \phi\rho_f - \rho_{22}, \quad \rho_{11} = (1-\phi)\rho_s - \rho_{12}, \tag{3.30}
$$

with ρ_s the density of the solid phase. The coefficients in Eq. (3.13) are thus

$$
\hat{A} = K_b - \frac{2}{3}N_s + \frac{K_{\mathrm{eq}}}{\phi}((1+(1-\phi))), \quad \hat{\rho} = \rho_1 + \phi\rho_f - 2\phi^2\rho_{eq} - \frac{\rho_f^2}{\rho_{eq}} \quad \beta = \frac{\rho_f}{\rho_{eq}} - 2\phi - 1, \tag{3.31}
$$

where $\rho_1 = \phi\rho_f + (1-\phi)\rho_s$ is the apparent density of the frame. Finally, the frame bulk modulus and shear modulus are linked by the Poisson ratio ν with $K_b = 2N_s(1+\nu)/3(1-2\nu)$.

3.6.B. Application to a single poroelastic layer surrounded by two fluid half-spaces

The dispersion relation of guided waves in a single layer of poroelastic material surrounded by two identical fluid half-spaces is considered (See Fig.3.5a). The poroelastic material properties are those considered in Section 3.4.2 and the layer thickness is $2h^{(1)} = 104$ mm, i.e. twice the thickness of the poroelastic layer considered in the two-layer system.

The discretized equations of motion is that provided in Eq.(3.16). $M^{(1)} = 11$ collocation points are considered. Only the boundary conditions at the lower, i.e. Γ_0 at $x_2 = 0$, and upper, i.e. Γ_1 at $x_2 = h$, interfaces are modified. They read as [95]

$$
\begin{aligned}
&\left(1+\phi+\phi\frac{\rho_{12}}{\rho_{22}}\right)u_2^{(1)} + \frac{\phi^2}{\rho_{22}\omega^2}\partial_2 p^{(1)} = u_2^{(0\pm)}, \qquad p^{(1)} = p^{(0\pm)},\\
&\hat{\sigma}_{22}^s = \left(1-\phi\left(1+\frac{Q}{R}\right)\right)p^{(0\pm)}, \qquad \hat{\sigma}_{12}^s = 0.
\end{aligned}
\tag{3.32}
$$

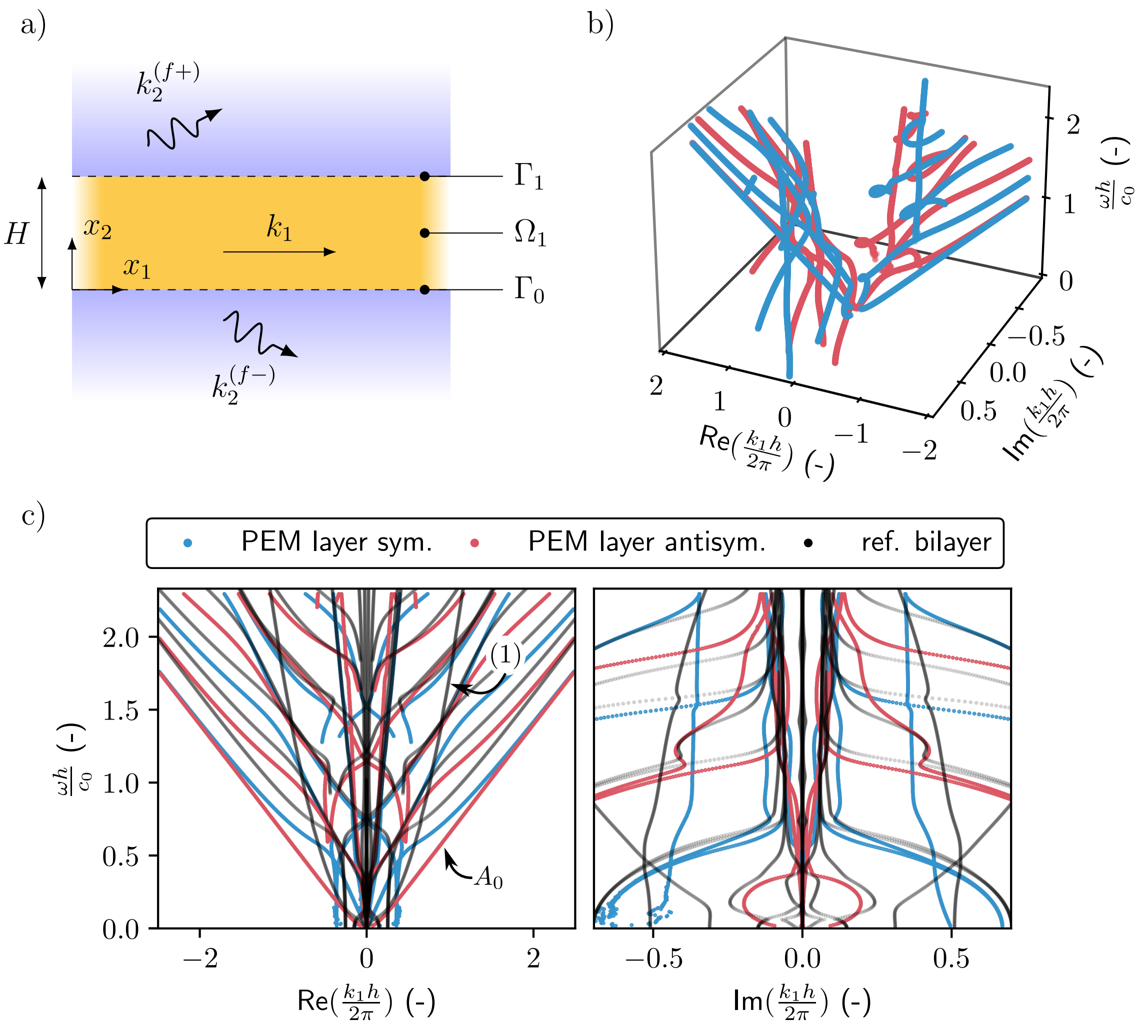


Figure 3.5: a) Sketch of the single layer poroelastic structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. The branches of the symmetric and antisymmetric modes are represented by blue and red curves respectively. The dispersion relation of the two-layer elatsic-poroelastic system studied in Section 3.4 are reminded with black lines.

The discretized form of these boundary conditions is

$$\underline{\underline{C}}_1 = \left(\begin{array}{cccc|c} \dfrac{\phi^2}{\rho_{22}} \underline{D}_2^+ & \underline{0} & \omega^2 \left(1 + \phi\left(1 + \dfrac{\rho_{12}}{\rho_{22}}\right)\right) \underline{I}^+ & & \dfrac{ik_2^{(f)}}{\rho_f} \\ \underline{I}^+ & \underline{0} & \underline{0} & & -1 \\ \underline{0} & \mathrm{i}k_1(\hat{P} - 2N_s)\underline{I}^+ & \hat{P}\underline{D}_2^+ & & 1 - \phi\left(1 + \dfrac{Q}{R}\right) \\ \underline{0} & N_s \underline{D}_2^+ & \mathrm{i}k_1 N_s \underline{I}^+ & & 0 \end{array} \right). \tag{3.33}$$

These matrices are then arranged in the same manner as Eq.(3.10) and the wavenumbers are calculated. Fig. 3.5b-c depicts the results.

In a similar way as Lamb modes, the different modes can be sorted in symmetric and antisymmetric modes. This is done by comparing the sign of the radiated amplitudes at each edge of the layer. Results are shown in Fig. 3.5b. All modes necessarily have an imaginary part, both because of the radiation condition, i.e. leakage, and because of viscothermal dissipation as can be noticed in Fig. 3.5c.

3.6.C. Evaluation of the physical fields from the SCM solutions

The discrete sets of eigenvectors resulting from the SCM are the spatial Laplace transforms of some physical fields at collocation points, i.e. the locations where the discrete equations of motion (residue) are exactly satisfied. These discrete fields are interpolated into a continuous form using

$$\boldsymbol{s}^{(n)}(x_2) = \left[\underline{\underline{\psi}}_m^{-1}(\underline{\xi}) \underline{s}_m^{(n)}(\underline{\xi}) \right] \psi\left(\xi^{(n)}\right) \tag{3.34}$$

where the terms in brackets correspond to the coefficients $\alpha_m^{(n)}$ given in Eq. (3.8). The Chebyshev polynomials ψ is the basis that maps from the discrete grid $\underline{\xi}$ back to the physical space.

The stress tensor components depend on the direct expressions of the fields and their first order derivative in x_2, denoted with a prime symbol in the following. The various fields, differentiated up to any order, can be calculated by replacing ψ by its derivative in Eq. (3.34). Explicit expressions are given in the following,

- in the elastic layer, the spatial Laplace transforms of stress tensor components are computed from $\boldsymbol{u}$, with

$$\begin{gathered} \sigma_{11} = (\lambda + 2\mu)\mathrm{i}k_1 \boldsymbol{u}_1 + \lambda \boldsymbol{u}_2', \quad \sigma_{12} = \mu\left(\boldsymbol{u}_1' + \mathrm{i}k_1 \boldsymbol{u}_2\right), \\ \sigma_{22} = \lambda \mathrm{i}k_1 \boldsymbol{u}_1 + (\lambda + 2\mu)\boldsymbol{u}_2'. \end{gathered} \tag{3.35}$$

- in the poroelastic layer, the full solid stress tensor is expressed as $\boldsymbol{\sigma}_s = \hat{\boldsymbol{\sigma}}_s - \phi\left(Q/R\right)\boldsymbol{p}$, with $\hat{\boldsymbol{\sigma}}_s$ the spatial Laplace transform of the *in vacuo* solid stress tensor introduced in Section 3.4. The components of this full solid stress tensor are,

$$\begin{gathered} \sigma_{s,11} = P\mathrm{i}k_1 \boldsymbol{u}_{s,1} + \hat{A}\boldsymbol{u}_{s,2}' - \phi\frac{Q}{R}\boldsymbol{p}, \quad \sigma_{s,12} = N_s\left(\boldsymbol{u}_{s,1}' + \mathrm{i}k_1 \boldsymbol{u}_{s,2}\right), \\ \sigma_{s,22} = \hat{A}\mathrm{i}k_1 \boldsymbol{u}_{s,2} + P\boldsymbol{u}_{s,2}' - \phi\frac{Q}{R}\boldsymbol{p}. \end{gathered} \tag{3.36}$$

  The Laplace transform of the fluid stress tensor is calculated as

$$\boldsymbol{\sigma}_f = -\phi \boldsymbol{p}\delta_{ij}, \tag{3.37}$$

  and the fluid displacement components are

$$\boldsymbol{u}_{f,1} = \frac{\phi}{\rho_{22}\omega^2}\mathrm{i}k_1 \boldsymbol{p} - \frac{\rho_{12}}{\rho_{22}}\boldsymbol{u}_{s,1}, \quad \boldsymbol{u}_{f,2} = \frac{\phi}{\rho_{22}\omega^2}\boldsymbol{p}' - \frac{\rho_{12}}{\rho_{22}}\boldsymbol{u}_{s,2}. \tag{3.38}$$

All physical fields $\theta(\boldsymbol{x})$ can finally be calculated by the inverse spatial Laplace transform with $\theta(\boldsymbol{x}) = \tilde{\theta}(k_1, x_2)e^{\mathrm{i}k_1x_1}$.

3.6.D. Energy flux in the multilayer structure - Poynting theorem in a poroelastic medium

Energy conservation is commonly conducted in the time domain. Let us consider a volum Ω, bounded by $\partial\Omega$, with $\boldsymbol{n}$ the outgoing normal vector. The energy balance or Poynting theorem in a poroelastic materials reads as [69, 102]

$$
\begin{aligned}
-\int_{\partial\Omega}(-\boldsymbol{\sigma}_s\cdot\boldsymbol{v}_s^\star)\cdot\boldsymbol{n}+(-\boldsymbol{\sigma}_f\cdot\boldsymbol{v}_f^\star)\cdot\boldsymbol{n}\,dS &= \frac{1}{2}\int_\Omega \rho_1\partial_t|\boldsymbol{v}_s|^2-\rho_{12}'\partial_t\left(\boldsymbol{v}_f-\boldsymbol{v}_s\right)^2+\rho_2\partial_t|\boldsymbol{v}_f|^2\,d\Omega \\
&+\int_\Omega \boldsymbol{\sigma}_s:\partial_t\boldsymbol{\epsilon}_s^\star+\boldsymbol{\sigma}_f:\partial_t\boldsymbol{\epsilon}_f^\star-b\left(\boldsymbol{v}_s-\boldsymbol{v}_f\right)^2\,d\Omega,
\end{aligned}
\tag{3.39}
$$

where $\boldsymbol{v}_s = \partial_t\boldsymbol{u}_s$ and $\boldsymbol{v}_f = \partial_t\boldsymbol{u}_f$ are the velocity fields, $\boldsymbol{\sigma}_s$ and $\boldsymbol{\sigma}_f$ are the stress tensors, $\boldsymbol{\epsilon}_s = 1/2\left(\boldsymbol{\nabla}\boldsymbol{u}_s+\boldsymbol{\nabla}^\mathsf{T}\boldsymbol{u}_s\right)$ and $\boldsymbol{\epsilon}_f = 1/2\left(\boldsymbol{\nabla}\boldsymbol{u}_f+\boldsymbol{\nabla}^\mathsf{T}\boldsymbol{u}_f\right)$ are the strain tensors in the solid frame and in the fluid phase respectively, and $\star$ denotes the complex conjugate. Note that a similar expression is provided in [100], where the alternative 1962 Biot formulation is used.

The left-hand side contains the energy density flux $\boldsymbol{P}_s = -\boldsymbol{\sigma}_s\cdot\boldsymbol{v}_s^\star$ and $\boldsymbol{P}_f = -\boldsymbol{\sigma}_f\cdot\boldsymbol{v}_f^\star$ in the elastic frame and fluid phase. The right-hand side contains the kinetic energy K [93] and the internal forces $P_{\text{int}} = \partial_t W + P_{dis}$. The latter is the sum of the time derivative of the strain energy W and the dissipated power P_{dis} by the elastic damping and the viscous and thermal losses [102]. In the frequency domain, the kinetic and strain energies read as

$$
\begin{aligned}
K &= \frac{1}{2}\rho_1\partial_t|\boldsymbol{v}_s|^2-\frac{1}{2}\rho_{12}'\partial_t\left(\boldsymbol{v}_f-\boldsymbol{v}_s\right)^2+\frac{1}{2}\rho_2\partial_t|\boldsymbol{v}_f|^2, \\
W &= \frac{1}{2}\hat{\boldsymbol{\sigma}}_s:\boldsymbol{\epsilon}_s^\star+\frac{1}{2}R\nabla\cdot(Q\boldsymbol{u}_s+R\boldsymbol{u}_f),
\end{aligned}
\tag{3.40}
$$

Finally, the time average total Poynting vector is $\langle\boldsymbol{P}_t\rangle = \langle\boldsymbol{P}_s\rangle+\langle\boldsymbol{P}_f\rangle$, with $\langle\boldsymbol{P}_s\rangle = -\mathrm{Re}\left(\boldsymbol{\sigma}_s\cdot\boldsymbol{v}_s^\star\right)/2$ and $\langle\boldsymbol{P}_f\rangle = -\mathrm{Re}\left(\boldsymbol{\sigma}_f\cdot\boldsymbol{v}_f^\star\right)/2$ and the total energy density is $\langle U\rangle = \langle K\rangle+\langle W\rangle = \mathrm{Re}(K)/2+\mathrm{Re}(W)/2$. Note that a similar procedure can be followed for isotropic elastic or fluid medium [103]. In particular, the elastic Poynting vector is $\langle\boldsymbol{P}\rangle = \mathrm{Re}\left(\boldsymbol{\sigma}\cdot\boldsymbol{v}^\star\right)/2$, where $\boldsymbol{\sigma}$ is the stress tensor and $\boldsymbol{v}$ is the velocity in the elastic medium.

The energy transport velocity V_e is evaluated from the latter expressions

$$
V_e = \frac{\langle\boldsymbol{P}_t\rangle\cdot\boldsymbol{n}}{\langle U\rangle}.
\tag{3.41}
$$

When a multi-layer configuration is considered, the latter velocity is averaged over the structure thickness $\overline{V_e} = \dfrac{\overline{P_t}}{\overline{U}}$, with the average Poynting vector and the average total energy defined as

$$
\overline{P_t} = \sum_n\frac{1}{h_n}\int_{h_{n-1}}^{h_n}\langle\boldsymbol{P}_t^{(n)}\rangle\cdot\boldsymbol{e}_{x_2}\,dx_2,\quad \overline{U} = \sum_n\frac{1}{h_n}\int_{h_{n-1}}^{h_n}\langle U^{(n)}\rangle\,dx_2.
\tag{3.42}
$$

3.6.E. Reminder of the SLaTCoW method for obtaining complex-valued wavenumbers.

The key point of the SLaTCoW method is the Laplace transform, $\mathfrak{L}_{\text{mes}}(z)$ with $z = z_r + \mathrm{i}z_i$ the Laplace space coordinate, of the data measured in space at a fixed frequency ω. In doing so, we

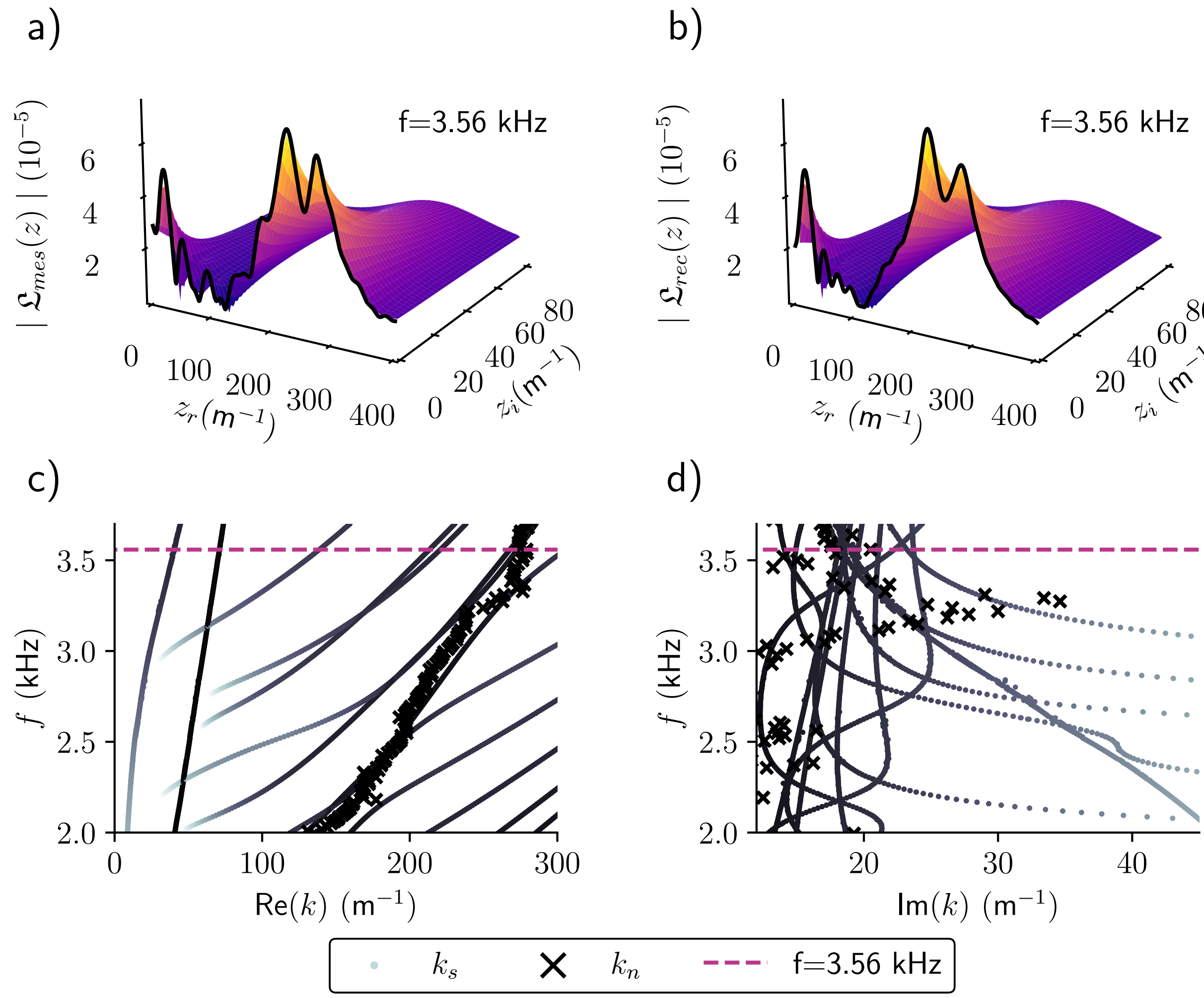


Figure 3.6: Description of the procedure of the SLaTCoW method: a) Spatial Laplace transform from the experimental data, b) Spatial Laplace transform of the reconstructed wave fields. Real c) and imaginary d) parts of the experimentally retrieved wavenumbers (crosses) compared to the SCM (black-to-grey dots). The frequency at which the Laplace transforms are represented corresponds to the magenta dashed line in the graph.

move from the spatial domain to the complex wavenumber domain. The difference between $\mathfrak{L}_{\text{mes}}(z)$ and the spatial Laplace transform of the displacement ansatz along the same line, $\mathfrak{L}_{\text{rec}}(z)$, is then minimized in the least square sense to recover both the complex amplitude (magnitude and phase) and the complex wavenumber of each mode assumed to be part of the displacement ansatz.

The experimental Laplace spatial transform $\mathfrak{L}_{\text{mes}}(z) = \int_0^L r(x)e^{-\text{i}zx}dx$ is computed at each frequency ω from the transfer function $r(x)$ between the excitation and the normal displacement measured along a line $x \in [0, L]$, with $L = 0.3$ m and spacing $d_x = 0.001$ m. The displacement anstaz is assumed to be of the form $w(x) = \sum_\beta A^\beta e^{ik^\beta x}$, with β the number of guided-waves, i.e., the number of modes, and where $A^\beta = |A^\beta|e^{\text{i}\phi^\beta}$ is the complex amplitude and k^β is the complex wavenumber of the β-th mode.. In practice, β is the smallest integer that gives a good fit between the two Laplace transforms and therefore depends on the frequency under consideration. Its spatial Laplace transform reads as $\mathfrak{L}_{\text{rec}}(z) = \sum_\beta A^\beta L e^{\text{i}(k^\beta - z)\frac{L}{2}} \operatorname{sinc}\left((k^\beta - z)\frac{L}{2}\right)$.

The cost function $\mathfrak{C} = (\|\mathfrak{L}_{\text{rec}}(z) - \mathfrak{L}_{\text{mes}}(z)\|) / \|\mathfrak{L}_{\text{mes}}(z)\|$ is finally minimized using the Matlab function *fminsearchbnd* for $\{|A^\beta|,\ \phi^\beta,\ \text{Re}\left(k^\beta\right),\ \text{Im}\left(k^\beta\right)\}, \forall\beta$. The initial guess are the recovered values of the minimization parameters at the previous frequency, with the exception of the first frequency. The method is graphically exposed in Figure 3.6. Figures 3.6 c) and d) depict the complex dispersion relation as already depicted in Fig. 3.4. Figure 3.6a) depicts the spatial Laplace transform of the experimental data and Figure 3.6b) depicts $\mathfrak{L}_{\text{rec}}(z)$ with the parameters derived from minimization of the cost function at the frequency $f = 3.56$ kHz. This frequency is highlighted by the purple dashed line in Figures 3.6 c) and d). The two Laplace transforms match almost perfectly.

3.7 abstract

A numerical method is developed in this paper allowing for the computation of leaky complex modes that exist in a metaporous layer with embedded rigid inclusions. The method is based on a finite element discretization of the unit cell along with a plane-wave modelling of the surrounding fluid half-spaces. The coupling of both models is imposed at the interfaces using a state vector formalism, leading to a matrix system that is solved using a root-finding procedure. The method is first applied to a homogeneous porous layer, allowing us to highlight the modal behaviour and propose a classification of the observed complex modes. Then, we investigate the case of a rigid inclusion and its modal behaviour. When tuned at a proper radius, a rigid inclusion provides a perfect absorption peak at subwavelength frequencies; the dispersion relation allows for the investigation of its modal behaviour. Further studies on the dispersion relations of split-ring resonators is also presented.

3.8 Introduction

Achieving the control of wave propagation is the main objective driving the development of phononic crystals and acoustic metamaterials. (These materials can achieve various effects, including significant sound attenuation, broadband absorption, and controlled diffusion due to their geometrical and material properties). Over the past decade, metaporous structures have been sought for their enhanced acoustic absorption properties. Notably it was shown that total absorption occured at a subwavelength frequency by tuning the inclusion radius [76, 104], from an effect emerging from the embedded periodic inclusion. Resonant inclusions have shown to be able to produce an additionnal absorption peak as well [105].

These configurations combine spatial periodicity by addition of a scatterer in the unit cell with the intrinsic material dissipations of porous and poroelastic materials, resulting in complex wave dispersion phenomena. While significant progress has been made in calculating the band structures for closed systems[106, 107]+autre citation ?, existing approaches often neglect the coupling between the structure and the surrounding medium that exist. Such coupling introduces energy leakage from the structure to the surrounding medium; modes in such configurations thus become leaky []. These leaky guided waves are modes that partially radiate energy into a surrounding medium, and are not addressed in the context of lossy periodic materials. This gap becomes critical when accurate modeling of dispersion is required to capture both material losses and radiation leakage, as encountered in real-world acoustic devices but also in the case of a computation of the acoustic response of the structure, where a plane wave originated from an air medium impinges on said structure.

Diverse methods are able to compute the acoustic response of such structure like the multiple scattering theory (MST), used for porous [104] and metaporoelastic [76] structures validated using classical finite elements method (FEM). The FEM with high-order finite elements [108, 109] has been addressed for various acoustic problems. Coupled to a transfer matrix formalism, it allows for the study of homogeneous multilayer configurations [110, 111] and periodic structures when coupled to Bloch waves [112].

Dispersion in such structures originates from the interaction of waves with geometrical obstacles and the inherent losses in these materials. The computation of the dispersion relations in periodic structure has long relied on Plane Wave Expansion (PWE) and its extended version (EPWE) [106, 113]. However, their use case is limited to materials of the same nature and to lossless materials respectively. FEM has been tackled as well to solve the dispersion relations of a 3D structure of porous material with rigid inclusions [107]. To our knowledge, using multiple scattering for the dispersion problem might be unfeasible since the Schlömilch series arising in the problem [76, 104] diverges when the longitudinal wavenumber becomes complex.

The dispersion of waves in periodic structure containing a porous or poroelastic host has been studied e.g. in a metaporous liner, highlighting the existence of an exceptional point in the presence of a resonant inclusion [114]. A formulation is presented in Ref. [115] for the computation of a dispersion relation in a 1D periodic poroelastic alternating medium. The complex dispersion curve of a 2D phononic crystal with an anisotropic poroelastic host has been studied [116], as well as with different interface effects [117].

Accounting for leaky modes leads to a more accurate modeling of the structure in the presence of a surrounding medium. Various numerical methods among which the spectral collocation method (SCM) or semi-analytical finite elements (SAFE) have been introduced to compute the complex dispersion curves in homogeneous layers with leakage in elastic layers [67, 68], viscoelastic [118] and in multilayer dissipative structures [0]. The discretization in 1D done in this framework cannot be generalized to periodic structures wherein a 2D discretization is required.

Structures composed of an array of scatterers have also been recently investigated [119–122] +détails sur Rayleigh-Bloch modes.

Leaky guided waves in lossy periodic layers has not been adressed in acoustics. Leaky waves in lossy structures was discussed in optics for plasmonic nanospheres [123]. However, related works in acoustics include Ref. [124] where material losses in phononic crystals are studied with dispersion diagrams, and Ref. [107] where the dispersion of waves in a 3D periodic porous layer is shown, but no leakage is accounted for in either of these papers. Computation of such dispersion relation using Comsol would result in a tedious formulation involving some perfectly-matched layer (PML) as done in homogenenous structures [] and thus the need of additionnaly discretizing the surrounding medium and would also lead to additionnal radiation modes and cavity resonances.

We propose a method that involves a finite element discretization of the unit cell coupled to a plane-wave description of its surrounding half-spaces. The coupling at the interfaces is imposed using a state vector formalism. The method will be first presented in Section. 3.9 with the analytical description of the air surrounding media and the procedure for its coupling with the periodic structure modeled using FEM. This problem is then solved using a root-finding method to retrieve the dispersion relation of said structure. Then, the case of an homogeneous porous layer will be discussed in Section. 3.10 in order to highlight on one hand the key modal behaviour of this simple geometry and on the other hand the classification proposed for these complex modes, which is not a trivial question. Finally, Section. 3.11 will show some results with a rigid inclusion, a configuration showing perfect acoustic absorption and a split-ring resonator. A comparison between the modal behaviour and the acoustic response of the structure is done, highlighting the specific modes involved in the scattering processes.

3.9 Method

3.9.A. Problem presentation

In all this paper, geometrical coordinates are denoted by $\boldsymbol{x} = (x_1, x_2)$ and a positive time dependance $e^{i\omega t}$ is considered. The unit cell, as depicted in Fig. 3.7 is a 2D rectangle of thickness h and width d (corresponding to the period of the unit cell). Hence, the considered layer is constituted by infinitely repeated cells in the x_1-direction. The host material Ω is a rigid-frame porous layer modeled as an equivalent fluid, which response is governed by Helmholtz equation with complex fluid properties, density ρ_{eq} and compressibility K_{eq}, both obtained from the JCA model [73, 74] as detailed in Appendix .1. A rigid inclusion of arbitrary shape is embedded in the cell. The top and bottom edges are respectivley denoted by Γ_t and Γ_b and they can be associated to a Neumann boundary condition (rigid wall) or to the interface with a semi-infinite air domain (in the x_2 direction). The method will focus on the transmission case, that is both top and bottom interfaces coupled to the semi-infinite air domain.

The proposed method is based on modelling the semi-infinite media using a plane wave formalism (presented in section 3.9.B) and a model of the periodic cell using FEM (presented in section 3.9.C). The final homogeneous system that describes the coupled problem for the transmission case is derived in section 3.9.D as well as the modification allowing to write the system for the rigidly-backed case, that is an addition of a rigid boundary condition at Γ_t.

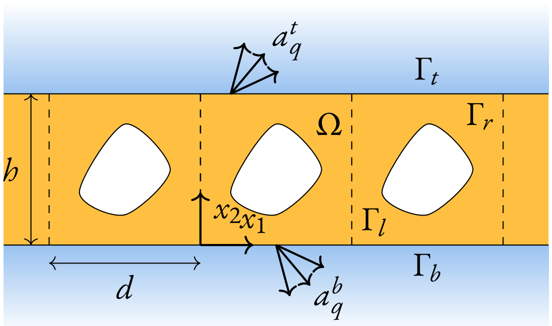


Figure 3.7: Diagram of the configuration

3.9.B. Wave formalism in the semi-infinite media

The pressure fields, solutions to the Helmholtz equation in the semi-infinite air domains, can be expressed as a countable superposition of Bloch modes. For each of them (indexed by $q \in \mathbb{Z}$), a decoupling of the space variables x_1 and x_2 is considered. The dependence in x_1 is thereby associated to a function,

$$\psi_q(x_1) = e^{-ik_{1,q}x_1}, \quad k_{1,q} = k_1 + 2\pi q/d. \tag{3.43}$$

The longitudinal wavenumber remains to be determined and can be written as $k_1 = k_1' + \mathrm{i}k_1''$. Its real part k_1' is related to the propagation of the phase flow while the imaginary part k_1'' represents the attenuation of the wave when negative. The amplitudes of each pressure field in the top and bottom semi-infinite media are respectively denoted as a_q^t and a_q^b. The expressions for the top and bottom pressures can be written in the form of outgoing waves,

$$\begin{aligned} p^t(\boldsymbol{x}) &= \sum_{q\in\mathbb{Z}} a_q^t \psi_q(x_1) e^{-\mathrm{i}k_{2,q}(x_2-h)}, \\ p^b(\boldsymbol{x}) &= \sum_{q\in\mathbb{Z}} a_q^b \psi_q(x_1) e^{+\mathrm{i}k_{2,q}x_2}, \end{aligned} \tag{3.44}$$

where the mode index q is associated to a transverse wavenumber $k_{2,q} = \sqrt{k_0^2 - k_{1,q}^2}$, with $k_0 = \omega/c_0$ the wavenumber in air and c_0 its wave velocity. This wavenumber can also be written in algebraic form as $k_{2,q} = k_{2,q}' + \mathrm{i}k_{2,q}''$. The assumptions on the signs that $k_{2,q}'$ and $k_{2,q}''$ should take depend on the choice of the square root branch cut and are discussed in Section 3.10. At this point, we only select forward wave solutions propagating towards positive x_1 direction, that is $k_1' > 0$ since the spectrum of solutions for backward waves is symmetric in this reciprocal system.

In order to couple the periodic layer and the semi-infinite layers, a state vector formalism is introduced. This vector gathers the normal displacement along x_2, u_2 and the pressure field linked to the latter by the Euler equation, $\rho_0\omega^2 u_2 = \partial_2 p$, with ρ_0, the density of air. At the top interface, the state vector can be decomposed on Bloch waves and reads as

$$\begin{aligned} \boldsymbol{S}^t(x_1) &= \begin{pmatrix} u_2(x_1, x_2 = h) \\ p(x_1, x_2 = h) \end{pmatrix} = \sum_{q\in\mathbb{Z}} \boldsymbol{S}_q^t \psi_q(x_1), \\ \boldsymbol{S}_q^t &= \begin{pmatrix} S_{1,q}^t \\ S_{2,q}^t \end{pmatrix} = \boldsymbol{\sigma}_q^t a_q^t = \begin{pmatrix} -\dfrac{ik_{2,q}}{\rho_0\omega^2} \\ 1 \end{pmatrix} a_q^t, \end{aligned} \tag{3.45}$$

where $\boldsymbol{S}_q^t$ is the state vector associated to the qth Bloch wave and $\boldsymbol{\sigma}_q^t$ its polarization vector. A similar expression can be obtained for the bottom interface,

$$\begin{aligned} \boldsymbol{S}^b(x_1) &= \begin{pmatrix} u_2(x_1, x_2 = 0) \\ p(x_1, x_2 = 0) \end{pmatrix} = \sum_{q\in\mathbb{Z}} \boldsymbol{S}_q^b \psi_q(x_1), \\ \boldsymbol{S}_q^b &= \begin{pmatrix} S_{1,q}^b \\ S_{2,q}^b \end{pmatrix} = \boldsymbol{\sigma}_q^b a_q^b = \begin{pmatrix} \dfrac{ik_{2,q}}{\rho_0\omega^2} \\ 1 \end{pmatrix} a_q^b. \end{aligned} \tag{3.46}$$

In the following, we will derive a state vector expression at both interfaces in the unit cell to couple both domains. This will be done using the fields involved in the FEM formulation.

3.9.C. Finite-element approach

The weak formulation of the Helmholtz equation [125] reads as

$$\begin{aligned} &\underbrace{\frac{1}{\rho_0\omega^2}\int_\Omega \nabla p \nabla \delta p{:}d\Omega - \frac{1}{\rho_0 c_0^2}\int_\Omega p\delta p{:}d\Omega}_{\mathcal{I}_\Omega(p,\delta p)} \\ &- \int_{\partial\Omega} u_n(M){:}\delta p(M){:}dS = 0, \quad \forall{:}\delta p \in V, \end{aligned} \tag{3.47}$$

with V is a set of admissible test functions. $\mathcal{I}_\Omega(p, \delta p)$ corresponds to the volume term. The pressure field p is associated to the unknown function while the normal outward displacement field u_n is associated to boundaries. The volume terms will be discretised using a classical FEM procedure [126, 127] and won't then be detailed in this paper. A focus is put on the boundary term, which is separated into five different regions denoted as Γ_l, Γ_r, Γ_t, Γ_b, and Γ_i respectively for the left, right, top, bottom and interior boundaries. The inclusion being rigid, the normal velocity at boundary Γ_i is zero and thus the associated boundary term is zero. We then finally focus on the the external boundary of the periodic cell, and the top interface is first considered. The associated boundary term $\mathcal{I}_t$ can be written as

$$\begin{aligned}\mathcal{I}_t &= \int_{\Gamma_t} u_2(x_1)\delta p(x_1)dx_1 \\ &= \int_{\Gamma_t}\left(\sum_{q\in\mathbb{Z}} :S^t_{1,q}\psi_q(x_1)\right)\delta p(x_1)dx_1,\end{aligned} \tag{3.48}$$

where $S^t_{1,q}$ is the first component of $\boldsymbol{S}^t_q$ defined in eq. (3.45). A boundary term for Γ_b denoted $\mathcal{I}_b$ is defined in a similar way as $\mathcal{I}_t$. Therefore, the weak form in Eq. (3.47) can be rewritten as,

$$\begin{aligned}\mathcal{I}_{\Omega_1}(p, \delta p) + \mathcal{I}_t + \mathcal{I}_b = &-\int_{\Gamma_r} u^r_2(x_2)\delta p(x_2):dx_2 \\ &-\int_{\Gamma_l} u^l_2(x_2)\delta p(x_2):dx_2.\end{aligned} \tag{3.49}$$

Since we introduced a state vector notation for the fields at the interface, we need also need to express an expansion on Bloch modes for the pressure field in order to provide the required new set of equations. Making use of the orthogonality properties of Bloch modes allows to write a relation such that, $\forall:q' \in \mathbb{Z}$,

$$\int_{\Gamma_t}\left(\sum_{q\in\mathbb{Z}} p(x_1)\psi^{-1}_{q'}(x_1)\right):dx_1 = S^t_{2,q'}d, \tag{3.50}$$

with d the periodicity and $S^t_{2,q'}$ the second component of $\boldsymbol{S}^t_{q'}$. In this expression, the orthogonality of Bloch modes is satisfied using $\psi^{-1}_{q'}(x_1) = e^{ik_{1,q'}x_1}$ instead of the usual complex conjugate, due to the complex nature of the Bloch wavenumber.

In summary, two main expressions were derived. The first one in Eq. (3.49) contains the weak form and expresses the first component of the state vector (displacement involved in the surface term) $S^t_{1,q}$ at the top and $S^b_{1,q}$ at the bottom interface, while Eq. (3.50) expresses the second component of the state vector (pressure involved in the volume term as nodal values) $S^t_{2,q}$ at the top interface $S^b_{2,q}$ at the bottom interface.

The discretization process involves multiple steps regarding both the Bloch wave basis and the finite element formulation. First, the infinite sums over $q \in \mathbb{Z}$ involved in the previous expressions are truncated so that $q = -Q, \ldots 0 \ldots Q$, with Q accounting for the necessary integer number of Bloch to take into the sum. This discrete basis is denoted $\boldsymbol{\Psi}(x_1) = \sum_{q=-Q}^{Q} \psi_q(x_1)$, and contains $\boldsymbol{Q} = 2Q + 1$ values. Additionnaly, using the standard procedure in finite elements, the nodal values and test functions are discretized using second-order Lagrange shape functions contained in matrix $\boldsymbol{N}(\boldsymbol{x})$. The pressure field is thus $p(x_1) = \boldsymbol{N}(x_1)\boldsymbol{p}$ the test function is $\delta p = \boldsymbol{N}(x_1)\delta\boldsymbol{p}$.

Finally, putting together the discretized equations allows to write a full system of equations for the discretized problem. It has three main rows; the first and last set of equations respectively contains the discrete Bloch orthogonality relation discretized from Eq. (3.50), while the second row contains the discretized weak form in Eq. (3.49). The system is composed of block matrices which indices relate to either the top or bottom interface and index i relates to the inner nodal values. Denoting $\boldsymbol{S}^t = (S^t_{-Q}, \dots, S^t_Q)^\mathsf{T}$ and $\boldsymbol{S}^b = (S^b_{-Q}, \dots, S^b_Q)^\mathsf{T}$, the system thus writes as

$$\begin{pmatrix} \boldsymbol{D}_{tt} & \boldsymbol{D}_{ti} & \boldsymbol{0} \\ \boldsymbol{D}_{it} & \boldsymbol{D}_{ii} & \boldsymbol{D}_{ib} \\ \boldsymbol{0} & \boldsymbol{D}_{bi} & \boldsymbol{D}_{bb} \end{pmatrix} \begin{pmatrix} \boldsymbol{S}^t \\ \boldsymbol{p} \\ \boldsymbol{S}^b \end{pmatrix} = \begin{pmatrix} 0 \\ \boldsymbol{f} \\ 0 \end{pmatrix}. \tag{3.51}$$

The first row contains matrices explicited as

$$\boldsymbol{D}_{ti} = \int_{\Gamma_t} \boldsymbol{\Psi}^{-1}(x_1)\boldsymbol{N}(x_1)\boldsymbol{:}dx_1, \quad \boldsymbol{D}_{tt} = -d\boldsymbol{I}. \tag{3.52}$$

Similar forms can be obtained for $\boldsymbol{D}_{bi}$ and $\boldsymbol{D}_{bb}$. In the second set of equations, $\boldsymbol{D}_{ii}$ is the discretized volume term with size (n_{dof}, n_{dof}) that only depends on the degrees of freedom (DOF) vector $\boldsymbol{p}$ and,

$$\boldsymbol{D}_{it} = \int_{\Gamma_t} \boldsymbol{N}^\mathsf{T}(x_1)\boldsymbol{\Psi}(x_1)\boldsymbol{:}dx_1, \tag{3.53}$$

is the discretized surface term for the top interface which is $(n_{dof}, 2\cdot Q)$. A similar form can be found for $\boldsymbol{D}_{ib}$. In Eq. 3.51, $\boldsymbol{p}$ contains the inner nodal pressure values and the right-hand side $\boldsymbol{f}$ contains the nodal force vector acting on the nodal values, the discrete form of the right-hand side of Eq. (3.49) related to the right and left boundaries surface terms. Each of the state vector $\boldsymbol{S}^t$, $\boldsymbol{S}^b$ has size $(2Q, 1)$.

In the cell, the displacement field from the right-hand side boundary u_r is related to that of the left side u_l, periodically modulated by $\Delta = e^{-ik_{1,q}d}$. The physical periodicity is thus written as $\boldsymbol{u}_r = \Delta\boldsymbol{u}_l$. Similarly, the periodicity relates the pressures fields between the left and right boundaries as $\boldsymbol{p}^r = \Delta\boldsymbol{p}^l$. Among other methods for the implementation of periodic boundary conditions in FEM [128, 129], we propose our own implementation. In the previously derived system, the nodal values vector can be written as $\boldsymbol{p} = \left(\boldsymbol{p}^l :: \boldsymbol{p}^i :: \boldsymbol{p}^r\right)^\mathsf{T}$ which contains respectively the pressure values at the left boundary, the inner DOFs and at the right boundary, respectively. Applying the periodicity relations, it follows that

$$\hat{\boldsymbol{P}}\boldsymbol{D}\boldsymbol{P}\begin{pmatrix} \boldsymbol{p}^l \\ \boldsymbol{p}^i \end{pmatrix} = \hat{\boldsymbol{P}}\begin{pmatrix} \boldsymbol{u}^l \\ \boldsymbol{0} \\ \Delta\boldsymbol{u}^l \end{pmatrix}, \tag{3.54}$$

where rows of $\boldsymbol{D}$ are multiplied by $\hat{\boldsymbol{P}}$ and its columns are multiplied by $\boldsymbol{P}$, both of which are expressed as

$$\boldsymbol{P} = \begin{pmatrix} \boldsymbol{I} & 0 \\ 0 & \boldsymbol{I} \\ \Delta\boldsymbol{I} & 0 \end{pmatrix}, \quad \hat{\boldsymbol{P}} = \begin{pmatrix} \boldsymbol{I} & 0 & -\Delta^{-1}\boldsymbol{I} \\ 0 & \boldsymbol{I} & 0 \end{pmatrix}. \tag{3.55}$$

Doing so allows the right-hand side from Eq. (3.54) to vanish, and yields an homogeneous linear system.

3.9.D. Final homogeneous linear system

The discrete system can be further manipulated so that most of the inner DOF related to p^i from the FE formulations are condensed. In a way, this comes back to writing a transfer matrix relating the top state vector to the bottom state vector. Further manipulation of the condensed matrix will therefore be much more computationnaly effective, due to the drastic reduction in matrix size. The matrix $\boldsymbol{D} = \hat{\boldsymbol{P}}\boldsymbol{D}\boldsymbol{P}$ denotes the full system and the $\boldsymbol{::}$ symbol denotes that the periodicity has been imposed on the matrix or vector. Let us write that the nodal values $\mathcal{p} = (\boldsymbol{p}_l, :\boldsymbol{p}_i)^{\mathsf{T}}$ can be written as a combination of $\boldsymbol{S}^t$ and $\boldsymbol{S}^b$ by manipulating Eq. (3.51) as,

$$\mathcal{p} = \boldsymbol{R}_t \boldsymbol{S}^t + \boldsymbol{R}_b \boldsymbol{S}^b, \tag{3.56}$$

with $\boldsymbol{R}_t = -\mathcal{D}_{ii}^{-1}\mathcal{D}_{it}$, $\boldsymbol{R}_b = -\mathcal{D}_{ii}^{-1}\mathcal{D}_{ib}$ and the indices used for those matrices are the same as the blocks of Eq. (3.51). The following system can thus be derived,

$$\begin{aligned} &\begin{pmatrix} \mathcal{D}_{tt} + \mathcal{D}_{ti}\boldsymbol{R}_t & \mathcal{D}_{ti}\boldsymbol{R}_b \\ \mathcal{D}_{bi}\boldsymbol{R}_t & \mathcal{D}_{bb} + \mathcal{D}_{bi}\boldsymbol{R}_b \end{pmatrix} \begin{pmatrix} \boldsymbol{S}^t \\ \boldsymbol{S}^b \end{pmatrix} \\ &= \boldsymbol{M}^t \boldsymbol{S}^t + \boldsymbol{M}^b \boldsymbol{S}^b = 0, \end{aligned} \tag{3.57}$$

which is an expression that relates the fields from the upper interface to those at the lower interface. Using the latter as well as the expressions relating the semi-infinite domain wave amplitudes with the state vector in the second line of Eq. (3.45) and Eq. (3.46), the final linear system is obtained as,

$$\underbrace{\begin{pmatrix} \boldsymbol{\sigma}^t & -\boldsymbol{I} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{M}^t & \boldsymbol{M}^b & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & -\boldsymbol{I} & \boldsymbol{\sigma}^b \end{pmatrix}}_{\boldsymbol{M}} \begin{pmatrix} \boldsymbol{a}^t \\ \boldsymbol{S}^t \\ \boldsymbol{S}^b \\ \boldsymbol{a}^b \end{pmatrix} = \boldsymbol{0}. \tag{3.58}$$

In this case, the final system matrix size does not depend upon the FE discretization number of DOFs. The matrices $\boldsymbol{M}^b$, $\boldsymbol{M}^t$ each have a size $(4Q, 2Q)$, both $\boldsymbol{\sigma}^t$ and $\boldsymbol{\sigma}^b$ have size $(Q, :Q)$, the identity matrix is $(Q, :2Q)$ and the state vectors are $(2Q, :1)$. Thus, the final system has a size $(6Q, :6Q)$.

In order to apply rigid boundary conditions on either one of the interface to obtain a rigidly-backed configuration, the same method can be applied by taking $\mathcal{I}_t = 0$ in Eq. (3.49) and thus the expression for $\boldsymbol{S}^t$ is removed.

The final matrix system that arise is iteratively solved using a root-finding method such that,

$$\det(\boldsymbol{M}(k_1, \omega)) = 0. \tag{3.59}$$

While an eigenvalue problem formulation would be much more efficient and reliable, this seems unfeasible due to the content of the block matrices of the system. More informations on the root-finding procedure is given in Appendix. .2.

An intermediate validation case is presented in Appendix .3. Neumann boundary conditions are applied at both top and bottom interfaces. The dispersion relation is presented for this simple case and is compared to both a generalized eigenvalue problem formulated from the same method (only valid in this particular case) and a Comsol model comparison. Presented results appear similar to those presented in Ref. [107]

The computation of the acoustic response of the considered structures can also be achieved by modifying slightly the method; the matrix system described Eq. (3.58) can be modified by adding a right-hand side state vector corresponding to the incident wave at the rows associated to the $\boldsymbol{a}^b$ wave amplitudes. Therefore, $\boldsymbol{a}^t$ can be understood as the transmission coefficients while the $\boldsymbol{a}^b$ becomes the reflection coeffcients.

3.9.E. Classification of proper solutions

All solutions obtained from Eq. (3.59) are mathematical solutions; only a subset of those are physical modes that actually represent the modal behaviour of the studied configuration. In order to sort out ”unphysical” solutions, a classification is therefore proposed in this section depending on their

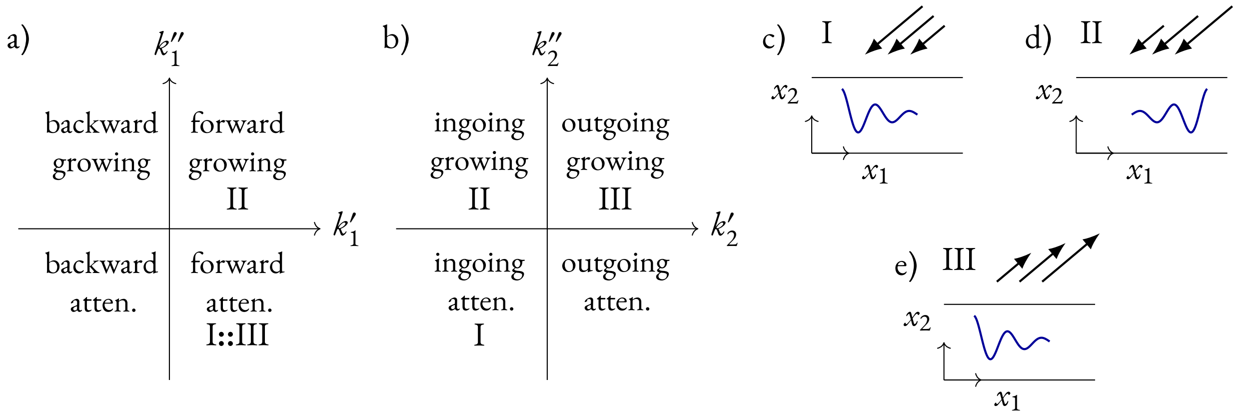


Figure 3.8: Classification of the solutions in the complex a) k_1 and b) k_{20} plane. Diagram for the form of the field for solution d) I, e) II, f) III.

locii in the complex plane. Considering the case of an homogeneous layer, there are no Bloch modes in this case. Thus we study the signs of real and imaginary component of $k_1 = k_1' + ik_1''$ and their corresponding transverse wavenumber in the fluid, $k_2 = \pm\sqrt{k_0^2 - k_1^2} = k_2' + \mathrm{i}k_2''$. As explained in Section 3.9.A, backward waves are not sought since the wavenumber spectrum is symmetric with respect to the $k_1' = 0$ plane.

In the case of leaky waves originated from a lossless layer, the selection of the branch cuts of the wavenumber is straightforward and has been widely discussed in the literature [130, 131]; they take the form of outgoing growing waves in the semi-infinite medium. However, the challenges faced with this configuration resides in the coupling in the system between two losses mechanisms that are viscothermal losses in the layer and radiation losses due to the fluid coupling.

The classification defined in Ref. [123], in which complex wavenumber solutions to a problem involving both material losses and radiation to a surrounding medium are categorized into either *proper* or *improper* modes, is adopted for the present work. Starting from the Snell-Descartes relation between the wavenumbers, stating that $k_1^2 + k_{2,0}^2 = k_0^2$ and taking the imaginary part of that equation yields,

$$k_1'k_1'' = -k_{2,0}'k_{2,0}'', \tag{3.60}$$

which should hold for all cases [57]. Restricting proper solutions $k_1' > 0$ and $k_1'' < 0$ (wave attenuated along the layer in the x_1 direction), leads to either $k_{2,0}'k_{2,0}'' < 0$ or $k_{2,0}'k_{2,0}'' > 0$. The latter is regarded as an improper solution due to the growing nature of the mode in the semi-infinite domain (positive imaginary part).

These categories of the encountered wave solutions are presented in diagrams in Fig. 3.8a and 3.8b, respectively for k_1 and k_2 complex plane. Each quadrant represents the solution whether it is forward or backward, growing or attenuated, outgoing or ingoing. Three types of solutions are also highlighted in Fig. 3.8c-e, of which only one is considered as a proper solution. In Fig. 3.8d, mode I corresponds to an attenuated field in the layer with ingoing attenuated radiation. The two other types of solutions are deemed improper: II is a growing field in the layer with ingoing growing radiation and III is an attenuated field in the layer with outgoing growing radiation respectively in Fig. 3.8e and Fig. 3.8f.

3.10 Case of an homogeneous layer with half-space coupling on one side: branch cut selection

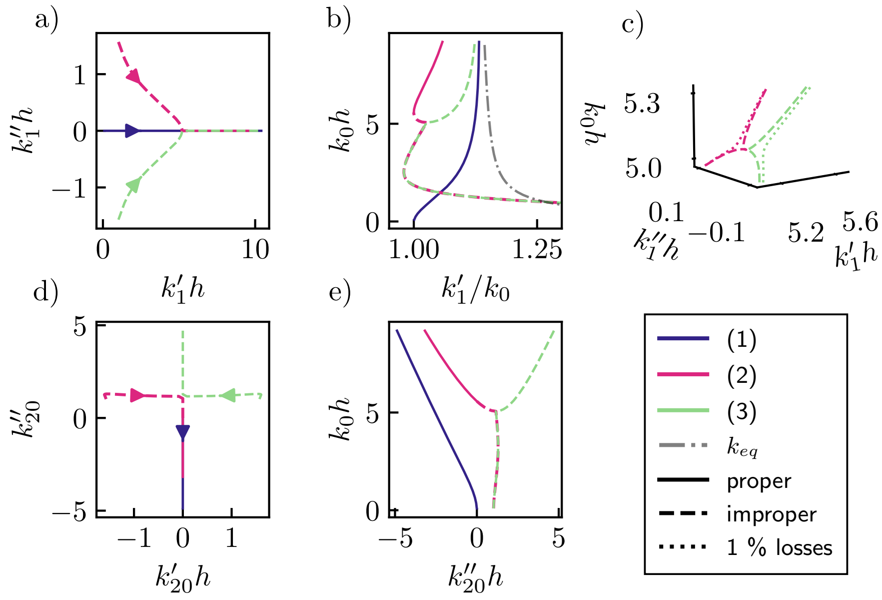


Figure 3.9: Lossless ($k_{eq} = k'_{eq}$) homogeneous porous dispersion relation: a) complex k_1 plane view; the arrows represents path with increasing frequencies. b) k'_1 against normalized frequency. c) 3d view of the crossing around $k_0h = 5$ with the lossless case and the porous layer with 1% losses. d) complex k_2 plane view. e) k''_{20} against normalized frequency.

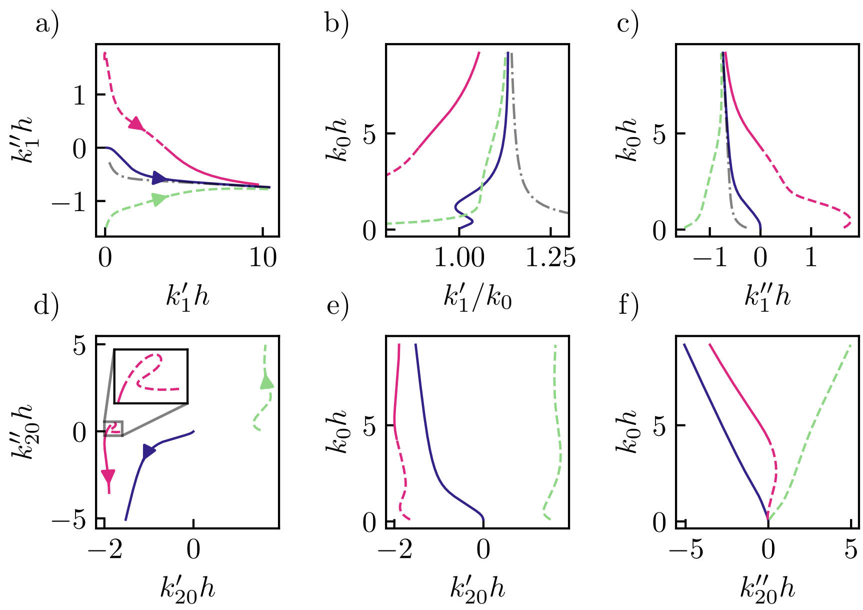


Figure 3.10: Homogeneous porous dispersion relation accounting material losses with views: a) complex k_1 plane, b) k'_1, c) k''_1, d) complex k_{20} plane, e) k'_{20} and f) k''_{20}. The figure shares the same legend as Fig. 3.9

To highlight the modal behaviour of the system and the coupling between the various loss mechanisms, we start by studying a rigidly-backed homogeneous porous layer of thickness h with a fluid half-space above. This configuration is described using its dispersion equation,

$$\frac{k_{2,0}}{\rho_0}\cos(k_{2,eq}h) + \mathrm{i}\frac{k_{2,eq}}{\rho_{eq}}\sin(k_{2,eq}h) = 0, \tag{3.61}$$

onto which solution for k_{20} are sought, with $k_1 = +\sqrt{k_0^2 - k_1^2}$, and $k_{2,eq} = +\sqrt{k_{eq}^2 - k_1^2}$. Here we impose $k_1' > 0$ for forward propagating waves and will retrieve all possible solutions for complex values of k_{20} in a given frequency range. The parameters for the JCA model for the porous layer are given in Table 3.2 and its effective parameters are given in Appendix .1. The saturating fluid and that occupying the half-space is air with density $\rho_f = 1.213$ kg.m^{-3}, the heat capacity ratio $\kappa = 1.4$, kinematic compressibility $\mu_f = 1.839 \times 10^{-5}$ Pa.s, the Prandtl number Pr $= 0.71$, and adiabatic bulk modulus κP_0, where the atmospheric pressure is $P_0 = 1.013 \times 10^5$ Pa.

Table 3.2: JCA parameters of the porous material (melamine foam)

Parameter	ϕ	R_f	Λ	Λ'	α_∞
Unit	-	kPa.s.m^{-2}	μm	μm	-
Melamine	0.99	10.567	240	490	1.2

The layer has a thickness $h = 50$ mm. At first, the dispersion relation is computed for a lossless porous layer, that is keeping only the real components of ρ_{eq} and K_{eq} and thus $k_{eq} = k_{eq}' \in \mathbb{R}$. In Fig. 3.9a and Fig. 3.9d, the non-zero imaginary part of the dispersion relation in k_1 or k_{20} are solely due to radiation losses as the dissipation in the porous layer is not accounted for. In Fig. 3.9b, mode I is always propagating ($k_1' \in (k_0, k_{eq})$) and always proper as it represents a purely guided wave radiating as a purely evanescent wave in the air domain. Modes II and III have a large k_1'' and are both improper below the cutoff frequency (around $k_0 h \approx 5$) as seen on Fig. 3.9e as they respectively correspond to a growing wave in the layer with exponentially growing radiation and an attenuated wave in the layer with exponentially growing radiation. Above the cutoff, mode III is still improper for the same reasons and mode II changes k_{20}'' sign, becoming proper with the same behaviour to that of mode I (purely guided wave).

For modes II and III, k_1'' is large at lower frequencies, meaning that radiation losses are prominent in this frequency range and that no energy is radiated from the layer past the cut-on frequency since $k_1'' = 0$. Additionnaly, the I mode is unaffected by these losses regardless of the frequency. As an amount of 1% of losses are introduced in the layer (that is $k_{eq} = k_{eq}' + 0.01 \cdot \mathrm{i}k_{eq}''$) in Fig. 3.9c, the point where the 2 modes coincide is lost and the 2 branches are fully separated, both wavenumbers k_1 and k_{20} become fully complex over the whole frequency range.

The next step is the addition of the full losses to the porous layer, with results presented in Fig. 3.10. In this case, the symmetries shown for modes II and III the lossless case in Fig. 3.9a and Fig. 3.9d with respect to the k_{20}' and k_1'' planes are lost. The introduction of the viscothermal losses in the porous layer leads all modes to be somewhat attenuated, as expected. It can first be observed that k_{20} is shifted for both the real and imaginary part for all branches. However, the proper part of mode II shifts towards negative values, indicating an attenuated field in the fluid half-space $k_{2,0}'' < 0$ but also an ingoing wave $k_{2,0}' < 0$. This shift direction is explained in greater details in Appendix .4. Mode I keeps the same trend to that of the lossless case with a small $k_1'' < 0$ value in Fig. 3.10c) and is shifted to negative k_{20}'. Mode II follows the same shift as mode I and remains *improper* below its cutoff frequency, identified in this case by a change of sign in k_1'' and k_{20}''.

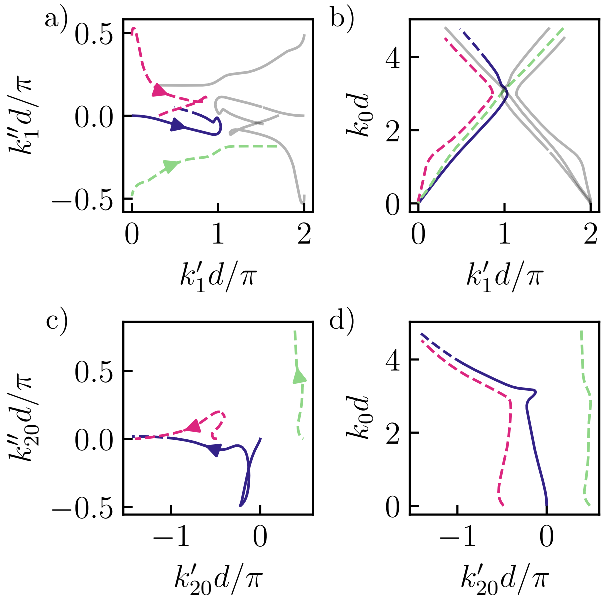


Figure 3.11: Dispersion relation for a periodic unit-cell with diameter $d/h = 1/2$, with views of: a) the complex k_1 plane view with the arrow indicating the direction for growing frequency, b) k_1' depending on the adimensional frequency, c) k_{20} complex plane view and d) view with k_{20}' depending on frequency. The grey lines denote the modes originating from $k_1' = 2\pi/d$, and are symmetric in the k_1 plane with those denoted in colors. The figure shares the same legend as Fig. 3.9.

In the study of leaky modes in lossless layers, as has been done for elastic materials in several papers, both II and III modes could be selected equivalently, and corresponds to either a growing solution in the layer which implies an ingoing growing wave in the fluid (backward leaky waves), or an attenuated solution in the layer with outgoing growing wave in the fluid (forward leaky waves). In this configuration, the latter is chosen for its more physically meaningful behaviour. However, the desymmetrization of these two modes in the case of a porous layer with losses means that the selection of the branch cut is not straightforward. The choice of the branch cut is made so that mode II is selected as a proper solution, with $k_1'' < 0$ and $k_{20}'' < 0$, which is considered the physically meaningful solution in our case. Mode III will be thus discarded as a solution in upcoming results as it is always improper.

3.11 Results

3.11.A. Dispersion relation of a unit cell with rigid inclusions

The first numerical results for the periodic geometry are presented in this section. A rigid cylindrical inclusion is embedded into the porous layer. The unit cell is considered with fluid radiation on the bottom side and a rigid-backing is imposed at the upper boundary. Results for an inclusion with radius $r = 5$ mm are presented in Fig. 3.11 and are significantly different from those presented for the homogeneous layer, due to the addition of periodicity and the scatterer in the system. The obtained dispersion relation has some of the classical features of periodic materials: the various branches in k_1' in Fig. 3.11 are π/d periodic and are folded at a frequency corresponding to the system's Bragg frequency. The same topology of modes as that of the homogeneous layer are observed but their shape is significantly changed by the addition of the rigid inclusion. No bandgap is opened for this configuration. III is still improper from the discussion in Section 3.10. Mode I is folded at a wavenumber $k_1' > \pi/d$ while Mode II is folded before reaching the end of the Brillouin zone. This seemingly appearing as a wavenumber bandgap around $k_0d = 3$ might rather be explained by a deformation of the complex plane of the solution due to the losses in the material. This unexpected interaction between the two modes I and II could be explained by the frequency dependance of the JCA model. In further results, the classification of the modes will not be detailed and mode III won't be represented in the dispersion relations as well as the proper and improper parts of each mode.

Transmission case described in Appendix. .5.A

3.11.B. Rigid inclusion tuned to perfect absorption

In order to study the following configurations, their scattering properties should be computed as well. Adapting the formulation of the problem to that of a forced configuration, a right-hand side is added to the matrix system in Eq. (3.58), corresponding to an incident plane wave. The a^b coefficients can become reflection coefficients denoted as R_q. The reflection coefficient is thus computed as follows,

$$\mathcal{R} = \sum_q \frac{k'_{2,q}}{k_{2,0}} |R_q|^2, \tag{3.62}$$

and the absorption coefficient is thus, $\alpha = 1 - \mathcal{R}$.

In the same 2 by 2 cm unit cell, a unitary absorption coefficient can be obtained by tuning the inclusion radius. Increasing the radius to $r = 7.3$ mm in this case leads to perfect absorption, at normal incidence. This result is presented in Fig. 3.12a, where the absorption coefficient is plotted depending on the frequency and the incident angle. The absorption peak is located at $f = 3.2$ kHz and $\theta = 0$. The in-plane mode can be excited in this configuration due to the proximity of k'_1 to the excitation line at 0 which has a steep slope in k'_1 at low frequency, remaining in the vicinity of 0. This corresponds also to a proximity of an hybridation of this mode to a trapped mode as shown in the inset of Fig. 3.12d. This mode is responsible for this behaviour as proven by the comparison between the field excited by the incident wave in Fig. 3.12e and the mode shapes in Fig. 3.12f and g.

With an increasing incident angle, the absorption peak is shifted towards high frequency as seen on Fig. 3.12a, due to the tilt of the excitation line which thus crosses the mode at higher frequencies, the higher the angle. The high absorption is still due to the same mode. Beyond $\theta = 50°$, a weaker peak at low frequencies can be observed: this one is due to an excitation line in the vicinity of the other mode. citer appendix E2

This total absorption case is illustrated by an exceptional point of k_1 as well as a zero of the scattering matrix as illustrated in Fig. 3.12h. Taking the absolute value of the reflection coefficient map depending on complex frequencies, it can be seen that a zero is reached at total absorption corresponding to $f'' = 0$ as investigated in previous papers +citer des trucs.

3.11.C. Split-ring resonators

Other inclusion geometries can also be investigated with the proposed method. In this section, we propose the case of a split-ring resonator (SRR) embedded in the porous layer. It was found to provide an additional absorption peak in metaporous surfaces [132, 133]. The SRR orientation towards the fluid or the rigid backing leads to different modal behaviours. Two configurations *up1* and *up2* are considered with the resonator slit towards the backing and *down1* with the slit towards the fluid. The results for the 3 configurations are presented in Fig. 3.13 and the parameters for the geometry for each configurations are given in Table 3 in Appendix .6.

In each case, the dispersion relation of a rigid inclusion with a radius r equal to that of the outer radius r_{out} of the SRR is also shown, as well as its absorption coefficient.

Oriented towards the rigid-backing, a second absorption seems to originate from some exceptional for each point. Coincidently, corresponding to the SRR resonance frequency. Towards the fluid, the resonance frequency appears strongly, the second absorption peak is close to that of the trapped mode, which seems slighltly shifted to higher frequency than with the rigid inclusion.

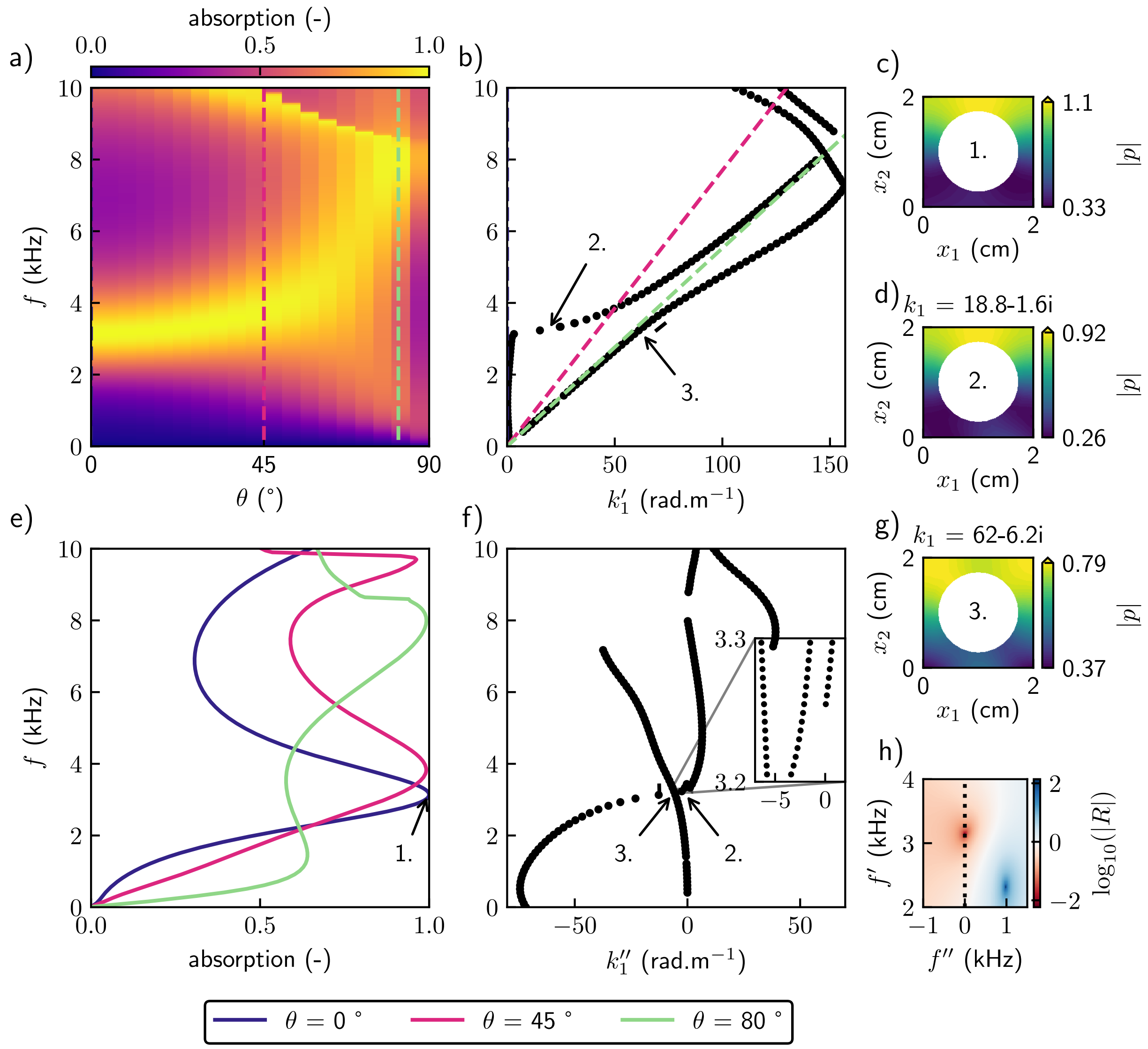


Figure 3.12: a) Absorption coefficient map depending on the incidence angle and the frequency. Dispersion relation b) for the real part of k_1 c) and the real part of k_{20}. d) Absorption taken at the angles corresponding to the dashed lines on a). Dispersion relation for e) the imag part of k_1 c) and the imag part of k_{20}. g) Field in the unit cell produced by the incident wave. h) i) Mode shapes of each of the two modes labelled as 2. and 3. with their location marked on b), e) The 3 snapshots are taken at 3.2 kHz, the peak absorption frequency.

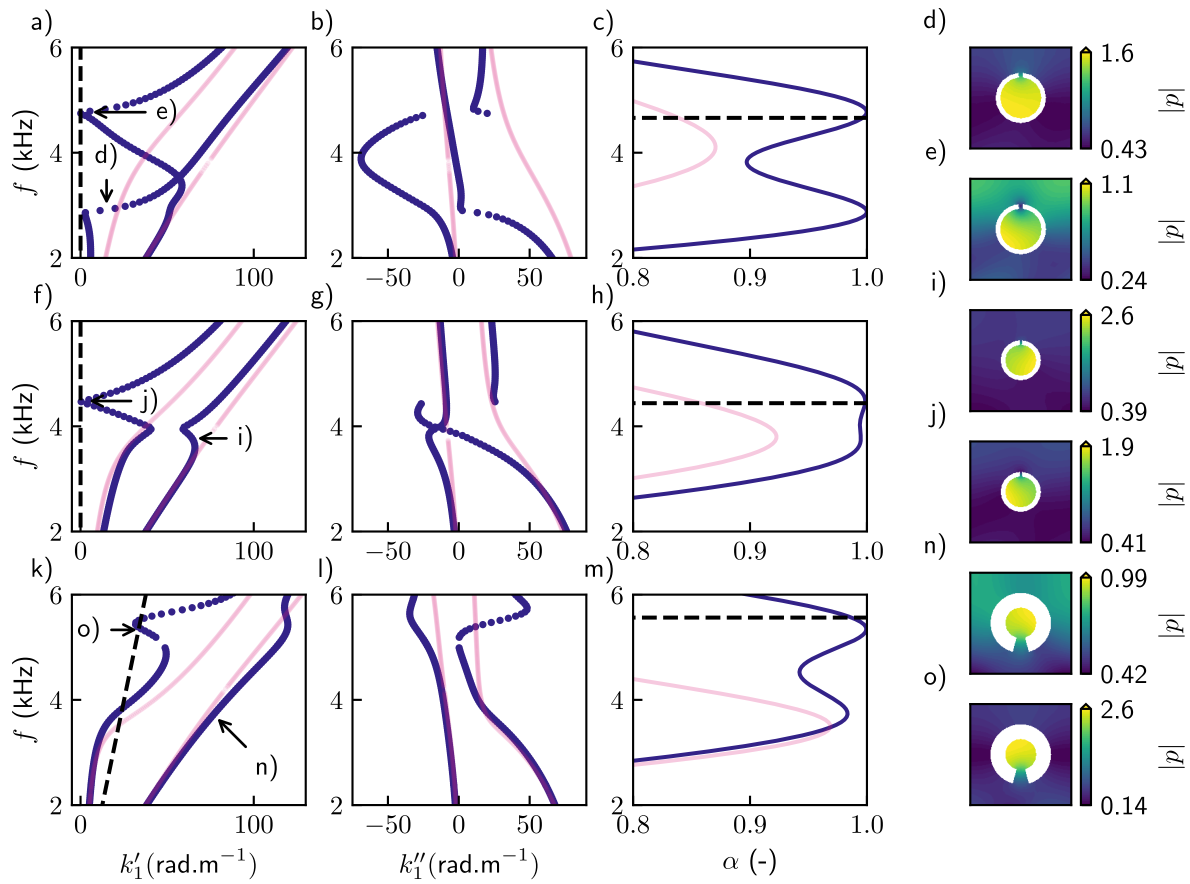


Figure 3.13: k_1' - f diagram, k_1'' - f diagram, absorption coefficient and mode shapes a-e) *up1* configuration, f-h) *up2* configuration, k-o) *down1* configuration. The mode shapes are taken at the frequency corresponding to the absorption peak and only one of the two modes is shown, as represented by the arrows.

For the *up1* configuration, there is a lowered branch with a k_1' crossing, analogous to the perfect absorption case. Compared to the latter, the inclusion size is smaller while the absorption peak is located at a lower frequency. A second peak due to the hybridation of the left-propagating mode and the right-propagating mode appears as well: change of sign in k_1'' so the branch just above e) originates from $k_1' < 0$. This second peak matches the computed SRR resonance, represented by the dashed lines (formula provided in Appendix .6). Mode shapes

The *up2* geometry reveals that only the second peak shows perfect absorption, and the one below seems to be in the vicinity of an avoided crossing of an exceptional point as shown by the dispersion relation. The two branches have a k_1'' crossing but the one at k_1' is avoided, hence the very high but not unitary absorption. The second peak is due to the same phenomenon as the *up1* configuration and mode shapes are alike in this case.

The *down1* configuration is excited at an incidence angle $\theta_0 = 10°$, allowing to locate the excitation line close to the trapped mode that is appearing at o). First peak is the excitation with the cut-offed mode, which has still an important attenuation hence, the non-unitary absorption. The second peak seems due to the SRR resonance, even though the modes seems to be quite different from the previous two cases as they are not hybridized, it is only the excitation of a mode with low k_1' and low attenuation.

3.12 Conclusion

In this work, we have developed a numerical procedure for the computation of the dispersion relation of the leaky surface waves in porous periodic layers with embedded rigid inclusions. The dispersion relation allow for an explanations for the modal behaviour of metaporous layers. Particular cases of a cylindrical inclusion tuned to a perfect absorption peak and split-ring resonators were studied and their scattering properties could be understood in a new light.

While the dispersion relation evaluation lies on a root-finding procedure, a costly and non robust method (in isolated cases and seemingly depending on the matrices conditioning, some roots can be missed, which is unavoidable), it is the only known way to solve the present system and the proposed method allows for the evaluation of both the dispersion relation and the scattering properties (reflection and transmission coefficients) when adding a right-hand side and solving directly the matrix system.

Next steps for this work include generalizing this approach to multilayer systems of arbitrary nature and an extension of poroelastic periodic structures. The investigation of the contributions of the skeleton motion to the modal behaviour of the structure could provide a better understanding of these structures.

.1 Expression of the effective parameters in porous materials with the JCA model.

The complex and frequency dependent density and bulk modulus of the fluid phase, which accounts respectively for the viscous and thermal losses are [73, 74] as,

$$
\begin{aligned}
\rho_{\text{eq}}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi}\left(1 + \frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty\omega}\sqrt{1 - \mathrm{i}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda}\right)^2}\right),\\
K_{\text{eq}} &= \frac{\kappa P_0}{\phi}\left(\kappa - (\kappa-1)\Big/\left(1 + \frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty \mathrm{Pr}\omega}\sqrt{1 - \mathrm{i}\mathrm{Pr}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda'}\right)^2}\right)\right)^{-1},
\end{aligned}
\tag{63}
$$

where ϕ is the porosity, α_∞ is the tortuosity, Λ and Λ' are respectively the viscous and thermal characteristic lengths, R_f is the flow resistivity, μ_f is the dynamic viscosity, κ is the heat capacity ratio, ρ_f the density of the saturating fluid, and P_0 is the ambient pressure.

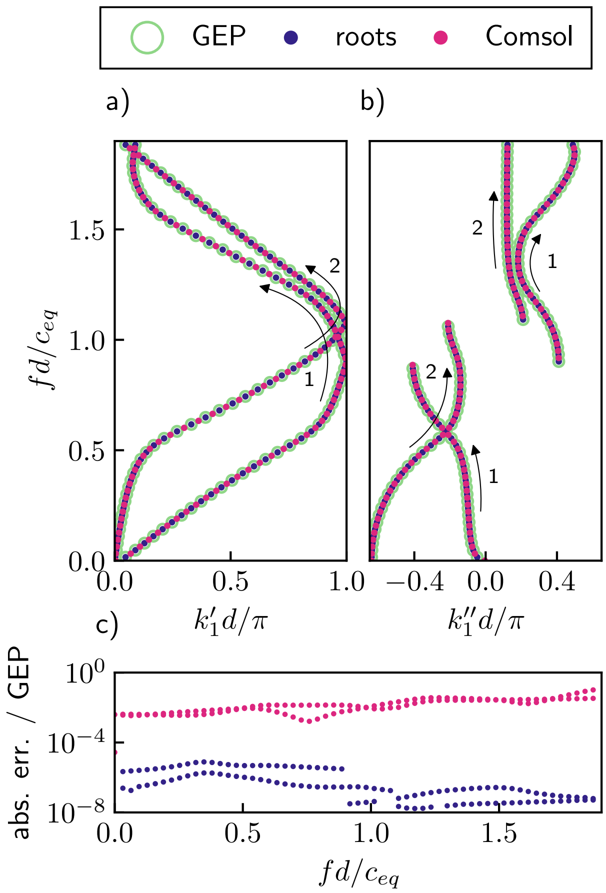


Figure 14: Complex dispersion relation of the rigid-rigid unit cell with a r = 8 mm rigid inclusion. a) k_1' b) k_1''. The two branches are denoted 1 and 2 and their path is highlighted with arrows. c) Absolute error of the Comsol (pink) and root-finding (blue) solutions w.r.t the GEP solution

.2 Root-finding: Müller method

In order to find solutions for modes in the form of complex wavenumbers $k_{1,q}$, we use the previously derived system from Eq. (3.58), that can also be expressed as $\boldsymbol{M}(k_1,\omega)\boldsymbol{S} = 0$. The modes are found by finding the discrete set of k_1 that satisfy the singularity of $\boldsymbol{M}$ using Eq. 3.59

In practice, these solutions are found using a root-finding procedure, our own implementation of the Müller method [53]. It is based on a parabolic interpolation to iteratively finds complex wavenumber solutions k_1 given a specific frequency ω from three initial guesses. Therefore, care has to be put into the initialization of the method. Since the system $\boldsymbol{M}$ is constructed from a finite elements scheme, computations will be all the more costly that a large number of iterations need to run before convergence is reached. The convergence criterion checks that threshold of the variation of the solution is below ε. In this paper, this criterion is taken with a magnitude around $\varepsilon = 10^{-13}$. Thus, the initialization of the method is done by evaluating the local minimas of a coarse mapping of the function in the complex plane. The latter is only computed every few frequency steps to save computation time. Additionnaly, the matrix $\boldsymbol{D}_{ii}$ from Eq. (3.51) does not depend on k_1 and can thus be only evaluated once per frequency step, thus saving valuable computing resourses.

Once the set of solutions has been obtained, each of the associated mode shapes can also be retrieved. Given a specific root $k_{1,j}$ that satisfies $\det(\boldsymbol{M}) = 0$, there is $\boldsymbol{M}(k_{1,j})\boldsymbol{S} = 0$, where the vector $\boldsymbol{S} = \ker(\boldsymbol{M}(k_{1,j}))$ is the nullspace of the matrix. In practice, the nullspace is computed using singular value decomposition (SVD), $\boldsymbol{M} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{\dagger}$. The last right singular vectors, that is the last row of $\boldsymbol{V}^{\dagger}$ provide an orthonormal basis for the null space of $\boldsymbol{M}$, associated to the singular value $\sigma = \min(\boldsymbol{\Sigma})$. All state vectors among which S^t and S^b are obtained by selecting the appropriate rows of $\boldsymbol{V}^*$. By doing the whole procedure backwards (using Eq. (3.56) and then Eq. (3.54)), we can retrieve the full degrees of freedom vector $\boldsymbol{p}$ (and normalized as $(p^i(x_1,x_2)/p^i(0,h))e^{ik_1x_1}$).

.3 Validation of the method in the case of a layer with rigid boundary conditions

.3.A. Generalized eigenvalue problem

With the full problem described in Section. 3.9, writing an eigenvalue problem proves to not be possible. However, in the case where no outgoing wave radiation are considered and by imposing Neumann boundary conditions at the interface instead, an eigenvalue problem can be obtained in this specific case. Equation 3.54 is rewritten with $\boldsymbol{P}$ decomposed as $\boldsymbol{P} = \boldsymbol{P}_m + \Delta\boldsymbol{P}_\Delta$ and $\hat{\boldsymbol{P}} = \hat{\boldsymbol{P}}_m + \Delta^{-1}\hat{\boldsymbol{P}}_\Delta$ such that

$$\left(\hat{\boldsymbol{P}}_m + \Delta^{-1}\hat{\boldsymbol{P}}_\Delta\right)\boldsymbol{D}\left(\boldsymbol{P}_m + \Delta\boldsymbol{P}_\Delta\right)\boldsymbol{p} = 0. \tag{64}$$

Developing the expression yields the form of a polynomial eigenvalue problem of order 2,

$$\left(\Delta^2\boldsymbol{A}_2 + \Delta\boldsymbol{A}_1 + \boldsymbol{A}_0\right)p_i = 0, \tag{65}$$

with matrices $\boldsymbol{A}_2 = \hat{\boldsymbol{P}}_m\boldsymbol{D}\boldsymbol{P}_\Delta$, $\boldsymbol{A}_1 = \hat{\boldsymbol{P}}_m\boldsymbol{D}\boldsymbol{P}_m + \hat{\boldsymbol{P}}_\Delta\boldsymbol{D}\boldsymbol{P}_\Delta$ and $\boldsymbol{A}_0 = \hat{\boldsymbol{P}}_\Delta\boldsymbol{D}\boldsymbol{P}_m$. This can be put into a generalized eigenvalue problem,

$$(\boldsymbol{A} - \Delta\boldsymbol{B})\,p = 0, \tag{66}$$

with $\boldsymbol{A} = \begin{pmatrix} -\boldsymbol{A}_1 & -\boldsymbol{A}_0 \\ I & 0 \end{pmatrix}$, $\boldsymbol{B} = \begin{pmatrix} \boldsymbol{A}_2 & 0 \\ 0 & I \end{pmatrix}$ and $\boldsymbol{p} = \begin{pmatrix}\Delta\boldsymbol{p}_l & \Delta\boldsymbol{p}_i & \boldsymbol{p}_l & \boldsymbol{p}_i\end{pmatrix}^{\mathsf{T}}$.

.3.B. Comsol model

Although it has been mentioned in the introduction that Comsol Multiphysics could not be used for modeling the full problem of the present work, it can still be used to validate the method in the case of a layer with rigid boundary conditions. The geometry is thus constituted of the sole unit cell, with Neumann boundary conditions at Γ_t, Γ_b. The Helmholtz equation is written as $(\nabla^2 + k_{eq}^2)\tilde{p}(x)e^{\kappa x_1} = 0$, with the eigenvalue of the problem denoted as $\kappa = -ik_{1,q}$. The coefficient form PDE interface is used for the model, which requires the Helmholtz equation to be written as,

$$\kappa^2\tilde{p} + \nabla\cdot\left(\boldsymbol{A}_0\nabla\tilde{p} + \boldsymbol{A}_1\tilde{p}\right) + \boldsymbol{A}_1'\nabla\tilde{p} + k_{eq}^2\tilde{p} = 0, \tag{67}$$

with $\nabla = \begin{pmatrix}\partial_1\\ \partial_2\end{pmatrix}$, $\boldsymbol{A}_0 = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$, $\boldsymbol{A}_1 = \begin{pmatrix}\kappa\\ 0\end{pmatrix}$ and $\boldsymbol{A}_1' = \begin{pmatrix}\kappa\\ 0\end{pmatrix}$. Around the inclusion, rigid boundary conditions are applied, as $-\boldsymbol{n}\cdot\nabla p - n_2\cdot\kappa p = 0$ with $\boldsymbol{n}$ the normal vector to the boundary and n_2 its x_2 component. At the top and bottom interfaces, the Neumann boundary conditions simply writes as $\partial_2 p(x_1, 0) = 0$ and $\partial_2 p(x_1, H) = 0$. The periodicity is imposed on the right and left boundaries with a continuity condition and since the pressure field $\tilde{p}$ is defined as a periodic function, the sole imposition of $\tilde{p}(0, x_2) = \tilde{p}(d, x_2)$ fully imposes the periodicity in the problem.

.3.C. Results

For the procedure that uses root-finding, the system is constructed following Eq. (3.57) and setting $Q = 0$, since there is no coupling to any semi-infinite medium and thus no need for a Bloch wave series. The system is thus,

$$\left(\begin{array}{cc|cc} 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\\ \hline \multicolumn{2}{c|}{\boldsymbol{I}} & \multicolumn{2}{c}{\boldsymbol{T}} \end{array}\right)\begin{pmatrix} S_{1,0}^t\\ S_{2,0}^t\\ S_{1,0}^b\\ S_{2,0}^b\end{pmatrix} = 0, \tag{68}$$

with $\boldsymbol{T} = \left(\boldsymbol{M}^t\right)^{-1}\boldsymbol{M}^b$ corresponds to the transfer matrix of the system. The two first rows correspond to the Neumann boundary conditions applied to the system.

For the rigid-rigid case, we therefore present results computed with Comsol as well as with our FEM approach solved using both root-finding and the eigenvalue formulation. All solutions were computed at the same frequencies with a mesh size $\lambda_{min}/10$. While the same size, the Comsol generated mesh might not be identical to that of the mesh generated for the root-finding and eigenproblem methods. In Fig. 14, two modes are obtained: one with lower attenuation is the first propagating mode denoted as 1, and the other one is in-plane with a cut-on frequency denoted as 2. All approaches yield similar results. Moreover, these results are comparable to those obtained in Ref. [107].

In Fig. 14c, the absolute error comparing Comsol results and the root-finding method with reference from the GEP is presented. It ensures that the method is able to converge properly to solutions coincinding with the eigenvalues. Note that the general matrices onto which both procedures are done are the same, hence the lower absolute error compared to Comsol which uses significantly different approach.

.4 Sign of the complex transverse wavenumber in the presence of losses and limit values at low frequencies

Keeping the focus on mode I and the proper mode II from the results for the homogeneous layer in the lossless case depicted in Fig. 3.9, we saw that $k_1 \to k_{eq}'$ at high frequencies. The expression for the transverse wavenumber becomes $k_{20} = -\sqrt{k_0^2 - k_{eq}'^2} \in -\mathrm{i}\mathbb{R}$ since $k_{eq}' > k_0$ and thus $\arg(k_{20}) = -\pi/2$. As depicted in Fig. 3.10 for the lossy case, the introduction of dissipation shifts the modes from zero real parts to negative real parts $k_{20}' < 0$ and $k_1' < 0$. In this lossy case, considering high frequencies again, $k_1 \to |k'_{eq}| - \mathrm{i}|k''_{eq}|$. Adding an imaginary part to k_1 leads to

$$k_{20} = -\sqrt{k_0^2 - |k'_{eq}|^2 + |k''_{eq}|^2 + 2i|k'_{eq}||k''_{eq}|}. \tag{69}$$

It can be represented by $k_{20} = |-k_{20}|e^{i(\theta/2+\pi)}$ where θ is the argument of the complex number under the square root. Looking for the value of θ, one has

$$\theta/2 = \pi + \arctan\left(\frac{|k'_{eq}||k''_{eq}|}{k_0^2 - |k'_{eq}|^2 + |k''_{eq}|^2}\right), \tag{70}$$

and $\theta/2 \to \pi/4$ because $|k'_{eq}||k''_{eq}| > 0, k_0^2 - |k'_{eq}|^2 + |k''_{eq}|^2 < 0)$. Finally, $\arg(k_{20}) \to -3\pi/4$ which corresponds to the quadrant with $k'_{20} < 0$ and $k''_{20} < 0$.

.5 Additionnal results

.5.A. Transmission case

The transmission case is taken so that its unit cell corresponds to two of the unit cells used for the rigid-backing stacked together, with twice the thickness and two rigid inclusions. Thus, several modes should be identical since the rigid-backing acts as a symmetry plane. Figure 15 shows the support for an additional propagation mode starting at wavenumber 0. It can be seen on Fig. 15c that the pressure field corresponds to a minimum along $x_2 = 2$ cm, at the half-width cell, unsuitable in the case of a rigidly-backed layer, which would be the symmetry plane for this configuration where a no displacement condition is imposed, implying non-zero pressure. This additional mode does not seem to be affected strongly by the inclusion and does not show any notable property.

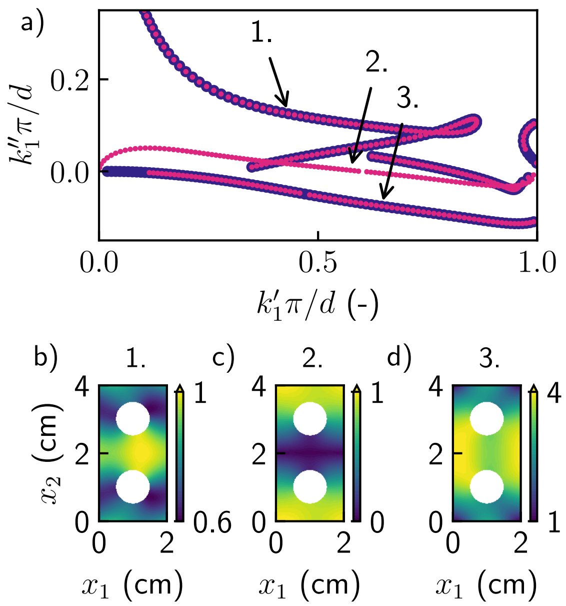


Figure 15: Dispersion relation for the symmetric configuration a) Overlaid rigid-backing h and symmetric $2h$ with two inclusions configurations b) c) d) Mode shapes of each of the three modes represented at $f = 5$ kHz.

.5.B. Effect of the cell stretching in the perfect absorption case

We want to highlight more precisely the identified behaviour for the perfect absorption configuration. A similar configuration can therefore be studied, by a stretching effect applied to the original geometry, the inclusion becoming an ellipse and the square unit cell, a rectangle. The stretching is defined as increasing the thickness and the ellipse semi-major axis, respectively as $h + 2\delta l$ and $b + \delta l$. Results are presented for the cases $\delta l = 0, 3$ and 5 mm in Fig. 16. Figure 16a to Fig. 16d shows results identical to Fig .3.12, Fig. 16e to Fig. 16h shows a cell with $h = 26$ mm and an ellipsoid inclusion with a semi-minor axis $a = 7.3$ mm, and a semi-major axis $b = 10.3$ mm, and Fig. 16i to l is a cell with $h = 30$ mm and an ellipsoid inclusion with $a = 7.3$ mm, $b = 13.3$ mm.. The trapped mode

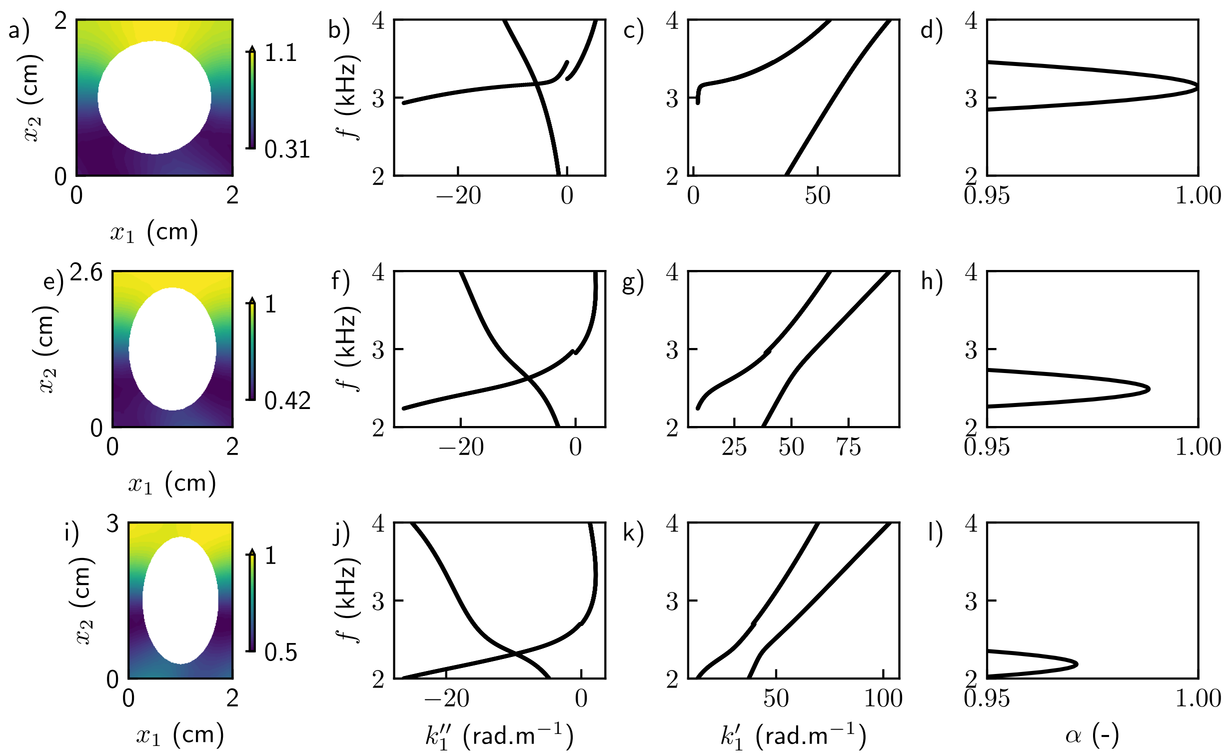


Figure 16: a-d) $\delta l = 0$ mm, e-h) $\delta l = 3$ mm, i-l) $\delta l = 5$ mm. For each row are the wavenumbers k_1', k_1'' and the absorption coefficient α.

shifts towards lower frequencies, explaining the frequency shift of the absorption coefficient. The mode shape remain similar, and the slope of the mode 1 becomes less steep and further from $k_1' = 0$, explaining the decrease in magnitude that appear for the absorption coefficient.

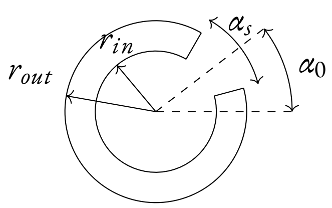


Figure 17: Diagram of the split-ring geometry

.6 Details on the split ring configurations

Parameter	Unit	Configuration name *up1*	*up2*	*down*
r_{out}	mm	5	4	6
r_{in}	mm	4	3	3
α_s	°	9	6	30
α_0	°	90	90	270
θ_0	°	0	0	20

Table 3: Parameters of the split-ring resonator configurations used in Section 3.11.C

Below is provided a figure describing the parametrization of the resonator geometry, as well the various configurations presented in the paper.

The resonance frequency is obtained by adapting the standard equation for the wavenumber at the resonance of an Helmholtz resonator [134] as $k_r = \sqrt{S/(LV)}$, where in this case $S = \theta_s/(2\pi)$ is the aperture width arc length, $L + \delta \approx r_{out} - r_{in}$ is the neck length and $V = \pi r_{in}^2$ is the cavity surface area. In the present case, the layer wave velocity depends on the frequency. The following expression for the resonance frequency is thus,

$$\begin{aligned} f_{res} &= \min\left|1 - \frac{c_{eq}(\omega)W}{\omega}\right|, \\ W &= \sqrt{\frac{\left(r_{out} - \frac{r_{in}}{2}\right)\alpha_s}{2\pi(r_{out} - r_{in})r_{in}^2}} \end{aligned} \tag{71}$$

Appendices

Contents

.1 Details on the Müller method

.1.1. The root-finding algorithm

equations, iterative scheme etc.
[135]

$$
\begin{aligned}
f(x_0) &= a(x_0 - x_2)^2 + b(x_0 - x_2) \\
f(x_1) &= a(x_1 - x_2)^2 + b(x_1 - x_2) \\
f(x_2) &= a(x_2 - x_2)^2 + b(x_2 - x_2)
\end{aligned}
\tag{72}
$$

Note that we have dropped the subscript"2" from the function for conciseness. Because we have three equations, we can solve for the three unknown coefficients, a, b,and c. Because two of the terms in Eq. (7.20) are zero, it can be immediately solved for $c = f(x_2)$. Thus, the coefficient c is merely equal to the function value evaluated at the third guess, x_2. This result can then be substituted into Eqs. (7.18) and (7.19) to yield two equations with two unknowns:

$$
\begin{aligned}
f(x_0) - f(x_2) &= a(x_0 - x_2)^2 + b(x_0 - x_2) \\
f(x_1) - f(x_2) &= a(x_1 - x_2)^2 + b(x_1 - x_2)
\end{aligned}
$$

Algebraic manipulation can then be used to solve for the remaining coefficients, a and b. One way to do this involves defining a number of differences,

$$
\begin{aligned}
h_0 &= x_1 - x_0 & h_1 &= x_2 - x_1 \\
\delta_0 &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} & \delta_1 &= \frac{f(x_2) - f(x_1)}{x_2 - x_1}
\end{aligned}
$$

These can be substituted into Eqs.(7.21) and (7.22) to give

$$
\begin{aligned}
(h_0 + h_1)b - (h_0 + h_1)^2 a &= h_0\delta_0 + h_1\delta_1 \\
h_1 \quad b - \quad h_1^2 \quad a &= \quad h_1\delta_1
\end{aligned}
$$

which can be solved for a and b.The results can be summarized as (7.24)

$$
\begin{aligned}
a &= \frac{\delta_1 - \delta_0}{h_1 + h_0} \\
b &= a h_1 + \delta_1 \\
c &= f(x_2)
\end{aligned}
$$

(7.25) (7.26) To find the root, we apply the quadratic formula to Eq. (7.17). However, because of potential round-off error, rather than using the conventional form, we use the alternative formulation [Eq.(3.13)] to yield

$$
x_3 - x_2 = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}
$$

or isolating the unknown x_3 on the left side of the equal sign,

$$
x_3 = x_2 + \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}
$$

Note that the use of the quadratic formula means that both real and complex roots can be located. This is a major benefit of the method. In addition, Eq. (7.27a) provides a neat means to determine the approximate error. Because the left side represents the difference between the present (x_3) and the previous (x_2) root estimate, the error can be calculated as

$$\varepsilon_a = \left|\frac{x_3 - x_2}{x_3}\right| 100\%$$

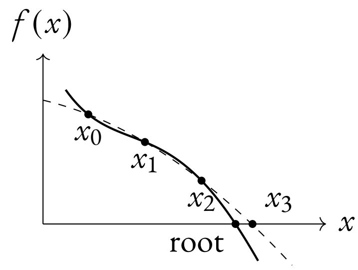


Figure 18: iteration muller

.1.2. A complete routine for the automated solving of dispersion relations

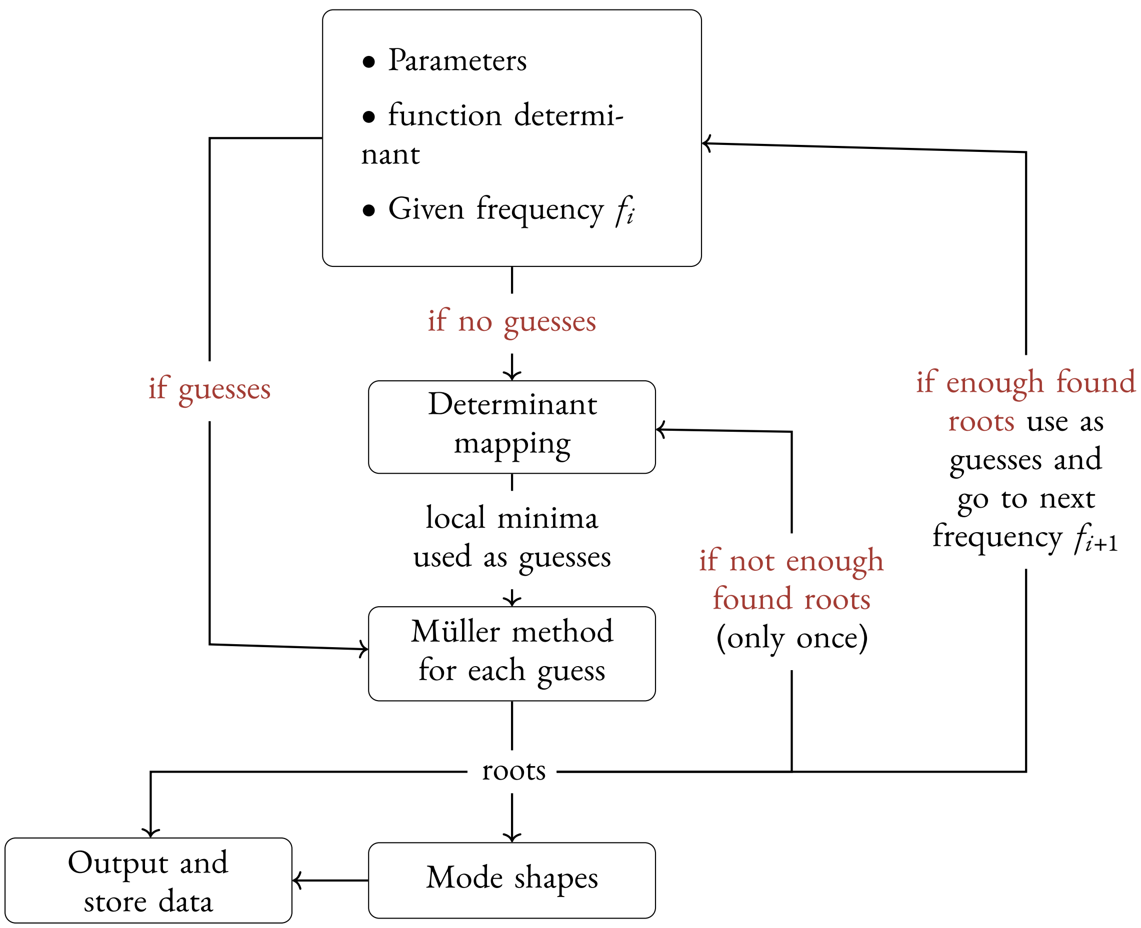


Figure 19: Description of the routine

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